

Notes on Complex Analysis
复变函数讲义

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Spring 2026
2026年春季学期

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Chapter 1

Complex Numbers

1.1 The genesis

1.1.1 Abstraction: the birth of $\mathbb{N}$

Definition 1.1. A *natural number system* is an ordered 3-tuple $(\mathbb{N}, 0, S)$ where $\mathbb{N}$ is a set whose elements are called the *natural numbers*, 0 is a distinguished element of $\mathbb{N}$ called *zero*, $S : \mathbb{N} \rightarrow \mathbb{N}$ is a 1-ary operation called the *successor function* such that the following *Peano's postulates* hold:

P1 $\forall a \in \mathbb{N}, S(a) \neq 0$;

P2 $S(a) = S(b) \implies a = b$;

P3 If $A \subseteq \mathbb{N}$ satisfies $0 \in A$ and $a \in A \implies S(a) \in A$, then $A = \mathbb{N}$.

Definition 1.2. An *isomorphism from $(\mathbb{N}, 0, S)$ to a 3-tuple (X, z, f)* where $z \in X$ and $f : X \rightarrow X$ is a bijection $h : \mathbb{N} \rightarrow X$ such that

- $h(0) = z$;
- $\forall a \in \mathbb{N}, h(S(a)) = f(h(a))$.

If h exists, we say that $(\mathbb{N}, 0, S)$ and (X, z, f) are *isomorphic*.

Theorem 1.3 (Dedekind). The natural number system in Definition 1.1 is unique.

Definition 1.4 (Von Neumann's natural numbers). *Von Neumann's natural numbers* is a natural number system constructively given by

$$0 := \emptyset; \quad S(a) = a \cup \{a\}. \quad (1.1)$$

1.1.2 Subtraction: the birth of $\mathbb{Z}$

Exercise 1.5. What would you do if you didn't know how to calculate the sum of two natural numbers, but had to compute $3+5=$?

Exercise 1.6. What is the relation of subtraction to addition?

Exercise 1.7. With respect to $\mathbb{N}$, what is the crucial difference between addition and subtraction?

Definition 1.8. A *negative integer* is a number that consists of the negative sign “ $-$ ” followed by a nonzero natural number.

Definition 1.9. The set of *integers*, denoted by $\mathbb{Z}$, is the set union of natural numbers and negative integers.

Lemma 1.10. Integers are closed under addition and subtraction.

Exercise 1.11. Prove Lemma 1.10.

1.1.3 Division: the birth of $\mathbb{Q}$

Exercise 1.12. What would you do if you did not know how to perform multiplication of natural numbers, but had to compute $3 \times 5 =$?

Exercise 1.13. What is the relation of division to multiplication?

Exercise 1.14. With respect to $\mathbb{Z}$, what is the crucial difference between multiplication and division?

Definition 1.15. A *common divisor* of two nonzero integers a, b is an integer c satisfying that there exist $d, e \in \mathbb{Z}$ such that $\frac{a}{d} = \frac{b}{e} = c$.

Definition 1.16. Two nonzero integers $a \neq b$ are said to be *relatively prime* if their largest common divisor is 1.

Definition 1.17. A *rational number* is either 0 or a number that can be expressed as the quotient $\frac{a}{b}$ of two relatively prime integers a and $b \neq 0$. The set of rational numbers is denoted by $\mathbb{Q}$.

Lemma 1.18. Nonzero rational numbers are closed under multiplication and division.

Exercise 1.19. Prove Lemma 1.18.

1.1.4 Power, root, and logarithm functions

Definition 1.20. The *n th power of a number c* is the multiplication of n copies of c , written c^n , where c is called the *base* and n the *exponent*.

Example 1.21. $5^2 = 5 \times 5 = 25$; $5^4 = 5^2 \times 5^2 = 25 \times 25 = 625$.

Exercise 1.22. What is the relation between power and multiplication?

Definition 1.23. The n th root of a number b , written $b^{\frac{1}{n}}$ or $\sqrt[n]{b}$, is the number c satisfying $c^n = b$.

Exercise 1.24. What is the relation between power and root?

Definition 1.25. The *logarithm of b with base c* , written $\log_c b$, is the number n satisfying $c^n = b$.

Exercise 1.26. What is the relation between power and logarithm?

Theorem 1.27. $\sqrt{2}$ is not a rational number.

Proof. Suppose $\sqrt{2} \in \mathbb{Q}$. Then $\exists p, q \in \mathbb{Z}$ s.t. $\sqrt{2} = \frac{p}{q}$, i.e. $p^2 = 2q^2$. Since p^2 is even, p is even. There exists $s \in \mathbb{Z}$ such that $p = 2s$, which implies $q^2 = 2s^2$, which further implies q is even. But the conclusion of both p and q being even contradicts Definitions 1.17 and 1.16. $\square$

1.1.5 Rational Cauchy sequences (RCSs): the birth of $\mathbb{R}$

Exercise 1.28. For a sequence of $\{x_n \in \mathbb{Q}\}$ with $x_0 = 2$, prove that

$$x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n} \Rightarrow \lim_{n \rightarrow \infty} x_n = \sqrt{2}. \quad (1.2)$$

Definition 1.29. Two RCSs $\{a_n\}$ and $\{b_n\}$ are *equivalent* iff $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$. In this case we write $\{a_n\} \sim \{b_n\}$.

Exercise 1.30. Prove that Definition 1.29 gives an equivalence relation (in the sense of Definition A.36) on the set of all RCSs.

Definition 1.31. The equivalence class of an RCS $\{s_n\}$ is the set of RCSs

$$\overline{\{s_n\}} := \{\{v_n\} : \{v_n\} \text{ is a RCS and } \{v_n\} \sim \{s_n\}\}. \quad (1.3)$$

Definition 1.32. Real numbers, denoted by $\mathbb{R}$, are equivalence classes that partition the set of all RCSs, i.e.,

$$\mathbb{R} := \{\overline{\{s_n\}} : \{s_n\} \text{ is a RCS}\}. \quad (1.4)$$

Definition 1.33. The *addition and multiplication of two real numbers* $s = \overline{\{s_n\}}$ and $v = \overline{\{v_n\}}$ are given by

$$s + v = \overline{\{s_n + v_n\}}, \quad s \cdot v = \overline{\{s_n \cdot v_n\}}. \quad (1.5)$$

Exercise 1.34. Prove that (1.5) is well defined. In particular, show that different representatives of the equivalence classes s and v lead to the same result.

Definition 1.35. The “less-than” relation on $\mathbb{R}$ is given by

$$s < v :\Leftrightarrow \exists \epsilon' > 0 \exists M \geq 1 \forall m \geq M, s_m \leq v_m - \epsilon', \quad (1.6)$$

where $s = \overline{\{s_n\}}$ and $v = \overline{\{v_n\}}$ are two real numbers. Then

$$s \leq v \Leftrightarrow s < v \text{ or } s = v. \quad (1.7)$$

Lemma 1.36. The relation “ $\leq$ ” in (1.7) is a total order, i.e., $\forall s, v \in \mathbb{R}$ with $s \neq v$, either $s < v$ or $v < s$.

Exercise 1.37. Prove Lemma 1.36.

Definition 1.38. The *absolute value of a real number* is given by

$$\forall s = \overline{\{s_n\}} \in \mathbb{R}, \quad |s| := \overline{\{|s_n|\}}. \quad (1.8)$$

Theorem 1.39 (Cauchy 1821). A sequence of real numbers is convergent (with a real number as its limit) if and only if it is a Cauchy sequence.

Exercise 1.40. Prove Theorem 1.39 via the viewpoint of Definition 1.32.

Lemma 1.41. $\mathbb{R}$ is closed under $+$, $-$, $\times$, and $\div$. $\mathbb{R}^+$ (but not $\mathbb{R}$) is closed under $\sqrt[n]{\cdot}$ if n is an even integer.

Exercise 1.42. Prove Lemma 1.41.

1.1.6 Cubic equations: the birth of $\sqrt{-1}$

Definition 1.43. The *quadratic equation of one unknown* is an equation of the form

$$ax^2 + bx + c = 0, \quad (1.9)$$

which, in the case of $a \neq 0$, is equivalent to

$$x^2 - mx - k = 0. \quad (1.10)$$

Exercise 1.44. The solution of the quadratic equation can be interpreted as the intersection of the parabola $y = x^2$ and the straight-line $y = mx + c$. What situation does $m^2 + 4c < 0$ correspond to?

Definition 1.45. The *cubic equation of one unknown* is an equation of the form

$$X^3 + AX^2 + BX + C = 0, \quad (1.11)$$

which, by the variable substitution $X = x - \frac{A}{3}$, is equivalent to

$$x^3 = 3px + 2q, \quad (1.12)$$

where p and q are algebraic expressions of A, B, C .

Formula 1.46. A solution of the cubic equation (1.12) can be found by steps as follows.

(i) $x = s + t$ is a solution of (1.12) if s, t satisfy

$$st = p, \quad s^3 + t^3 = 2q;$$

(ii) the above equation yields a quadratic equation on s^3 ;

(iii) we solve the quadratic equation to get s and t ;

(iv) a solution of (1.12) is then

$$x = \sqrt[3]{q + \sqrt{q^2 - p^3}} + \sqrt[3]{q - \sqrt{q^2 - p^3}}.$$

Example 1.47. Find a solution for $x^3 = 6x + 6$.

Exercise 1.48. What is the crucial difference between the two cubic equations $x^3 = 6x + 6$ and $x^3 = 15x + 4$?

Exercise 1.49. For (1.12), what situation does $q^2 - p^3 < 0$ correspond to? What is the crucial difference between this situation and that of $m^2 + 4c < 0$ for quadratic equations?

1.2 The field $\mathbb{C}$ of complex numbers

Theorem 1.50. There exists a field $\mathbb{C}$ such that

- (a) the field $\mathbb{R}$ of real numbers is a subfield of $\mathbb{C}$, i.e., $\mathbb{R} \subset \mathbb{C}$ and addition and multiplication in $\mathbb{R}$ are the restrictions to $\mathbb{R}$ of the addition and multiplication in $\mathbb{C}$;
- (b) in $\mathbb{C}$ the equation $X^2 + 1 = 0$ has exactly two solutions $+\mathbf{i}$ and $-\mathbf{i}$;
- (c) the following map is a bijection:

$$(x, y) \mapsto x + \mathbf{i}y : \quad \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}. \quad (1.13)$$

Proof. Define addition and multiplication on $\mathbb{R} \times \mathbb{R}$ as

$$(x, y) + (u, v) := (x + u, y + v), \quad (1.14a)$$

$$(x, y) \cdot (u, v) := (xu - yv, xv + yu). \quad (1.14b)$$

It follows from Definition A.32, Definition C.1, and (1.14) that $\mathbb{R} \times \mathbb{R}$ is a field with its additive identity as $(0, 0)$ and its multiplicative identity as $(1, 0)$. Denote $\mathbf{i} := (0, 1)$. Then $\mathbf{i}$ and $-\mathbf{i}$ are the two solutions of $X^2 + 1 = 0$. The last statement of (c) holds because $(1, 0)$ and $\mathbf{i}$ form an orthogonal basis of the linear space $\mathbb{R}^2$. $\square$

Definition 1.51. In the unique representation of a complex number $z = x + \mathbf{i}y$ in (1.13), $x \in \mathbb{R}$ and $y \in \mathbb{R}$ are respectively the *real part* and the *imaginary part* of z , respectively denoted by $\operatorname{Re} z$ and $\operatorname{Im} z$.

Definition 1.52. The field of *complex numbers* is the field $\mathbb{C}$ in Theorem 1.50, with their addition and multiplication given by

$$(x + \mathbf{i}y) + (u + \mathbf{i}v) := (x + u) + \mathbf{i}(y + v), \quad (1.15a)$$

$$(x + \mathbf{i}y)(u + \mathbf{i}v) := (xu - yv) + \mathbf{i}(xv + yu). \quad (1.15b)$$

1.2.1 $\mathbb{C}$ is not an ordered field

Definition 1.53. A field $\mathcal{F}$ is said to be *ordered* iff there exists a subset $\mathcal{P} \subset \mathcal{F}$, called *the set of positive elements* of $\mathcal{F}$, satisfying

- (ODF-1) closure: $\forall x, y \in \mathcal{P}, x + y \in \mathcal{P}$ and $xy \in \mathcal{P}$.
- (ODF-2) trichotomy: for each $x \in \mathcal{F}$, one and only one of $x = 0$, $x \in \mathcal{P}$, and $x \in -\mathcal{P}$ is true.

Lemma 1.54. An ordered field $\mathcal{F}$ satisfies

$$\forall x \in \mathcal{F}, \quad x^2 := x \cdot x \geq 0, \quad (1.16)$$

where the equality hold only when $x = 0$. In particular, we have $1 > 0$.

Proof. Once the set $\mathcal{P}$ of positive elements in a field has been specified, it is easier to work with inequalities than with $\mathcal{P}$: we write $x > y$ if and only if $x - y \in \mathcal{P}$; $x \geq y$ and $x < y$ are also defined similarly. The two conditions in Definition 1.53 are equivalent to

- $x > 0, y > 0$ implies $x + y > 0$ and $xy > 0$;

- for each $x \in \mathcal{F}$, one and only one of $x = 0$, $x > 0$, and $x < 0$ is true,

which completes the proof. $\square$

Corollary 1.55. $\mathbb{R}$ is an *ordered field* while $\mathbb{C}$ is not.

Exercise 1.56. Prove Corollary 1.55.

Corollary 1.57. No real number satisfies $x^2 = -1$.

Proof. This follows directly from Corollary 1.55 and Lemma 1.54. $\square$

1.2.2 Solving cubic and quartic equations

Example 1.58. Equipped with complex numbers, Formula 1.46 always yields a solution for any cubic equations. Take $x^3 = 15x + 4$ for an example, we have

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}.$$

By (1.15), it is straightforward to verify

$$(11\mathbf{i})^2 = -121, \quad (2 \pm \mathbf{i})^3 = 2 \pm 11\mathbf{i}.$$

Hence we have $\sqrt{-121} = \pm 11\mathbf{i}$ and

$$x = \sqrt[3]{2 + 11\mathbf{i}} + \sqrt[3]{2 - 11\mathbf{i}} = (2 + \mathbf{i}) + (2 - \mathbf{i}) = 4,$$

which is indeed a solution of $x^3 = 15x + 4$. Therefore, the definition of “complex numbers” is useful as it helps to solve realworld problems that are difficult to solve without complex numbers.

Formula 1.59. The following generic quartic equation

$$ax^4 + bx^3 + cx^2 + dx + e = 0 \quad (1.17)$$

is solved by steps as follows.

WLOG, we can assume $a = 1$. Move the quadratic, linear, and constant terms to the right-hand side (RHS), add $(\frac{b}{2}x)^2$ to both sides, and we have

$$x^4 + bx^3 + \frac{1}{4}b^2x^2 = \frac{1}{4}(b^2 - 4c)x^2 - dx - e.$$

Introduce another parameter $y \in \mathbb{R}$, add $(x^2 + \frac{b}{2}x)y + \frac{1}{4}y^2$ to both sides of the above equation, and we have

$$\left[x^2 + \frac{b}{2}x + \frac{1}{2}y\right]^2 = \frac{1}{4}(b^2 - 4c + 4y)x^2 + \frac{1}{2}(by - 2d)x + \frac{1}{4}y^2 - e.$$

The RHS is a quadratic form of x and it can be factored into a square if

$$\frac{1}{4}(by - 2d)^2 - (b^2 - 4c + 4y)\left(\frac{1}{4}y^2 - e\right) = 0,$$

which is a cubic equation of y and can be solved by Formula 1.46. Then we can get two quadratic equations, from which we obtain the four solutions of (1.17).

Exercise 1.60. Use Formula 1.59 to solve the equation

$$x^4 + 2x^3 + 2x + 1 = 0. \quad (1.18)$$

1.2.3 Complex arithmetic

Definition 1.61. The *complex conjugate* of $z = x + iy$ is defined as $\bar{z} = x - iy$.

Corollary 1.62. $\forall z, w \in \mathbb{C}$, we have

- (1) $\bar{\bar{z}} = z$,
- (2) $\overline{z \pm w} = \bar{z} \pm \bar{w}$, $\overline{zw} = \bar{z} \cdot \bar{w}$,
- (3) $z \in \mathbb{R} \Leftrightarrow z = \bar{z}$, $z \in i\mathbb{R} \Leftrightarrow z = -\bar{z}$.
- (4) $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$, $\operatorname{Im} z = \frac{1}{2i}(z - \bar{z})$,

Definition 1.63. The *absolute value* or *modulus* of a complex number $z = x + iy$ is defined by

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \quad (1.19)$$

Corollary 1.64. $\forall z \in \mathbb{C}$, $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$.

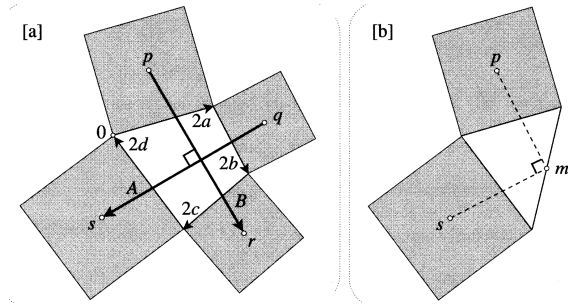
Corollary 1.65. The *inverse* of a complex number $z \neq 0$ is

$$z^{-1} = \frac{\bar{z}}{|z|^2}. \quad (1.20)$$

Corollary 1.66. $\forall z, w \in \mathbb{C}$, we have

- (1) $|z \cdot w| = |z| \cdot |w|$,
- (2) $|z \pm w| \leq |z| + |w|$,
- (3) $||z| - |w|| \leq |z \pm w|$.

Exercise 1.67. Construct four squares for a quadrilateral, as shown in [a].



Prove that the line-segments joining the centers of opposite squares are perpendicular and of equal length; see [b].

1.3 Complex sequences and series

Notation 1. The *open disk* and the *punctured open disk* centered at c with radius r are, respectively,

$$\mathcal{U}_r(c) := \{x \in \mathbb{C} : |x - c| < r\}, \quad (1.21)$$

$$\dot{\mathcal{U}}_r(c) := \{x \in \mathbb{C} : 0 < |x - c| < r\}. \quad (1.22)$$

The closure and the boundary of $\mathcal{U}_r(c)$ are denoted by $\overline{\mathcal{U}_r(c)}$ and $\partial\mathcal{U}_r(c)$, respectively.

1.3.1 Sequences of complex numbers

Definition 1.68. A sequence $(z_n)_{n \in \mathbb{N}}$ of complex numbers is called a *null sequence* iff

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, |z_n| < \epsilon. \quad (1.23)$$

Definition 1.69. A sequence $(z_n)_{n \in \mathbb{N}}$ of complex numbers *converges* to the complex number z iff the sequence of differences $(z_n - z)_{n \in \mathbb{N}}$ is a null sequence.

Lemma 1.70. Let (z_n) be a sequence of complex numbers and z another complex number. The following statements are equivalent:

- $\lim_{n \rightarrow \infty} z_n = z$,
- $\lim_{n \rightarrow \infty} \operatorname{Re} z_n = \operatorname{Re} z$; $\lim_{n \rightarrow \infty} \operatorname{Im} z_n = \operatorname{Im} z$.

Lemma 1.71. Suppose two complex sequences converge: $\lim_{n \rightarrow \infty} z_n = z$ and $\lim_{n \rightarrow \infty} w_n = w$. Then we have

- (1) $\lim_{n \rightarrow \infty} (z_n \pm w_n) = z \pm w$,
- (2) $\lim_{n \rightarrow \infty} z_n \cdot w_n = z \cdot w$,
- (3) $\lim_{n \rightarrow \infty} |z_n| = |z|$,
- (4) $\lim_{n \rightarrow \infty} \overline{z_n} = \bar{z}$,
- (5) $z \neq 0, z_n \neq 0 \Rightarrow \lim_{n \rightarrow \infty} z_n^{-1} = z^{-1}$.

1.3.2 Complex power series

Definition 1.72. A *complex power series* (with center zero) is a series of the form

$$p(z) = \sum_{n=0}^{\infty} a_n z^n, \quad (1.24)$$

where the coefficients $a_j \in \mathbb{C}$ are given.

Definition 1.73. A complex series $\sum_{n=0}^{\infty} f_n$ is *absolutely convergent* if the sequence $(\sum_{i=0}^n |f_i|)_{n \in \mathbb{N}}$ converges.

Definition 1.74. A series $\sum_{n=0}^{\infty} f_n$ of complex functions $f_n : \mathcal{D} \rightarrow \mathbb{C}, \mathcal{D} \subset \mathbb{C}$ is said to *converge normally* on $\mathcal{D}$ iff for each $a \in \mathcal{D}$ there exist an open ball $\mathcal{U}_\epsilon(a)$ and a sequence of non-negative constants $\{M_n\}$ such that

- $\forall x \in \mathcal{D} \cap \mathcal{U}_\epsilon(a), \forall n \in \mathbb{N}, |f_n(x)| \leq M_n$,
- the series $\sum_{n=0}^{\infty} M_n$ converges.

Theorem 1.75 (Weierstrass M-test). Normal convergence implies absolute and locally uniform convergence.

Proof. Let $S_n(x) = \sum_{j=0}^n f_j(x)$. By normal convergence of S_∞ and the Cauchy criterion (Lemma B.43), we have

$$\forall a \in \mathcal{D}, \exists \epsilon > 0 \text{ s.t. } \mathcal{U}_\epsilon(a) \subset \mathcal{D}, \text{ and } \exists (M_n), \exists N \in \mathbb{N}, \text{ s.t.}$$

$$\forall n > N, \forall x \in \mathcal{U}_\epsilon(a), |S_n(x) - S_N(x)| \leq \sum_{j=N+1}^n M_j,$$

which shows the local uniform convergence of $(S_n(x))$ since $\lim_{n, N \rightarrow \infty} \sum_{j=N+1}^n M_j = 0$.

The absolute convergence follows directly from Definition B.46, the two conditions of normal convergence in Definition 1.74, and the comparison test (Theorem B.44). $\square$

Corollary 1.76. A normally convergent series of functions can be arbitrarily re-ordered without changing its convergence or its limit.

Proof. This follows from absolute convergence. $\square$

Theorem 1.77 (Abel 1826). For each power series (1.24), there exists a number r uniquely determined from a_j ,

$$r := \sup\{t \geq 0 : (a_n t^n) \text{ is a bounded sequence}\}, \quad (1.25)$$

such that

- (a) the series (1.24) converges normally in the open disk $\mathcal{U}_r(0)$,
- (b) the series (1.24) diverges in the open set $|z| > r$ for each $z \in \mathbb{C}$.

Proof. Conclusion (b) follows directly from the definition of r as a supremum in (1.25). If $r = 0$, conclusion (a) holds vacuously. For $r > 0$, it suffices to show that $\forall \rho \in (0, r)$, the series (1.24) satisfies the two conditions in Definition 1.74. Choose $\rho_1 \in (\rho, r)$ and we know from (1.25) that

$$\exists M \in \mathbb{R}^+ \text{ s.t. } \forall n \in \mathbb{N}, |a_n \rho_1^n| < |a_n r^n| < M < +\infty.$$

It follows that

$$\forall n \in \mathbb{N}, \forall z \in \mathcal{U}_\rho(0), \quad |a_n z^n| = \left| a_n \rho_1^n \frac{z^n}{\rho_1^n} \right| \leq M \frac{\rho^n}{\rho_1^n}.$$

Hence the first condition in Definition 1.74 holds by setting $M_n = M \frac{\rho^n}{\rho_1^n}$. The second condition follows from the convergence of the geometric series $\sum M_n$. $\square$

Definition 1.78. For the power series (1.24), the number r in (1.25) and the open disk $\mathcal{U}_r(0)$ are respectively called its *convergence radius* and its *convergence disk*.

Theorem 1.79. If the power series (1.24) satisfies $a_n \neq 0$ for all $n \in \mathbb{N}$, then its convergence radius r is determined by

$$\frac{1}{r} = L := \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|, \quad (1.26)$$

where $L \in [0, +\infty]$ and we adopt the convention that $\frac{1}{0} = +\infty$ and $\frac{1}{\infty} = 0$.

Proof. As a property of limit superior, the existence of $\lim_{n \rightarrow \infty} v_n$ implies

$$\limsup_{n \rightarrow \infty} (u_n v_n) = (\limsup_{n \rightarrow \infty} u_n) (\lim_{n \rightarrow \infty} v_n). \quad (1.27)$$

The rest of the proof follows from applying the ratio test to the power series (1.24). $\square$

Example 1.80. For $\sum_{n=1}^{\infty} \cos(ni)(z-1)^n$, its radius of convergence is e^{-1} .

Theorem 1.81 (Cauchy-Hadamard). The radius of convergence r of the power series (1.24) is given by

$$\frac{1}{r} = L := \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}. \quad (1.28)$$

Proof. If the power series (1.24) converges for $|z| = \rho$, we have

$$\limsup_{n \rightarrow \infty} |a_n z^n|^{\frac{1}{n}} = \rho L.$$

The root test then completes the proof. $\square$

Exercise 1.82. Determine the convergence radius of the series $\sum_{n=1}^{\infty} a_n z^n$ where

(a) $a_n = (\log n)^2$;

(b) $a_n = n!$;

(c) $a_n = \frac{n^2}{4^n + 3^n}$;

(d) $a_n = \frac{(n!)^3}{(3n)!}$.

Hint: Use Stirling's formula $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

1.4 Fundamental complex functions

Notation 2. $\mathbb{C}^\bullet := \mathbb{C} \setminus \{0\}$.

1.4.1 The exponential function

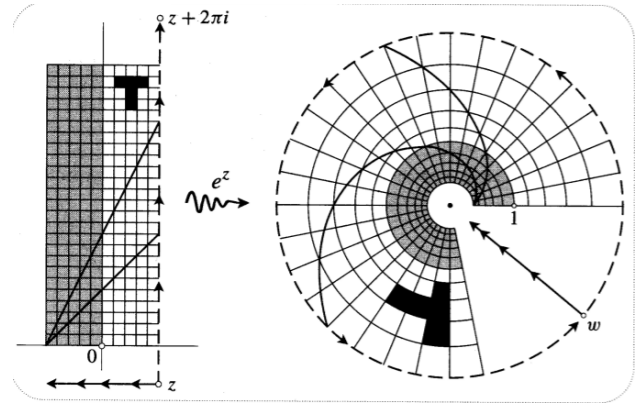
Definition 1.83. The *exponential function* $\exp : \mathbb{C} \rightarrow \mathbb{C}$ is defined as the power series

$$e^z = \exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}. \quad (1.29)$$

Definition 1.84. The *Euler's number*, or the *number e* , is $e := \exp(1)$.

Example 1.85. There are three common ways to visualize complex functions. In the first type, a z -plane (or $x-y$ -plane) that includes the domain of f is drawn separately from a w -plane (or $u-v$ -plane) that includes the range of f . Simple subsets of $\mathbb{C}$ are drawn on the z -plane with their corresponding images under f drawn on the w -plane.

Example 1.86. The following dual-plane plots visualize the transformative actions of $z \mapsto \exp(z)$.



Lemma 1.87 (Cauchy multiplication theorem). Two absolutely convergent series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ satisfy

$$\sum_{n=0}^{\infty} \sum_{k=0}^n a_k b_{n-k} = \left(\sum_{n=0}^{\infty} a_n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \right). \quad (1.30)$$

Proof. From the following table,

	b_0	b_1	$\dots$	b_m
a_0	a_0b_0	a_0b_1	$\dots$	a_0b_m
a_1	a_1b_0	a_1b_1	$\dots$	a_1b_m
$\vdots$	$\dots$	$\dots$	$\dots$	$\dots$
a_m	a_mb_0	a_mb_1	$\dots$	a_mb_m

we define

$$A_m = \sum_{n=0}^m a_n, \quad B_m = \sum_{n=0}^m b_n, \quad (1.31)$$

$$c_n = \sum_{k=0}^n a_k b_{n-k}, \quad C_m = \sum_{n=0}^m c_n. \quad (1.32)$$

Clearly, $A_m B_m$ is the sum of all entries in the table, c_n is the sum of entries on an antidiagonal, and C_m that for all antidiagonals $n \leq m$. Since $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are absolutely convergent, we have

$$\sum_{n=0}^{\infty} |a_n| = A < \infty, \quad \sum_{n=0}^{\infty} |b_n| = B < +\infty,$$

$$\left(\sum_{n=0}^{\infty} |a_n| \right) \left(\sum_{n=0}^{\infty} |b_n| \right) = AB < \infty.$$

By Theorem B.14, the sequence of partial sums $(\sum_{n=0}^m d_n)_m$ with $d_n = \sum_{k=0}^n |a_k| |b_{n-k}|$ converges because it is monotone and bounded above by AB . Hence $\lim_{m \rightarrow \infty} \sum_{n=m+1}^{\infty} d_n = 0$. It follows that

$$\lim_{m \rightarrow \infty} |A_m B_m - C_m| \leq \lim_{m \rightarrow \infty} \sum_{n=m+1}^{\infty} |d_n| = 0,$$

which yields (1.30). $\square$

Theorem 1.88 (Cauchy's product formula). If the convergence radius for both series $A = \sum_{n=0}^{\infty} a_n z^n$ and $B = \sum_{n=0}^{\infty} b_n z^n$ is at least $R > 0$, then we have

$$\forall z \in \mathcal{U}_R(0), \quad \left(\sum_{n=0}^{\infty} a_n z^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n z^n \right) = \sum_{n=0}^{\infty} c_n z^n, \quad (1.33)$$

where $c_n := \sum_{k=0}^n a_k b_{n-k}$.

Proof. For $|z| < R$, both A and B are normally convergent, hence absolute convergent by Theorem 1.75. Then the Cauchy multiplication theorem (Lemma 1.87) completes the proof. $\square$

Theorem 1.89. $\forall z, w \in \mathbb{C}$, $\exp(z+w) = \exp(z) \cdot \exp(w)$.

Proof. This follows from Lemma 1.87, (1.29), and the binomial theorem. $\square$

Corollary 1.90. $\forall z \in \mathbb{C}$, $\exp(z) \neq 0$.

Proof. This follows from Theorem 1.89 with $w = -z$. $\square$

Lemma 1.91. The exponential function satisfies

(EXP-1) The restriction of $\exp$ to $\mathbb{R}$ is a monotonically increasing positive function with $\lim_{x \rightarrow -\infty} e^x = 0$ and $\lim_{x \rightarrow +\infty} e^x = +\infty$.

(EXP-2) There exists a positive number π such that $e^{\frac{\pi}{2}\mathbf{i}} = \mathbf{i}$ and such that $e^z = 1$ if and only if $\frac{z}{2\pi\mathbf{i}} \in \mathbb{Z}$.

(EXP-3) $\exp$ is a periodic function with period $2\pi\mathbf{i}$.

(EXP-4) The mapping $\theta \mapsto e^{i\theta}$ maps $\mathbb{R}$ onto the unit circle.

(EXP-5) $w \in \mathbb{C}^\bullet$ implies $w = e^z$ for some $z \in \mathbb{C}$.

Exercise 1.92. Prove Lemma 1.91.

Hint: For (EXP-2), one can define $\cos(\theta) := \operatorname{Re}(e^{i\theta})$, $\sin(\theta) := \operatorname{Im}(e^{i\theta})$, and look for the zeros of these functions.

1.4.2 Trigonometric functions

Definition 1.93. The *complex sine* and *complex cosine* are defined as the complex series

$$\sin(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}, \quad (1.34)$$

$$\cos(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}. \quad (1.35)$$

Corollary 1.94. $\forall z \in \mathbb{C}$, we have

$$\exp(\mathbf{i}z) = \cos z + \mathbf{i} \sin z, \quad (1.36)$$

$$\cos(z) = \frac{\exp(\mathbf{i}z) + \exp(-\mathbf{i}z)}{2}, \quad (1.37)$$

$$\sin(z) = \frac{\exp(\mathbf{i}z) - \exp(-\mathbf{i}z)}{2\mathbf{i}}. \quad (1.38)$$

Proof. These follow from Definitions 1.83 and 1.93. $\square$

Corollary 1.95. $\forall z \in \mathbb{C}$ and $\forall k \in \mathbb{Z}$, we have

$$\sin z = 0 \Leftrightarrow z = k\pi, \quad (1.39a)$$

$$\cos z = 0 \Leftrightarrow z = k\pi + \frac{\pi}{2}. \quad (1.39b)$$

1.4.3 Euler's formula

Corollary 1.96 (Euler's formula).

$$\forall \theta \in \mathbb{R}, \quad \exp(\mathbf{i}\theta) = \cos \theta + \mathbf{i} \sin \theta. \quad (1.40)$$

Corollary 1.97. $\forall z = x + \mathbf{i}y$, $e^z = e^x(\cos y + \mathbf{i} \sin y)$.

Corollary 1.98. $\forall z, w \in \mathbb{R}$, we have

$$\cos(z+w) = \cos z \cos w - \sin z \sin w, \quad (1.41)$$

$$\sin(z+w) = \sin z \cos w + \cos z \sin w. \quad (1.42)$$

Proof. We have

$$\begin{aligned} \cos(\theta + \phi) + \mathbf{i} \sin(\theta + \phi) &= e^{\mathbf{i}(\theta + \phi)} = e^{\mathbf{i}\theta} e^{\mathbf{i}\phi} \\ &= (\cos \theta + \mathbf{i} \sin \theta)(\cos \phi + \mathbf{i} \sin \phi) \\ &= (\cos \theta \cos \phi - \sin \theta \sin \phi) + \mathbf{i}(\sin \theta \cos \phi + \cos \theta \sin \phi), \end{aligned}$$

where the first and the third step follow from Euler's formula, the second from Theorem 1.89, and the fourth step from Definition 1.52. The proof is completed by the fact that the equality of two complex numbers implies the equality of both the real parts and the imaginary parts. $\square$

Exercise 1.99. Prove $\cos(3\theta) = C^3 - 3CS^2$ and $\sin(3\theta) = 3SC^2 - S^3$, where $C = \cos \theta$ and $S = \sin \theta$.

Exercise 1.100. Prove $\tan(3\theta) = \frac{3T - T^3}{1 - 3T^2}$ where $T = \tan \theta$.

1.4.4 Hyperbolic sine and cosine

Definition 1.101. The complex functions of *hyperbolic sine* and *hyperbolic cosine* are defined as

$$\cosh(z) := \frac{\exp(z) + \exp(-z)}{2}, \quad (1.43)$$

$$\sinh(z) := \frac{\exp(z) - \exp(-z)}{2}. \quad (1.44)$$

Corollary 1.102. Hyperbolic sine and hyperbolic cosine functions have period $2\pi i$ and satisfy

$$\cosh(z + w) = \cosh z \cosh w + \sinh z \sinh w, \quad (1.45)$$

$$\sinh(z + w) = \sinh z \cosh w + \cosh z \sinh w, \quad (1.46)$$

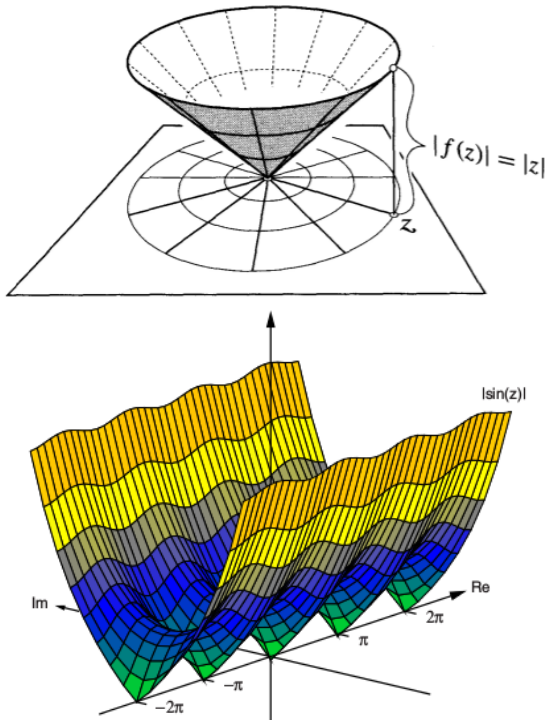
$$\cosh z = \cos(iz), \quad \sinh z = -i \sin(iz), \quad (1.47)$$

$$\cos z = \cosh(iz), \quad \sin z = -i \sinh(iz), \quad (1.48)$$

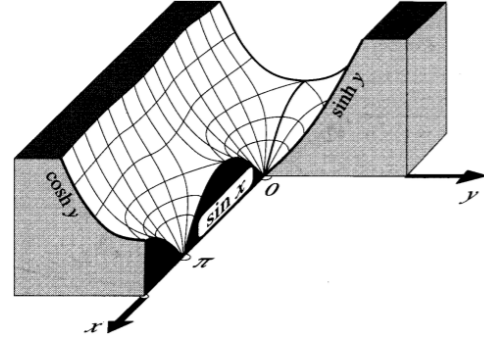
$$\cosh^2 z - \sinh^2 z = 1. \quad (1.49)$$

Definition 1.103. A *surface patch* is (the image of) a continuous map $\sigma : (0, 1)^2 \rightarrow \mathbb{R}^3$.

Example 1.104. The second way to visualize a complex function $f(z)$ is to plot the *modular surface*, or *analytic mountainscape* or *analytic landscape*, where the argument of $f(z)$ is discarded and $|f(z)|$ is plotted against each point z to obtain a surface patch $z \mapsto (z, |f(z)|)$. Two examples are as follows. These mountains have no peaks, only valley bottoms that reach down to the complex plane. So it is not fun to climb these mountains!



Example 1.105. Corollary 1.102 unites sine, cosine, hyperbolic sine and cosine functions. As illustrated in the following figure, taking different cross sections of the modular surface of $f(z) = \sin(z)$ yields these different functions.



Exercise 1.106. What are the shapes of images of straight lines $x = \text{const}$ and $y = \text{const}$ under $z = x + iy \mapsto \cos(z)$? Draw a dual-plane plot similar to that in Example 1.86 to answer this question.

Hint: Use Corollary 1.102.

1.4.5 The natural logarithm

Theorem 1.107. The following map is surjective but not injective,

$$(r, \varphi) \mapsto r(\cos \varphi + i \sin \varphi) : \mathbb{R}^+ \times \mathbb{R} \rightarrow \mathbb{C}^\bullet. \quad (1.50)$$

Definition 1.108. In the polar coordinate representation of $z \in \mathbb{C}^\bullet$,

$$z = |z|(\cos \varphi + i \sin \varphi), \quad (1.51)$$

each $\varphi \in \mathbb{R}$ is called an *argument* of z . The unique argument within the interval $(-\pi, \pi]$ is called the *principal value of the argument* of z and denoted by $\text{Arg}(z)$.

Theorem 1.109 (Addition theorem for circular functions).

$$(\cos \varphi + i \sin \varphi)(\cos \phi + i \sin \phi) = \cos(\varphi + \phi) + i \sin(\varphi + \phi).$$

Theorem 1.110. The *principal branch of the logarithm* $\text{Log} : \mathbb{C}^\bullet \rightarrow \mathbb{C}$ is uniquely determined by

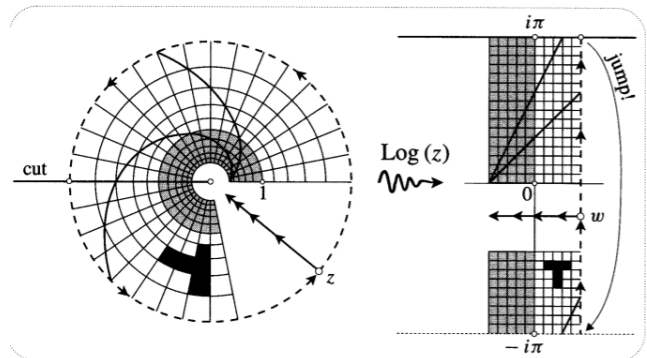
- $\exp(\text{Log } z) = z$,
- $\forall z \neq 0, \text{Im } \text{Log } z \in (-\pi, \pi]$.

Theorem 1.111. $\forall z \in \mathbb{C}^\bullet, \text{Log } z = \log |z| + i \text{Arg } z$.

Proof. This follows directly from Definition 1.108 and Theorem 1.110. $\square$

Example 1.112. As negative consequences for forcing the logarithm to be single-valued, a discontinuity shows up at the boundary of $(-\pi, \pi]$ and familiar rules for real numbers may not hold any more, e.g.

$$\exists a, b \in \mathbb{C} \text{ s.t. } \text{Log}(ab) \neq \text{Log}(a) + \text{Log}(b).$$



Example 1.113. The principal value of a^b for $a, b \in \mathbb{C}$ can be defined as the element with $k = 0$ in the following set

$$\{a^b\} := \{\exp(b \operatorname{Log} a) \exp(2\pi i b k) : k \in \mathbb{Z}\}.$$

1.4.6 The power function

Definition 1.114. The *integer power of a complex number* z is

$$z^n = \begin{cases} n \text{ copies of } z \text{ multiplied} & \text{if } n \in \mathbb{Z}^+; \\ 1 & \text{if } n = 0; \\ (z^{-1})^{-n} & \text{otherwise.} \end{cases} \quad (1.52)$$

Theorem 1.115. For each $n \in \mathbb{N}^+$ there exist exactly n different n th roots of unity, namely

$$\zeta_k := \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad 0 \leq k < n, \quad (1.53)$$

each of which satisfies the *cyclotomic equation* $z^n = 1$.

Exercise 1.116. With $\omega = se^{i\varphi}$, where $s \geq 0$ and $\varphi \in \mathbb{R}$, solve the equation $z^n = \omega$ in $\mathbb{C}$ where n is a natural number. How many solutions are there?

Definition 1.117. The *complex power function* is

$$z^w = \exp(w \operatorname{Log} z). \quad (1.54)$$

Exercise 1.118. Calculate i^i .

Exercise 1.119. We have been augmenting sets of numbers: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. For $n \in \mathbb{Z}$ and $x, z \in \mathbb{X}$ where $\mathbb{X}$ is a set in the first column, is $\mathbb{X}$ closed under the operations in the first row?

	$+/-$	$\times/\div$	$\cdot^n$	$\sqrt[n]{\cdot}$	e^x	z^x
$\mathbb{N}$						
$\mathbb{Z}$						
$\mathbb{Q}$						
$\mathbb{R}$						
$\mathbb{C}$						

Use the above table to tell a story of marching from $\mathbb{N}$ to $\mathbb{C}$.

1.5 The group isomorphism of $\mathbb{C}^\bullet$ to similarity transformations on $\mathbb{R}^2$

Lemma 1.120. Define

$$\mathbb{R}_N^{2 \times 2} = \left\{ \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix} : \alpha, \beta \in \mathbb{R}; \alpha^2 + \beta^2 \neq 0 \right\}. \quad (1.55)$$

Then the following algebra is an Abelian group,

$$\mathbf{M} = (\mathbb{R}_N^{2 \times 2}, *, ^{-1}, I), \quad (1.56)$$

where $*$ denotes matrix multiplication, $^{-1}$ matrix inverse, and I the 2-by-2 identity matrix.

Proof. It is straightforward to verify the conditions in Definition G.6, e.g., the invertibility holds because

$$\begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}^{-1} = \frac{1}{\alpha^2 + \beta^2} \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}. \quad \square$$

Theorem 1.121 (The isomorphism of $\mathbb{C}^\bullet$ and similarities). Define an Abelian group

$$\mathbf{C}_m = (\mathbb{C}^\bullet, *, ^{-1}, 1), \quad (1.57)$$

where $*$ denotes complex multiplication, $^{-1}$ the inverse of a complex number, and 1 the multiplication identity of $\mathbb{C}^\bullet$. Then $\mathbf{C}_m$ is isomorphic to (1.56) with the isomorphism as

$$h(\alpha + i\beta) = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix} \quad (1.58)$$

Proof. We verify that (1.58) is a homomorphism by

$$\begin{aligned} h((x + iy)(u + iv)) &= h((xu - yv) + i(xv + yu)) \\ &= \begin{bmatrix} x & -y \\ y & x \end{bmatrix} \cdot \begin{bmatrix} u & -v \\ v & u \end{bmatrix} \\ &= h(x + iy)h(u + iv). \end{aligned}$$

Then the proof is completed by Lemma 1.120 and the bijectiveness of (1.58). $\square$

Lemma 1.122. For an $\mathbb{R}$ -linear map $A : \mathbb{C} \rightarrow \mathbb{C}$, the following statements are equivalent:

- (a) $\exists \ell \in \mathbb{C}$ s.t. $\forall z \in \mathbb{C}, A(z) = \ell z$.
- (b) A is $\mathbb{C}$ -linear.
- (c) $A(i) = iA(1)$.
- (d) The matrix of A with respect to the canonical basis $1 = (1, 0)$ and $i = (0, 1)$ has the special form

$$\begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}, \quad \alpha, \beta \in \mathbb{R}. \quad (1.59)$$

Exercise 1.123. Prove Lemma 1.122.

Chapter 2

Complex Differentiation

Notation 3. The n -sphere, the open n -ball, and the n -ball are, respectively,

$$\mathbb{S}^n := \{\mathbf{x} \in \mathbb{R}^{n+1} : \|\mathbf{x}\| = 1\}, \quad (2.1)$$

$$\mathbb{D}^n := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| < 1\}, \quad (2.2)$$

$$\overline{\mathbb{D}^n} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq 1\}, \quad (2.3)$$

where the norm is the Euclidean norm. We will often use the shorthand notation $\mathbb{E} := \mathbb{D}^2$.

2.1 Continuity of complex functions

Definition 2.1. A function $f : \mathcal{D} \rightarrow \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ is said to be *continuous at a point* $a \in \mathcal{D}$ iff

$$\forall \epsilon > 0, \quad \exists \delta > 0 \text{ s.t. } \forall z \in \mathcal{D} \setminus \{a\}, \quad |z - a| < \delta \implies |f(z) - f(a)| < \epsilon, \quad (2.4)$$

where $|\cdot|$ denotes the Euclidean norm. f is *continuous* on $\mathcal{D}$ iff it is continuous at each point $a \in \mathcal{D}$.

Lemma 2.2. A function $f : \mathcal{D} \rightarrow \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ is continuous at a if and only if $\lim_{n \rightarrow +\infty} f(a_n) = f(a)$ for *each* sequence (a_n) that converges to a .

Proof. This follows from Definitions 2.1 and B.3. $\square$

Lemma 2.3. The sum, difference, and product of two continuous functions are continuous.

Proof. This follows from Definition 2.1. $\square$

Example 2.4. A complex polynomial of degree $n \in \mathbb{N}$

$$p_n(z) = \sum_{j=0}^n a_j z^j, \quad a_j \in \mathbb{C}. \quad (2.5)$$

is continuous.

Lemma 2.5. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ is continuous if and only if its real and imaginary parts

$$(\operatorname{Re} f)(z) := \operatorname{Re} f(z), \quad (\operatorname{Im} f)(z) := \operatorname{Im} f(z) \quad (2.6)$$

are continuous.

Proof. This follows from Definition 2.1 and Corollaries 1.64 and 1.66. $\square$

Corollary 2.6. The modulus of a continuous complex function is continuous.

Proof. This follows from Definition 1.63, Lemma 2.5 and Definition 2.1. $\square$

Corollary 2.7. The function $f(z) = z^{-1}$ is continuous.

Proof. This follows from Corollary 1.65, Definition 1.61, Lemma 2.5, and Corollary 2.6. $\square$

Lemma 2.8. The composition of two continuous functions are continuous.

Proof. This follows from Definition 2.1. $\square$

Corollary 2.9. If $f : \mathcal{D} \rightarrow \mathbb{C}^\bullet$ is continuous, so is the function $\frac{1}{f} : \mathcal{D} \rightarrow \mathbb{C}^\bullet$.

Proof. The conclusion follows from setting $g(z) = z^{-1}$ and applying Corollary 2.7 and Lemma 2.8 to $g \circ f$. $\square$

Definition 2.10. The *inverse function* of an injection $f : \mathcal{D} \rightarrow \mathbb{C}$ is a function $f^{-1} : f(\mathcal{D}) \rightarrow \mathcal{D}$ satisfying

$$\forall z \in \mathcal{D}, \quad f^{-1}(f(z)) = z, \quad (2.7a)$$

$$\forall w \in f(\mathcal{D}), \quad f(f^{-1}(w)) = w. \quad (2.7b)$$

Lemma 2.11. The function $z \mapsto \operatorname{Arg} z$ is not continuous along the negative real axis.

Proof. By Definition 2.1, we need to show that the negation of (2.4) holds with $a = -1$, i.e.,

$$\exists \epsilon > 0, \text{ s.t. } \forall \delta > 0, \exists z \in \mathbb{C}^\bullet, \text{ s.t. } |z + 1| < \delta \implies |\operatorname{Arg}(z) - \operatorname{Arg}(a)| \geq \epsilon,$$

which holds by setting $\epsilon = \frac{\pi}{2}$ and $z = -1 - \mathbf{i}\frac{\delta}{2}$. $\square$

Exercise 2.12. The function $z \mapsto \operatorname{Arg} z$ is continuous on the complex plane slit along the negative real axis

$$\mathbb{C}_- := \mathbb{C} \setminus \{t \in \mathbb{R} : t \leq 0\}. \quad (2.8)$$

Corollary 2.13. The principal value of the logarithm is discontinuous along the negative real axis. However, it is continuous on $\mathbb{C}_-$.

2.2 Complex derivatives

Definition 2.14. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open is *complex differentiable* or has a *complex derivative* at the point $a \in \mathcal{D}$ iff the limit

$$\lim_{z \rightarrow a; z \in \mathcal{D} \setminus \{a\}} \frac{f(z) - f(a)}{z - a} \quad (2.9)$$

exists, in which case we denote the limit by $f'(a)$.

Lemma 2.15. Consider a function $f(x) : \mathbb{R} \rightarrow \mathbb{C}$. It is complex differentiable if and only if its real and imaginary parts, $u(x)$ and $v(x)$, are differentiable. If so we have

$$f'(x) = u'(x) + i v'(x). \quad (2.10)$$

Definition 2.16. A point $a \in \mathbb{C}$ is called an *accumulation point* or a *limit point* or a *cluster point* of $\mathcal{D} \subset \mathbb{C}$ iff

$$\forall \epsilon > 0, \exists z \in \mathcal{D} \text{ s.t. } |z - a| \in (0, \epsilon). \quad (2.11)$$

Definition 2.17. An *adherent point*, or a *closure point* or a *point of closure* or a *contact point*, of a subset A of a topological space $\mathcal{X}$, is a point $x \in \mathcal{X}$ such that every open set containing x contains at least one point of A .

Lemma 2.18. Let $a \in \mathcal{D}$ be an accumulation point of $\mathcal{D} \subset \mathbb{C}$, $f : \mathcal{D} \rightarrow \mathbb{C}$ a function, and $\ell \in \mathbb{C}$. The following statements are equivalent:

- (a) f is complex differentiable at a with $f'(a) = \ell$.
- (b) There exists a function $\varphi : \mathcal{D} \rightarrow \mathbb{C}$ which is continuous at a such that $\varphi(a) = \ell$ and

$$f(z) = f(a) + \varphi(z)(z - a).$$

- (c) There exists a function $\rho : \mathcal{D} \rightarrow \mathbb{C}$ which is continuous at a such that $\rho(a) = 0$ and

$$f(z) = f(a) + \ell(z - a) + \rho(z)(z - a).$$

- (d) If one defines $r : \mathcal{D} \rightarrow \mathbb{C}$ by

$$f(z) = f(a) + \ell(z - a) + r(z),$$

then we have $\lim_{z \rightarrow a; z \in \mathcal{D} \setminus \{a\}} \frac{r(z)}{z - a} = 0$.

Proof. (a) $\implies$ (b). (a) and Definition 2.14 imply

$$\lim_{z \rightarrow a; z \in \mathcal{D} \setminus \{a\}} \frac{f(z) - f(a)}{z - a} = \ell.$$

By Definition 2.1, the function $\varphi : \mathcal{D} \rightarrow \mathbb{C}$ below is continuous at a :

$$\varphi(z) := \begin{cases} \ell & \text{if } z = a; \\ \frac{f(z) - f(a)}{z - a} & \text{otherwise.} \end{cases}$$

- (b) $\implies$ (c) is given by

$$\rho(z) := \begin{cases} 0 & \text{if } z = a; \\ \varphi(z) - \ell & \text{otherwise.} \end{cases}$$

- (c) $\implies$ (d) is given by the continuity of ρ and

$$r(z) := f(z) - f(a) - \ell(z - a) = (z - a)\rho(z).$$

- (d) $\implies$ (a) is given by Definition 2.14. $\square$

Corollary 2.19. A function that is differentiable at a is continuous at a .

Theorem 2.20. Let the functions $f, g : \mathcal{D} \rightarrow \mathbb{C}$, $\mathcal{D} \subseteq \mathbb{C}$ be complex differentiable at $a \in \mathcal{D}$. Then the functions λf with $\lambda \in \mathbb{C}$, $\frac{1}{f}$ with $f(a) \neq 0$, $f + g$, and $f \cdot g$ are complex differentiable at a , and we have

$$(\lambda f)'(a) = \lambda f'(a), \quad (2.12)$$

$$\left(\frac{1}{f}\right)'(a) = -\frac{f'}{f^2}(a), \quad (2.13)$$

$$(f + g)'(a) = f'(a) + g'(a), \quad (2.14)$$

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a). \quad (2.15)$$

Exercise 2.21. Prove Theorem 2.20.

Exercise 2.22. Show that the function $f : \mathcal{D} \rightarrow \mathbb{C}$ given by

$$f(z) = z^n \text{ with } \begin{cases} \mathcal{D} = \mathbb{C} & \text{if } n \in \mathbb{N}, \\ \mathcal{D} = \mathbb{C}^\bullet & \text{if } n \in -\mathbb{N}^+ \end{cases}$$

is complex differentiable and $f'(z) = nz^{n-1}$. Then deduce that the complex derivative of the complex polynomial (2.5) is

$$p'_n(z) = \sum_{j=1}^n j a_j z^{j-1}. \quad (2.16)$$

Example 2.23. For complex polynomials $p, q : \mathbb{C} \rightarrow \mathbb{C}$ with the affine variety $\mathcal{N}(q) := \{z \in \mathbb{C} : q(z) = 0\}$, the function

$$z \mapsto \frac{p(z)}{q(z)} : \mathbb{C} \setminus \mathcal{N} \rightarrow \mathbb{C} \quad (2.17)$$

is complex differentiable and its derivative follows directly from (2.13), (2.15), and (2.16).

Theorem 2.24. The complex exponential function is complex differentiable on $\mathbb{C}$ with its derivative as itself.

Proof. By Definition 1.83, we have

$$\frac{\exp z - 1}{z} - 1 = \sum_{n=1}^{+\infty} \frac{z^n}{(n+1)!}; \quad \frac{\exp |z| - 1}{|z|} - 1 = \sum_{n=1}^{+\infty} \frac{|z|^n}{(n+1)!},$$

which, together with Corollary 1.66 (1,2), imply

$$\left| \frac{\exp z - 1}{z} - 1 \right| \leq \frac{\exp |z| - 1}{|z|} - 1. \quad (2.18)$$

Take the limit $\lim_{z \rightarrow 0}$ on both sides of (2.18), apply the known result on $\exp : \mathbb{R} \rightarrow \mathbb{R}$, and we have

$$\lim_{z \rightarrow 0; z \in \mathbb{C}} \frac{\exp z - \exp(0)}{z - 0} = 1, \quad (2.19)$$

which, by Definition 2.14, implies that $z \mapsto \exp(z)$ is differentiable at $z = 0$ with its derivative as 1. Consequently, for any fixed $w \in \mathbb{C}^\bullet$, Theorem 1.89 yields

$$e^{-w} \lim_{z \rightarrow w} \frac{e^z - e^w}{z - w} = \lim_{z \rightarrow w} \frac{e^{z-w} - e^0}{z - w} = \lim_{y \rightarrow 0} \frac{e^y - e^0}{y - 0} = 1,$$

where the second step follows from the variable substitution $y = z - w$ and the last from (2.19). Multiplying both sides by e^w completes the proof. $\square$

Corollary 2.25. In addition to Corollary 2.13, the principal value of the logarithm is complex differentiable on $\mathbb{C}_-$ with its derivative as

$$(\text{Log } z)' = \frac{1}{z}. \quad (2.20)$$

Proof. This follows directly from Theorems 2.24, 2.74, and 1.110. $\square$

Theorem 2.26 (Chain rule). Suppose two functions $f : \mathcal{D} \rightarrow \mathbb{C}$ and $g : \Omega \rightarrow \mathbb{C}$ satisfy $f(\mathcal{D}) \subseteq \Omega$. If f is complex differentiable at $a \in \mathcal{D}$ and g is complex differentiable at $f(a) \in \Omega$, then their composition $g \circ f : \mathcal{D} \rightarrow \mathbb{C}$ is differentiable at $z = a$ and

$$(g \circ f)'(a) = g'(f(a))f'(a). \quad (2.21)$$

Proof. By Lemma 2.18 (b) and the condition of f and g being complex differentiable, there exist functions φ and ψ respectively continuous at a and $b = f(a)$ such that

$$\begin{aligned} f(z) - f(a) &= \varphi(z)(z - a), \quad \varphi(a) = f'(a); \\ g(w) - g(b) &= \psi(w)(w - b), \quad \psi(b) = g'(b). \end{aligned}$$

It follows that $\frac{g(f(z)) - g(f(a))}{z - a} = \psi(f(z)) \frac{f(z) - f(a)}{z - a}$. Take the limit $z \rightarrow a$ and we have

$$(g \circ f)'(a) = \psi(f(a))f'(a) = g'(f(a))f'(a). \quad \square$$

Corollary 2.27. The sine and cosine functions are complex differentiable and

$$\forall z \in \mathbb{C}, \quad \sin' z = \cos z, \quad \cos' z = -\sin z. \quad (2.22)$$

Proof. This follows directly from Theorem 2.24, Theorem 2.26, and Corollary 1.94. $\square$

Corollary 2.28. For any given $s \in \mathbb{C}$, the power function $f : \mathbb{C}_- \rightarrow \mathbb{C}$ defined as

$$f(z) = z^s := \exp(s \text{Log } z) \quad (2.23)$$

is complex differentiable with its derivative as

$$f'(z) = \exp(s \text{Log } z) \cdot s \frac{1}{z} = sz^{s-1}. \quad (2.24)$$

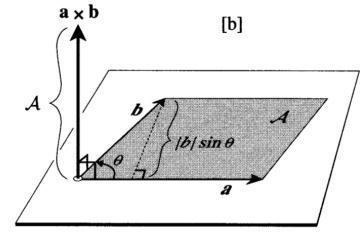
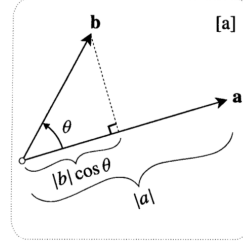
Proof. This follows directly from Theorem 2.26, Theorem 2.24, and Corollary 2.25. $\square$

2.3 Conformal = analytic + ($f' \neq 0$)

Definition 2.29. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is called *analytic* or *holomorphic* or *regular* in $\mathcal{D}$ iff it is complex differentiable at every point of $\mathcal{D}$. In particular, f is called *analytic at a point* $a \in \mathcal{D}$ iff there exists an open neighborhood $\mathcal{U} \subset \mathcal{D}$ of a such that f is analytic in $\mathcal{U}$.

Definition 2.30. For $z = x + iy$, $w = u + iv$ with $x, y, u, v \in \mathbb{R}$, the *standard scalar product* in $\mathbb{C}$ with respect to the basis $(1, i)$ is defined by

$$\langle z, w \rangle := \text{Re}(\bar{z}w) = xu + yv. \quad (2.25)$$



Exercise 2.31. Define a Hermitian inner product $(\cdot, \cdot)$ in $\mathbb{C}$ by

$$(z, w) := \bar{z}w.$$

The term Hermitian is used to describe the fact that $(\cdot, \cdot)$ is not symmetric, but rather satisfies the relation

$$\forall z, w \in \mathbb{C}, \quad (z, w) = \overline{(w, z)}.$$

Show that

$$\langle z, w \rangle = \frac{1}{2}[(z, w) + (w, z)] = \text{Re}(z, w).$$

Lemma 2.32. For each pair $(z, w) \in \mathbb{C}^\bullet \times \mathbb{C}^\bullet$, there exists a unique real number $\theta := \theta(z, w) \in (-\pi, \pi]$, called the *oriented angle* from z to w , such that

$$\begin{cases} \cos \theta = \cos \theta(z, w) = \frac{\langle z, w \rangle}{|z||w|}, \\ \sin \theta = \sin \theta(z, w) = \frac{\langle iz, w \rangle}{|z||w|}. \end{cases} \quad (2.26)$$

Proof. Definition 2.30 yields

$$\forall z, w \in \mathbb{C}, \quad \langle z, w \rangle^2 + \langle iz, w \rangle^2 = |z|^2 |w|^2, \quad (2.27)$$

hence both $\frac{\langle z, w \rangle}{|z||w|}$ and $\frac{\langle iz, w \rangle}{|z||w|}$ are in the range of $[-1, 1]$. It can also be verified by direct calculation that

$$\frac{\bar{z}w}{|z||w|} = \frac{\langle z, w \rangle}{|z||w|} + i \frac{\langle iz, w \rangle}{|z||w|}. \quad (2.28)$$

By Definitions 1.61 and 1.108, $\text{Arg} \frac{w}{z}$ is the oriented angle from z to w and is uniquely determined by (2.26). $\square$

Corollary 2.33. The *law of cosines* in trigonometry can be expressed as

$$|z + w|^2 = |z|^2 + 2 \langle z, w \rangle + |w|^2, \quad (2.29a)$$

$$|z - w|^2 = |z|^2 - 2 \langle z, w \rangle + |w|^2, \quad (2.29b)$$

where z, w are arbitrary nonzero complex numbers.

Definition 2.34. An $\mathbb{R}$ -linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ is *orientation preserving* iff $\det T > 0$.

Definition 2.35. An $\mathbb{R}$ -linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ is *angle-preserving* iff it satisfies

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \quad |T\mathbf{x}||T\mathbf{y}| \langle \mathbf{x}, \mathbf{y} \rangle = |\mathbf{x}||\mathbf{y}| \langle T\mathbf{x}, T\mathbf{y} \rangle, \quad (2.30)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product in Definition 2.30 and $|\cdot|$ the Euclidean norm.

Example 2.36. The linear map $z \mapsto \bar{z}$ is angle-preserving but not orientation preserving.

Lemma 2.37. A linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ preserves oriented angles if and only if it is a normal operator as defined in (1.55).

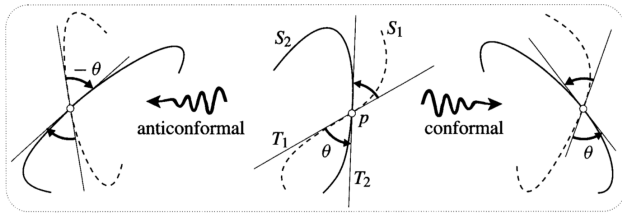
Proof. Suppose the matrix of T is an element of $\mathbb{R}_N^{2 \times 2}$ as defined in (1.55). It is then straightforward to verify Definitions 2.34 and 2.35.

Now suppose T is both angle and orientation preserving. Write the matrix of T as

$$M(T) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Set $\mathbf{x} = (1, 0)^\top$ and $\mathbf{y} = (0, 1)^\top$, plug them into (2.30), and we have $ab + cd = 0$. Set $\mathbf{x} = (1, 1)^\top$ and $\mathbf{y} = (1, -1)^\top$, plug them into (2.30), and we have $a^2 + c^2 = b^2 + d^2$, the substitution of $a = -\frac{cd}{b}$ into which yields $a^2 = d^2$ and $b^2 = c^2$. Suppose $b = c$, then $a = -d$ and $\det T < 0$, a contradiction to T being orientation preserving. Hence we have $b = -c$, $a = d$, and $a^2 + b^2 \neq 0$, which complete the proof. $\square$

Definition 2.38 (Conformal and anticonformal maps). Let $\mathcal{D}, \Omega \subseteq \mathbb{R}^n$ be open. Let $f : \mathcal{D} \rightarrow \Omega$ be a continuously differentiable map. f is said to be (locally) *conformal* iff its Jacobian matrix $J(f; a)$ is both angle and orientation preserving at each point $a \in \mathcal{D}$. f is *anticonformal* iff it preserves the magnitude of the angle but reverses the orientation at each point $a \in \mathcal{D}$. f is *globally conformal* if it is bijective and conformal.



Theorem 2.39. A C^1 function $f : \mathcal{D} \rightarrow \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{R}^2$ open is conformal if and only if f is analytic and $f' \neq 0$ when $\mathcal{D}, \Omega$ are interpreted as open subsets of $\mathbb{C}$.

Proof. Suppose f is conformal. By the continuous differentiability of f , Definition 2.38 and Lemmas 2.37 and 1.122, the action of the Jacobian $J(f; a)$ is to multiply the infinitesimal vector $\lim_{z \rightarrow a} (z - a)$ with a constant $c \in \mathbb{C}$:

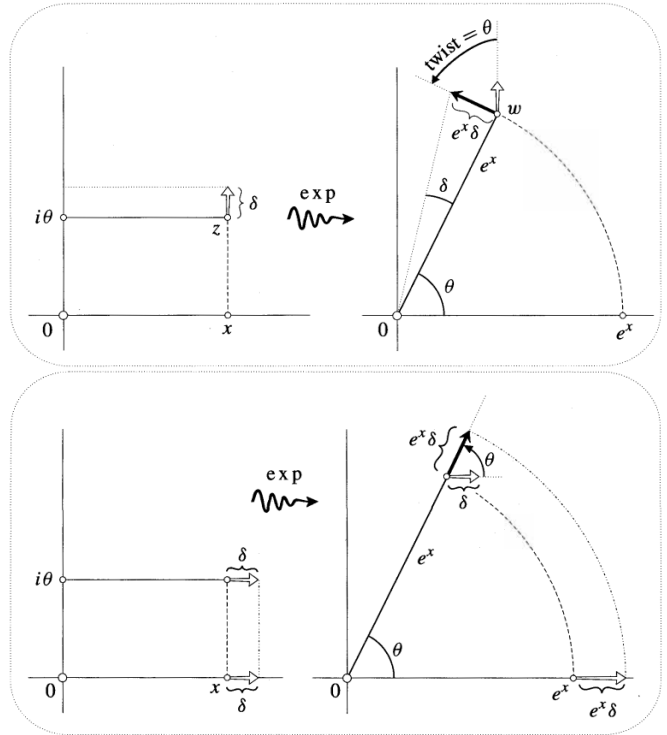
$$\lim_{z \rightarrow a} (f(z) - f(a)) = \lim_{z \rightarrow a} J(f; a)(z - a) = c \lim_{z \rightarrow a} (z - a),$$

which implies that f is analytic with $f'(a) = c$.

The sufficiency follows directly from Lemma 1.122 and the fact that a $\mathbb{C}$ -linear map can always be interpreted as an $\mathbb{R}$ -linear map. $\square$

Example 2.40. Consider the exponential function $z \mapsto e^z$ in Definition 1.83. By Theorem 2.24 and Definition 2.29, $z \mapsto e^z$ is analytic. By Theorem 2.39 and Definition 2.38, $z \mapsto e^z$ is locally conformal from $\mathbb{C}$ to $\mathbb{C}$ and globally conformal from $\{z \in \mathbb{C} : \operatorname{Im}(z) \in (-\pi, \pi)\}$ to $\mathbb{C}_+$. The behavior of $z \mapsto e^z$ in Example 1.86 is well understood by Definitions 2.34, 2.35, and 2.38.

Example 2.41 (Visual differentiation of the exponential function). The derivative of the exponential function $z \mapsto e^z$ can be obtained via a geometric reasoning: it equals the complex number ℓ that satisfies $\exp(z + \delta) - \exp(z) = \ell\delta$.



Exercise 2.42. Give a visual differentiation of the principle branch of the logarithm function defined in Theorem 1.110. You must draw two plots similar to those in Example 2.41 and elaborate on how the derivative is derived from the plots.

Exercise 2.43. Repeat Exercise 2.42 for the power function with $n \in \mathbb{Z}^+$, as defined in (1.52).

Exercise 2.44. The family of mappings introduced here plays an important role in complex analysis. These mappings, sometimes called *Blaschke factors*, will reappear in various applications in later chapters.

(a) Let z, w be two complex numbers such that $\bar{z}w \neq 1$. Prove that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| < 1 \quad \text{if } |z| < 1 \text{ and } |w| < 1,$$

and also that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| = 1 \quad \text{if } |z| = 1 \text{ or } |w| = 1.$$

(b) Prove that for a fixed w in the open unit disk $\mathbb{E}$, the mapping

$$F : z \mapsto \frac{w - z}{1 - \bar{w}z}$$

satisfies the following conditions:

(i) F maps the unit disk to itself (that is, $F : \mathbb{E} \rightarrow \mathbb{E}$), and is analytic.

- (ii) F interchanges 0 and w , namely $F(0) = w$ and $F(w) = 0$.
- (iii) $|F(z)| = 1$ if $|z| = 1$.
- (iv) $F : \mathbb{E} \rightarrow \mathbb{E}$ is bijective. *Hint:* Calculate $F \circ F$.

2.4 Critical points

Definition 2.45. A critical point of a continuous function of a single real variable is a point in the domain of f at which either $f' = 0$ or f' is not differentiable.

Example 2.46. A regular value of a curve $\gamma : (0, 1) \rightarrow \mathbb{R}^n$ is a regular point of the curve, cf. Definition C.379. For a parameter $t \in [0, 1]$ to be a critical point of γ , the tangent vector must be either zero or undefined.

Definition 2.47. A critical point of a differentiable function $f : \Omega \rightarrow \mathbb{R}^n$ with $\Omega \subset \mathbb{R}^m$ is a point in Ω where the rank of the Jacobian matrix of f is not maximal; a regular point of f is a point in Ω that is not a critical point of f . A point $p \in \mathbb{R}^n$ is a regular value of f if none of its preimages is critical; otherwise it is a critical value of f .

Example 2.48. The continuous function $h(x + iy) = x - iy^3$ satisfies $\det J_h = -3y^2$. All points on the real axis are critical points as a result of crushing in the vertical direction.

Theorem 2.49 (Sard). For a C^k function $f : \Omega \rightarrow \mathbb{R}^n$ with $\Omega \subseteq \mathbb{R}^m$ and $k \geq \max(m - n + 1, 1)$, the set of critical values of f has Lebesgue measure 0 in $\mathbb{R}^n$.

Lemma 2.50. Suppose an analytic function $f : \Omega \rightarrow \mathbb{C}$ with $\Omega \subseteq \mathbb{C}$ open satisfies $f'(z_0) \neq 0$ at $z_0 \in \Omega$. Then its Jacobian determinant $\det f'(z_0)$ is positive at z_0 .

Proof. By Theorem 2.39, f is conformal at any z_0 satisfying $f'(z_0) \neq 0$. Then Definitions 2.38, 2.34, and 2.35 imply that the Jacobian of $f(x + iy) = u(x, y) + iv(x, y)$, i.e.,

$$f' = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}, \quad (2.31)$$

preserves oriented angles. By Lemma 2.37, f' must be a normal operator in (1.55), which implies $\det f'(z_0) > 0$. $\square$

Corollary 2.51. Any critical point $a \in \Omega$ of an analytic function $f : \Omega \rightarrow \mathbb{C}$ satisfies $f'(a) = 0$.

Proof. This follows directly from Lemma 2.50. $\square$

Example 2.52. For the analytic function $g(z) = z^2$, we have $g' = 2z$ and thus the origin $z = 0$ is the only critical point of g , where any angle of z is doubled by g .

2.5 Cauchy–Riemann equations

Definition 2.53. A function $f : \mathcal{D} \rightarrow \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ being open is totally differentiable at a point $a \in \mathcal{D}$ iff there exists an $\mathbb{R}$ -linear map $A : \mathbb{R}^p \rightarrow \mathbb{R}^q$ such that the remainder r introduced by the equation

$$f(x) - f(a) = A(x - a) + r(x)$$

satisfies

$$\lim_{x \rightarrow a} \frac{r(x)}{|x - a|} = 0,$$

where $|x - a|$ is the Euclidean distance between x and a . A is called the Jacobian or total differential of f at a .

Definition 2.54. The partial derivatives of a complex function $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ are

$$\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}, \quad \frac{\partial f}{\partial y} = \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y}, \quad (2.32)$$

Lemma 2.55. A function $\mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is complex differentiable at $a \in \mathcal{D}$ if and only if f is totally differentiable at a (in the sense of Definition 2.53) and its Jacobian $J(f) : \mathcal{D} \rightarrow \mathbb{C}$ is of the form

$$J(f; a)(z - a) = \ell(z - a), \quad (2.33)$$

where ℓ , the derivative of f , is a constant complex number.

Proof. For necessity, f being complex differentiable, Definition 2.14, Lemma 2.18(a,d), and the isomorphism $\mathbb{C} \simeq \mathbb{R}^2$ in Theorem 1.50 imply that f is totally differentiable in the sense of Definition 2.53. In particular, the $\mathbb{C}$ -linear map ℓ in Lemma 2.18(a,d) can be regarded as the $\mathbb{R}$ -linear map A in Definition 2.53. In addition, (2.33) follows trivially from f being complex differentiable as $\ell = f'(a)$.

For sufficiency, $J(f; a)$ being an $\mathbb{R}$ -linear map, (2.33), and Lemma 1.122 implies that $J(f; a)$ is also a $\mathbb{C}$ -linear map. Then the proof is completed by the isomorphism in Theorem 1.121. $\square$

Theorem 2.56 (Cauchy–Riemann). Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $a \in \mathcal{D}$. The following statements are equivalent for a function $f : \mathcal{D} \rightarrow \mathbb{C}$:

- (a) f is complex differentiable at a .
- (b) f is totally differentiable at a in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$), and for $u := \operatorname{Re} f$ and $v := \operatorname{Im} f$ the following differential equations hold

$$\frac{\partial u}{\partial x}(a) = \frac{\partial v}{\partial y}(a), \quad \frac{\partial u}{\partial y}(a) = -\frac{\partial v}{\partial x}(a). \quad (2.34)$$

In case that (a) or (b) holds one has

$$f'(a) = \frac{\partial u}{\partial x}(a) + i \frac{\partial v}{\partial x}(a) = \frac{\partial v}{\partial y}(a) - i \frac{\partial u}{\partial y}(a). \quad (2.35)$$

Proof. This follows from Lemmas 2.55 and 1.122. $\square$

Notation 4. Instead of $\frac{\partial u}{\partial x}(a)$ and $\frac{\partial u}{\partial y}(a)$, one often writes $\partial_x u(a)$ and $\partial_y u(a)$, or, $\partial_1 u(a)$ and $\partial_2 u(a)$.

Corollary 2.57. A complex function $f : \mathcal{D} \rightarrow \mathbb{C}$ differentiable at $a \in \mathcal{D}$ satisfies

$$\det J(f; a) = |f'(a)|^2 = (\partial_x u)^2 + (\partial_x v)^2 = (\partial_y u)^2 + (\partial_y v)^2. \quad (2.36)$$

Exercise 2.58. Prove Corollary 2.57.

Example 2.59. The complex conjugate function $f(z) = \bar{z}$ is not complex differentiable anywhere since $1 = \partial_1 u \neq \partial_2 v = -1$.

Example 2.60. Theorem 2.24 is easily verified by the Cauchy–Riemann conditions.

Example 2.61. The function $f : \mathbb{C} \rightarrow \mathbb{C}$ given by

$$f(z) = \begin{cases} 0 & \text{if } z = 0, \\ \frac{x^3 y(y - ix)}{x^6 + y^2} & \text{otherwise.} \end{cases} \quad (2.37)$$

satisfies the Cauchy–Riemann equations, but is not totally differentiable: taking $z = t + it^3$ and $z = te^{i\theta}$ for $t \in \mathbb{R}$ and fixed $\theta \in \mathbb{R}$ yields different limits. Hence $f(z)$ is not analytic at $z = 0$.

Corollary 2.62. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is locally constant in $\mathcal{D}$ if and only if f is complex differentiable on $\mathcal{D}$ and $f'(z) = 0$ for any $z \in \mathcal{D}$.

Proof. The necessity follows directly from Definition 2.14. For sufficiency, write $f(x + iy) = u(x, y) + iv(x, y)$. Then $f'(z) = 0$ and Theorem 2.56 imply that $\partial_x u(z) = \partial_x v(z) = 0$ and $\partial_y u(z) = \partial_y v(z) = 0$. Hence both $u(x, y)$ and $v(x, y)$ are locally constant, so is f . $\square$

Corollary 2.63. Any complex differentiable function in $\mathcal{D}$ with only real (or purely imaginary) values is locally constant in $\mathcal{D}$.

Exercise 2.64. Prove Corollary 2.63.

Example 2.65. It follows from Corollary 2.63 that the functions $f(z) = |\sin z|$ and $f(z) = \operatorname{Re} z$ are not complex differentiable in $\mathbb{C}$.

Corollary 2.66. A subset $\mathcal{D} \subseteq \mathbb{C}$ is connected if and only if each locally constant function $f : \mathcal{D} \rightarrow \mathbb{C}$ is constant.

Exercise 2.67. Prove Corollary 2.66.

Corollary 2.68. The real part of an analytic function in a connected open set $\mathcal{D} \subseteq \mathbb{C}$ is uniquely determined by its imaginary part, up to an additive constant.

Exercise 2.69. Prove Corollary 2.68.

Exercise 2.70. Suppose an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being connected and open satisfies $|f|$ is constant in $\mathcal{D}$. Prove that f is constant in $\mathcal{D}$.

Definition 2.71. For a vector space of analytic functions, the *linear operators of partial differentiation with respect to* $z = x + iy$ and $\bar{z} = x - iy$ are

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right). \quad (2.38)$$

Theorem 2.72. A complex function $f(z) = u(x, y) + iv(x, y)$ is analytic on an open set $\mathcal{D} \subseteq \mathbb{C}$ if and only if $(u, v)^\top : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is totally differentiable and

$$\forall z \in \mathcal{D}, \quad \frac{\partial f}{\partial \bar{z}}(z) = 0, \quad (2.39)$$

in which case $f' = \frac{\partial f}{\partial z}$.

Proof. (2.39) follows from Definition 2.71 and Theorem 2.56 while $f' = \frac{\partial f}{\partial z}$ from (2.35) and (2.38). $\square$

Exercise 2.73. Show that in polar coordinates, the Cauchy–Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\operatorname{Log} z = \log r + i\theta, \quad z = re^{i\theta},$$

is analytic in the region $r > 0$ and $-\pi < \theta < \pi$.

Theorem 2.74 (Inverse function). Suppose $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D}$ open is complex differentiable and $f'(a) \neq 0$ at a point $a \in \mathcal{D}$. Then there exist open neighborhoods $\mathcal{U}_r(a)$ and $\mathcal{U}_R(f(a))$ such that f is locally bijective at a , its local inverse g is complex differentiable, and

$$g'(f(a)) = \frac{1}{f'(a)}. \quad (2.40)$$

Proof. The Cauchy–Riemann theorem 2.56 implies that f is totally differentiable in the sense of real analysis and Lemma 2.50 implies that its Jacobian determinant $J(f; a)$ is nonzero. By the real inverse function theorem D.122, there exists an open ball $\mathcal{U}_r(a)$ such that f is locally bijective at a and the corresponding real inverse function $g : \mathcal{U}_R(f(a)) \rightarrow \mathcal{U}_r(a)$ is totally differentiable. Lemma 2.55 implies $J(f; a)(z - a) = f'(a)(z - a)$ where $f'(a) \in \mathbb{C}$ and then Lemma 1.122 implies that $J(f; a)$ is a normal matrix of the form (1.59). By Lemma 1.120, these normal matrices form a group and thus the inverse of $J(f; a)$ is also in the group and its determinant is nonzero. Hence the Cauchy–Riemann theorem 2.56 implies that g is complex differentiable at a . Finally, Definition 2.14 gives

$$\begin{aligned} \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} &= \lim_{z \rightarrow a} \frac{f(z) - f(a)}{g(f(z)) - g(f(a))} \\ &= \frac{1}{\lim_{z \rightarrow a} \frac{g(f(z)) - g(f(a))}{f(z) - f(a)}} = \frac{1}{g'(f(a))}. \quad \square \end{aligned}$$

Corollary 2.75. If $\varphi : \mathcal{D} \rightarrow \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{C}$ open is globally conformal, then so is its inverse function $\varphi^{-1} : \Omega \rightarrow \mathcal{D}$.

Proof. This follows directly from Definition 2.38 and Theorems 2.39 and 2.74. $\square$

2.6 Harmonic functions

Definition 2.76. A $\mathcal{C}^2$ function $u : \mathcal{D} \rightarrow \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ being open is called a *potential function* or a *harmonic function* if it satisfies the *Laplace equation*

$$\Delta u := \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u = 0, \quad (2.41)$$

where Δ is called the *Laplace operator*.

Exercise 2.77. Show that

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta.$$

Definition 2.78. A harmonic function $v : \mathcal{D} \rightarrow \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ being open is called the *conjugate harmonic function* to a harmonic function $u : \mathcal{D} \rightarrow \mathbb{R}$ iff the complex function $f = u + \mathbf{i}v$ is analytic in $\mathcal{D}$.

Theorem 2.79. If $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is an analytic function whose real and imaginary parts are continuously partial differentiable at least twice, then both the real part and the imaginary part are harmonic functions.

Proof. By the Cauchy–Riemann Theorem 2.56, we have $\partial_1 u = \partial_2 v$ and $\partial_2 u = -\partial_1 v$, which imply $\partial_1^2 u = \partial_1(\partial_2 v)$ and $\partial_2^2 u = -\partial_2(\partial_1 v)$. Partial derivatives commute and hence u is harmonic. Similarly, v is also harmonic. $\square$

Corollary 2.80. If a function $u : \mathcal{D} \rightarrow \mathbb{R}$ is harmonic, then the following function $g : \mathcal{D} \rightarrow \mathbb{C}$ is analytic:

$$g = \partial_1 u - \mathbf{i} \partial_2 u. \quad (2.42)$$

Proof. This follows directly from Definition 2.78 and Theorem 2.56. $\square$

Theorem 2.81. Suppose $u : \mathcal{D} \rightarrow \mathbb{R}$ is a harmonic function where $\mathcal{D} := (a, b) \times (c, d) \subseteq \mathbb{C}$ is an open rectangle whose sides are parallel to the coordinate axes. Then there exists an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with u as its real part and with the conjugate harmonic function of u uniquely determined up to an additive constant as

$$v(x, y) := \int_{y_0}^y \partial_x u(x, t) dt - \int_{x_0}^x \partial_y u(t, y_0) dt + v(x_0, y_0), \quad (2.43)$$

where $x_0 \in (a, b)$ and $y_0 \in (c, d)$.

Proof. Define $f = u + \mathbf{i}v$. We claim that f is analytic and that $g = u + \mathbf{i}(v + c)$ is also analytic for any constant $c \in \mathbb{C}$.

By (2.43), we have $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$. Since $u \in \mathcal{C}^2$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is totally differentiable. Then Theorem 2.56 implies that f is analytic.

By construction, we have $g - f = \mathbf{i}c$. Then the proof is completed by Corollary 2.62. $\square$

Example 2.82. The function

$$u(x, y) := \log \sqrt{x^2 + y^2} \quad (2.44)$$

is harmonic on $\mathbb{C}^\bullet$, but there is no analytic function $f : \mathbb{C}^\bullet \rightarrow \mathbb{C}$ that satisfies

$$\operatorname{Re} f(z) = \log |z|.$$

We can prove this by contradiction. Suppose there is such a function f . Then by Theorem 1.111 f would have to agree with $g(z) = \operatorname{Log} z + \mathbf{i}c$, where $c \in \mathbb{R}$ is a constant in $\mathbb{C}_-$, but this contradicts Corollary 2.13, which states that $g(z)$ is not even continuous in $\mathbb{C}^\bullet$.

Exercise 2.83. For an analytic function $f : \mathbb{C}_- \rightarrow \mathbb{C}$, prove that $\operatorname{Re} f = u$ in (2.44) implies $f(z) = \operatorname{Log}(z) + \mathbf{i}c$ for some constant $c \in \mathbb{R}$.

Exercise 2.84. Find $a \in \mathbb{R}$ such that the function $u_a : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $u_a(x, y) = x^3 + axy^2$ is harmonic. Furthermore, determine all conjugate harmonic functions to u_a or all analytic functions $f : \mathbb{C} \rightarrow \mathbb{C}$ with $\operatorname{Re} f = u_a$.

Chapter 3

Complex Integration over Paths

Definition 3.1. The *primitive* of a function $f : \mathcal{D} \rightarrow \mathbb{C}$ is an analytic function $F : \mathcal{D} \rightarrow \mathbb{C}$ such that $F' = f$ for any $z \in \mathcal{D}$.

Motivation 3.2. In the case of $\mathcal{D} \subseteq \mathbb{R}$, the first fundamental theorem of calculus (Theorem B.127) guarantees the existence of the primitive of any function $f : \mathcal{D} \rightarrow \mathbb{R}$, since we can construct a suitable integral as the primitive of f . However, this is not true in the case of $\mathcal{D}$ being an open subset of $\mathbb{C}$: a function that has a primitive must be analytic, c.f. Theorem 3.84. Therefore, it is natural to ask the question whether the primitive of a function $f : \mathcal{D} \rightarrow \mathbb{C}$ exists; this question is a key problem in our extension of $\mathbb{R}$ to $\mathbb{C}$.

Roadmap 3.3. This chapter concerns most of the equivalence of the following statements for an analytic function $f : \Omega \rightarrow \mathbb{C}$ where Ω is an open set.

- (a) f has a primitive at a local neighborhood of each point.
- (b) The complex integral of f on a closed curve vanishes.
- (c) The complex line integral of f is path independent.
- (d) f is analytic.
- (e) f is representable as a power series in each open disk fully contained in its domain.

After examining a stepping-stone integral in Definition 3.4, we formally define the concept of complex line integrals in Section 3.2 and prove Theorem 3.16 that states (a) $\Rightarrow$ (b). One key result of Section 3.3 is Theorem 3.42, which states that (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c). The central result of Section 3.3 is the Cauchy theorem, which states (d) $\Rightarrow$ (b) in Theorems 3.47, 3.55, 3.59, 3.64, and 3.73 on various domains. In Section 3.4, the generalized Cauchy integral formula (Theorem 3.84) gives (a) $\Rightarrow$ (d); this is utilized in the proof of the Morera theorem 3.92, which states (b) $\Rightarrow$ (d). The equivalence of (e) to (d) will be proved in Section 4.1.

3.1 A stepping stone

Definition 3.4. A function $f : [a, b] \rightarrow \mathbb{C}$ with $a, b \in \mathbb{R}$ is called *integrable* iff $\operatorname{Re} f, \operatorname{Im} f : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable as in Definition B.106; then

$$\int_a^b f(x) dx = \int_a^b \operatorname{Re} f(x) dx + i \int_a^b \operatorname{Im} f(x) dx. \quad (3.1)$$

Corollary 3.5. Denote by $\mathcal{V}$ the set of continuous functions $[a, b] \rightarrow \mathbb{C}$. Then $\mathcal{V}$ is a vector space and the integral $\int_a^b$ defined in (3.1) is a linear functional on $\mathcal{V}$.

Proof. We need to show that the integration is a linear map from $\mathcal{V}$ to $\mathbb{C}$, which holds because

$$\begin{aligned} \int_a^b (f(x) + g(x)) dx &= \int_a^b f(x) dx + \int_a^b g(x) dx; \\ \forall \lambda \in \mathbb{C}, \int_a^b \lambda f(x) dx &= \lambda \int_a^b f(x) dx. \quad \square \end{aligned}$$

Lemma 3.6. If $f : [a, b] \rightarrow \mathbb{C}$ is continuous and F is a primitive of f , i.e. $F' = f$, then

$$\int_a^b f(x) dx = F(b) - F(a). \quad (3.2)$$

Proof. This follows from (3.1) and the second fundamental theorem of calculus (Theorem B.128). $\square$

Lemma 3.7. If $f : [a, b] \rightarrow \mathbb{C}$ is integrable and satisfies $|f(x)| \leq C$ for all $x \in [a, b]$, then we have

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \leq (b-a)C. \quad (3.3)$$

Proof. The second inequality follows from the integral mean value theorem B.124. To prove the first inequality, define $\int_a^b f(x) dx = re^{i\theta}$ and we have

$$\begin{aligned} \left| \int_a^b f(x) dx \right| &= r = e^{-i\theta} \int_a^b f(x) dx = \int_a^b e^{-i\theta} f(x) dx \\ &= \operatorname{Re} \int_a^b e^{-i\theta} f(x) dx = \int_a^b \operatorname{Re} [e^{-i\theta} f(x)] dx \\ &\leq \int_a^b |e^{-i\theta} f(x)| dx = \int_a^b |f(x)| dx, \end{aligned}$$

where all steps follows from complex arithmetic. $\square$

Theorem 3.8. If $f : [a, b] \rightarrow \mathbb{R}^k$ is integrable, then $\|f\|$ is integrable and

$$\left\| \int_a^b f dx \right\| \leq \int_a^b \|f\| dx, \quad (3.4)$$

where $\|\cdot\|$ denotes the Euclidean norm in Definition C.176.

Proof. Write $f(x) := (f_1(x), \dots, f_k(x))$, $y_i := \int_a^b f_i(x)dx$, and we have $y := \int_a^b f(x)dx = (y_1, \dots, y_k)$ and

$$\|y\|^2 = \sum y_j^2 = \sum y_j \int f_j dx. \quad (3.5)$$

For a fixed $x \in [a, b]$, the Cauchy-Schwarz inequality yields

$$\langle y, f(x) \rangle = \sum y_i f_i(x) \leq \|y\| \|f(x)\|.$$

Integrate the above inequality, apply (3.5), and we have

$$\|y\|^2 = \sum y_j \int_a^b f_j dx \leq \|y\| \int_a^b \|f(x)\| dx.$$

If $\|y\| \neq 0$, we divide both sides of the above inequality by $\|y\|$ to obtain (3.4). If $\|y\| = 0$, each y_i must be zero, hence (3.4) also holds trivially. $\square$

Lemma 3.9. For continuously differentiable functions $u, v : [a, b] \rightarrow \mathbb{C}$, we have

$$\int_a^b u(x)v'(x)dx = (uv)\Big|_a^b - \int_a^b u'(x)v(x)dx. \quad (3.6)$$

Proof. This follows from Lemma 3.6 and the product rule $(uv)' = uv' + u'v$ in Theorem 2.20. $\square$

Lemma 3.10. Let $I_1, I_2 \subset \mathbb{R}$ be intervals, $a, b \in I_1$ and $\varphi : I_1 \rightarrow I_2$ continuously differentiable and $f : I_2 \rightarrow \mathbb{C}$ continuous. Then

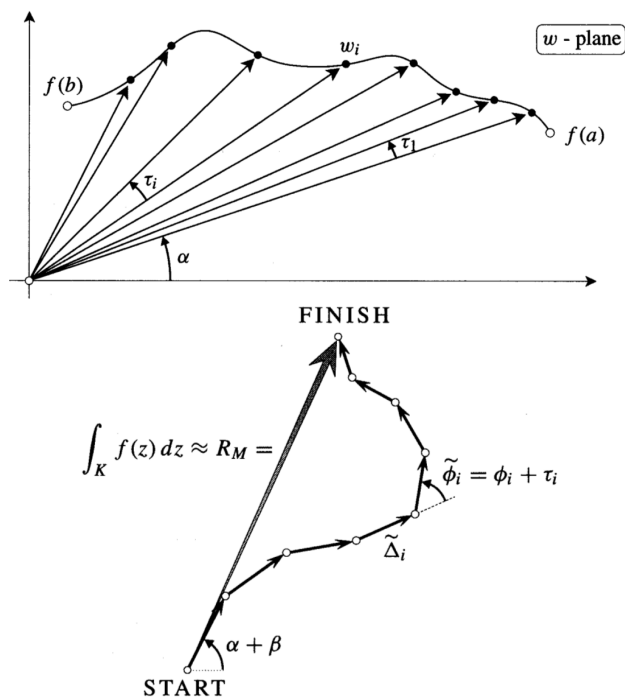
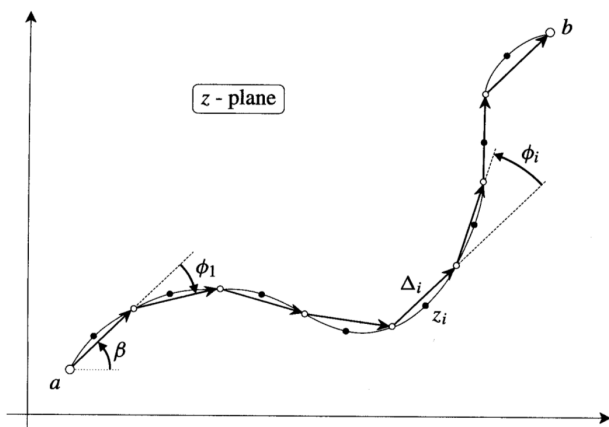
$$\int_{\varphi(a)}^{\varphi(b)} f(y)dy = \int_a^b f(\varphi(x))\varphi'(x)dx. \quad (3.7)$$

Proof. This follows directly from Theorem B.130 and Definition 3.4. $\square$

3.2 Complex line integrals

Definition 3.11. For a piecewise smooth curve $K : [a, b] \rightarrow \mathbb{C}$ and a continuous function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \supset K([a, b])$, the *complex line integral* or *complex contour integral* of f along K , written $\int_K f$, is given by

$$\int_K f(z)dz := \int_a^b f(K(t))K'(t)dt. \quad (3.8)$$



Lemma 3.12. Let $\alpha : [c, d] \rightarrow \mathbb{C}$ be a piecewise smooth curve and $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\alpha([c, d]) \subset \mathcal{D} \subset \mathbb{C}$. Let $\varphi : [a, b] \rightarrow [c, d]$ be a continuously differentiable function with $\varphi(a) = c$ and $\varphi(b) = d$. Then we have

$$\int_{\alpha} f(\zeta)d\zeta = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad (3.9)$$

Proof. Definition 3.11 yields

$$\begin{aligned} \int_{\alpha} f(\zeta)d\zeta &= \int_c^d f(\alpha(t))\alpha'(t)dt \\ &= \int_a^b f(\alpha(\varphi(s)))\alpha'(\varphi(s))d\varphi(s) \\ &= \int_a^b f[(\alpha \circ \varphi)(s)](\alpha \circ \varphi)'(s)ds = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad \square \end{aligned}$$

Lemma 3.13. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function and $\alpha : [a, b] \rightarrow \mathbb{C}$ with $\alpha([a, b]) \subset \mathcal{D} \subset \mathbb{C}$. If $|f(z)| \leq C$ for all $x \in \mathcal{D}$, then we have

$$\left| \int_{\alpha} f(z)dz \right| \leq \int_{\alpha} |f(z)||dz| \leq C\ell(\alpha), \quad (3.10)$$

where $\ell(\alpha)$ is the arc length of α as in Definition C.382.

Proof. Definition 3.11 and Lemma 3.7 yield

$$\begin{aligned} \left| \int_{\alpha} f(\zeta)d\zeta \right| &= \left| \int_a^b f(\alpha(t))\alpha'(t)dt \right| \\ &\leq \int_a^b |f(\alpha(t))||\alpha'(t)|dt \\ &\leq C \int_a^b |\alpha'(t)|dt = C\ell(\alpha). \end{aligned}$$

Then the proof is completed by

$$\int_{\alpha} |f(z)||dz| = \int_a^b |f(\alpha(t))||\alpha'(t)|dt. \quad \square$$

Lemma 3.14. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F , i.e. $F' = f$, then for any piecewise smooth curve $\alpha \subset \mathcal{D}$ we have

$$\int_{\alpha} f(\zeta)d\zeta = F(\alpha(b)) - F(\alpha(a)). \quad (3.11)$$

Proof. Let $F = U + \mathbf{i}V$, then the Cauchy–Riemann equations yield $f = \frac{\partial U}{\partial x} + \mathbf{i}\frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} - \mathbf{i}\frac{\partial U}{\partial y}$. Write $\alpha(t) = x(t) + \mathbf{i}y(t)$ and we have

$$\begin{aligned} f(\alpha(t))\alpha'(t) &= \frac{\partial U}{\partial x}x' + \frac{\partial U}{\partial y}y' + \mathbf{i}\left(\frac{\partial V}{\partial x}x' + \frac{\partial V}{\partial y}y'\right) \\ &= \frac{dU}{dt} + \mathbf{i}\frac{dV}{dt}. \end{aligned}$$

Then (3.11) follows from integrating the above identity over α and applying (3.8) and the fundamental theorem of calculus (Theorem B.128). $\square$

Corollary 3.15. $\forall z \in \mathbb{C}_-, \text{Log}(z) = L(z) := \int_1^z \frac{1}{\zeta} d\zeta$.

Proof. Corollary 2.25 states that $\text{Log}(z)$ is analytic in $\mathbb{C}_-$ and $(\text{Log}(z))' = \frac{1}{z}$. The rest follows from Lemma 3.14. $\square$

Theorem 3.16. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F , i.e. $F' = f$, then for any closed piecewise smooth curve γ in $\mathcal{D}$ we have

$$\oint_{\gamma} f(\zeta) d\zeta = 0. \quad (3.12)$$

Proof. Decompose γ into two curves $\alpha, \beta : [0, 1] \rightarrow \mathbb{C}$ such that $\gamma = \alpha \cup \beta$, $\alpha(0) = \beta(1)$, and $\alpha(1) = \beta(0)$. The proof is completed by $\oint_{\gamma} f = \int_{\alpha} f + \int_{\beta} f$ and Lemma 3.14. $\square$

Corollary 3.17. For a counterclockwise oriented circle $\gamma(t) = r \exp(\mathbf{i}t)$ where $r > 0$ and $t \in [0, 2\pi]$, we have

$$\forall n \in \mathbb{Z}, \quad \oint_{\gamma} \zeta^n d\zeta = \begin{cases} 2\pi\mathbf{i} & \text{if } n = -1, \\ 0 & \text{otherwise.} \end{cases} \quad (3.13)$$

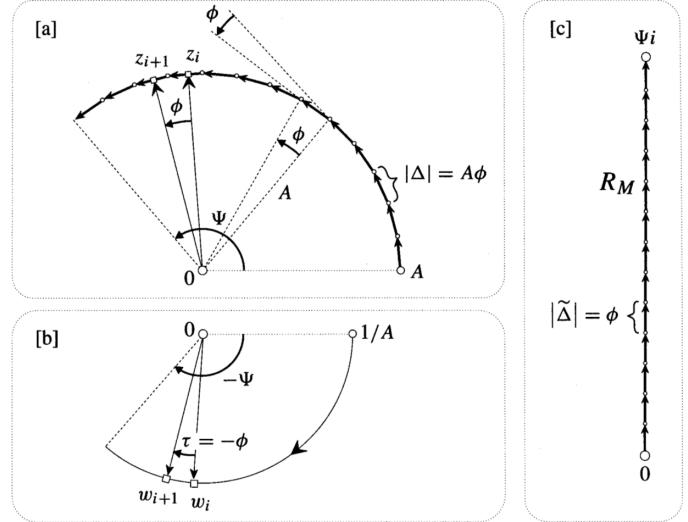
Proof. For $n \neq -1$, the function $f(z) = z^n$ has the primitive $F(z) = \frac{z^{n+1}}{n+1}$, c.f. Exercise 2.22. For $n = -1$,

$$\oint_{\gamma} \zeta^{-1} d\zeta = \int_0^{2\pi} (re^{\mathbf{i}t})^{-1} r \mathbf{i} e^{\mathbf{i}t} dt = 2\pi\mathbf{i}. \quad \square$$

Exercise 3.18 (An alternative proof of Corollary 3.17 via Corollary 3.15). By Corollary 2.25, we have $(\text{Log}z)' = \frac{1}{z}$ for $z \in \mathbb{C}_-$. By Definition 3.28, the unit circle $\mathbb{S}^1 = \alpha \oplus \beta$ is the concatenation of two curves $\theta \mapsto e^{\mathbf{i}\theta}$,

$$\alpha := ((-\pi)^+, 0) \rightarrow \mathbb{C}, \quad \beta := (0, \pi) \rightarrow \mathbb{C}.$$

Put $z_{-1}^- := \alpha((-\pi)^+)$ and $z_{-1}^+ := \beta(\pi)$. Based on the above constructions, starting from the variable substitution $\zeta = rz$, calculate $\oint_{\gamma} \zeta^{-1} d\zeta$.



Corollary 3.19. The function $f(z) = \frac{1}{z}$ does not have a primitive if its domain is $\mathbb{C}^\bullet$.

Proof. Otherwise it would contradict Corollary 3.17. $\square$

Exercise 3.20. Draw plots similar to those of $\int_K \frac{1}{z} dz$ to give a geometric interpretation of $\int_A^B z^m dz$ where $m \in \mathbb{N}^+$ and the integral is on the arc $\gamma(t) = r \exp(\mathbf{i}t)$ with $t \in [\text{Arg}A, \text{Arg}B]$. Then deduce the formula

$$\int_A^B z^m dz = \frac{1}{m+1} (B^{m+1} - A^{m+1}). \quad (3.14)$$

Exercise 3.21. Give a geometric interpretation of $\int_A^B e^z dz$ and the integral is on the vertical segment $\gamma(t) = A + \mathbf{i}t$ with $t \in [0, \theta]$, where $B = A + \mathbf{i}\theta$ and $\theta > 0$. Then deduce the formula

$$\int_A^B e^z dz = e^B - e^A. \quad (3.15)$$

Exercise 3.22. In Corollary 3.17, we computed the integral $\oint_{\gamma} \zeta^n d\zeta$ for all integers n , where γ is a circle centered at the origin with positive orientation. Without invoking the Cauchy integral theorem, calculate $\oint_{\gamma} \zeta^n d\zeta$ where

- (a) γ is any circle not enclosing the origin;
- (b) γ is a positively oriented square with vertices at $\pm 1 \pm \mathbf{i}$.

Hint: For each of the four line segments, use the parameterization and calculate the real integral $\int_{-1}^1 \frac{1}{1+t^2} dt$.

Exercise 3.23. For the counterclockwise oriented circle $\gamma(t) = r \exp(\mathbf{i}t)$ with $t \in [0, 2\pi]$ and $r \in (|a|, |b|)$, prove

$$\oint_{\gamma} \frac{1}{(\zeta - a)(\zeta - b)} d\zeta = \frac{2\pi\mathbf{i}}{a - b}.$$

Hint: Express $\frac{1}{\zeta - a}$ as the geometric series expansion $\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n$ for $|z| < 1$.

3.3 The Cauchy integral theorem

Definition 3.24. A set $\mathcal{D} \subset \mathbb{C}$ is called *arcwise connected* or *path connected* iff for any two points $z, w \in \mathcal{D}$ there is a piecewise smooth curve $\alpha : [a, b] \rightarrow \mathcal{D}$ such that $\alpha(a) = z$ and $\alpha(b) = w$.

Definition 3.25. A *domain* is an arcwise connected nonempty open set.

Example 3.26. The point set $\mathcal{D} = \{z \in \mathbb{C} : |z^2 - 3| < 1\}$ is not a domain because it is not connected: $\pm\sqrt{3} \in \mathcal{D}$ whereas $\{yi : y \in \mathbb{R}\} \cap \mathcal{D} = \emptyset$.

Example 3.27. The set $\mathcal{D} \setminus \mathcal{D}'$ is a domain where

$$\mathcal{D} = \{x + iy : x \in (0, 1), y \in (0, 1)\};$$

$$\mathcal{D}' = \bigcup_{n=2}^{\infty} \left\{ x + iy : x = \frac{1}{n}, y \in \left(0, \frac{1}{2}\right] \right\},$$

because $\mathcal{D} \setminus \mathcal{D}' = \mathcal{D} \setminus \overline{\mathcal{D}'} = \mathcal{D} \cap (\mathbb{C} \setminus \overline{\mathcal{D}'}): \overline{\mathcal{D}'}$ is closed, $(\mathbb{C} \setminus \overline{\mathcal{D}'})$ is open, and thus $\mathcal{D} \setminus \mathcal{D}'$ is open. $\mathcal{D} \setminus \mathcal{D}'$ is also path connected.

3.3.1 Simply connected domains

Definition 3.28. The *concatenation of two curves* $\alpha : [a, b] \rightarrow \mathbb{C}$ and $\beta : [b, c] \rightarrow \mathbb{C}$ with $\alpha(b) = \beta(b)$ is another curve $\alpha \oplus \beta : [a, c] \rightarrow \mathbb{C}$ given by

$$(\alpha \oplus \beta)(t) := \begin{cases} \alpha(t) & \text{if } t \in [a, b], \\ \beta(t) & \text{if } t \in [b, c]. \end{cases} \quad (3.16)$$

Example 3.29. $\alpha(t) = \sqrt{1-t^2}$, $t \in [-1, 0]$ and $\beta(t) = 1$, $t \in [0, 2]$ are two smooth curves. The concatenation $\alpha \oplus \beta$ is not smooth but piecewise smooth: the second derivatives of α and β are different at $t = 0$. Indeed, the curvature of a unit circle and a straight line is 1 and 0, respectively.

Definition 3.30. The *reciprocal curve* of a curve $\alpha : [a, b] \rightarrow \mathbb{C}$ is the curve $\alpha^-(t) : [a, b] \rightarrow \mathbb{C}$ defined by $\alpha^-(t) = \alpha(b + a - t)$.

Definition 3.31. A *path* in a topological space $\mathcal{X}$ is a continuous map $\zeta : [0, 1] \rightarrow \mathcal{X}$.

Definition 3.32. A *loop* or *cycle* is a path whose endpoints coincide; this common endpoint is the *basepoint* of the loop.

Definition 3.33. A *homotopy of paths* in $\mathcal{X}$ is a family of paths $h_t : [0, 1] \rightarrow \mathcal{X}$, $t \in [0, 1]$, such that $h_t(0)$, $h_t(1)$ are independent of time and the associated map $H : [0, 1]^2 \rightarrow \mathcal{X}$ defined by $H(s, t) = h_t(s)$ is continuous. Two paths are *homotopic* if they are connected by a homotopy.

Example 3.34. The family of flow maps of a well-posed ODE $\frac{dx}{dt} = \mathbf{u}(\mathbf{x}, t)$ that act on a path at $t = 0$ is a homotopy of paths in the phase space of the ODE.

Lemma 3.35. The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

Proof. This follows from Definitions 3.33 and A.36. $\square$

Definition 3.36. A *free homotopy* between two loops γ_1 and γ_2 in $\mathcal{X}$ is a function $H_f : [0, 1]^2 \rightarrow \mathcal{X}$ such that $H_f(s, 0) = \gamma_1(s)$ and $H_f(s, 1) = \gamma_2(s)$ for all s , and $H_f(0, t) = H_f(1, t)$ for all t . Then γ_1 and γ_2 are said to be *freely homotopic* in $\mathcal{X}$.

Example 3.37. If the basepoint of a free loop homotopy is further required to be fixed, i.e., $H_f(0, t) = H_f(1, t) = x_0$ for all $t \in [0, 1]$, we get a *based homotopy* or a *path homotopy*, in which case we also say that the two loops are freely homotopic *relative to their endpoints*.

Example 3.38. In the space of $\mathbb{S}^1 = [0, 1]/\{0, 1\}$, the loops $\alpha(s) := e^{2\pi is}$ and $\beta(s) := 1$ are freely homotopic through $H_f(s, t) = e^{2\pi is(1-t)}$. If the basepoint at $t = 0$ is chosen to be $x_*(0) = \alpha(1)$, then the expression of the basepoint at time t is $x_*(t) = e^{2\pi i(1-t)}$. During this homotopy, x_* covers $\mathbb{S}^1$ once. However, if x_* is required to be fixed at $\alpha(1)$, the loop α cannot be deformed into the constant loop β .

Definition 3.39. A closed curve $\alpha : [0, 1] \rightarrow \Omega$ with its basepoint at $z_0 := \alpha(0) = \alpha(1)$ is *null-homotopic* in Ω iff it is homotopic to the constant curve $\beta(t) = z_0$.

Definition 3.40. A domain is *simply connected* iff any closed curve in it is null-homotopic in it.

Exercise 3.41. Prove that Theorem 2.81 holds not only for rectangular domains but also for simply connected domains.

3.3.2 A proof by Riemann

Theorem 3.42. For a continuous function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D}$ a domain, the following statements are equivalent:

- (a) f has a primitive.
- (b) The integral of f along any piecewise smooth closed curve in $\mathcal{D}$ vanishes.
- (c) The integral of f along any piecewise smooth curve in $\mathcal{D}$ depends only on the beginning and end points of the curve.

Proof. By Theorem 3.16, (a) implies (b).

To show (b) implies (c), let $\alpha : [a, b] \rightarrow \mathcal{D}$ and $\beta : [c, d] \rightarrow \mathcal{D}$ be two curves with the same starting and ending points. WLOG we can assume $b = c$. Otherwise set $\varphi(t) = t + c - b$ in Lemma 3.12 and we have $\int_{\beta} f(\zeta) d\zeta = \int_{\beta \circ \varphi} f(\zeta) d\zeta$. It follows that

$$0 = \oint_{\alpha \oplus \beta^-} f = \int_{\alpha} f - \int_{\beta} f.$$

To show (c) implies (a), fix a point $z_* \in \mathcal{D}$ and construct the integral

$$F(z) = \int_{z_*}^z f(\zeta) d\zeta, \quad (3.17)$$

where the integral is along a piecewise smooth curve connecting z_* with z in $\mathcal{D}$. It follows from Lemma 2.18 that $F'(z_0) = f(z_0)$ holds for any given $z_0 \in \mathcal{D}$ if the residue

$$r(z) := F(z) - F(z_0) - f(z_0)(z - z_0)$$

satisfies

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.18)$$

Since $z_0 \in \mathcal{D}$, there exists an open ball $\mathcal{U}_\rho(z_0)$ that is a subset of $\mathcal{D}$. For all $z \in \mathcal{U}_\rho(z_0)$, we have

$$F(z) = \int_{z_*}^{z_0} f(\zeta) d\zeta + \int_{z_0}^z f(\zeta) d\zeta = F(z_0) + \int_{z_0}^z f(\zeta) d\zeta,$$

where the last integral can be taken from z_0 to z along the line segment $\sigma : [0, 1] \rightarrow \mathcal{D}$ given by $\sigma(t) := z_0 + t(z - z_0)$. Definition 3.11 yields $\int_\sigma d\zeta = z - z_0$. Hence we have

$$r(z) = \int_\sigma (f(\zeta) - f(z_0)) d\zeta$$

and the continuity of f gives

$$\forall \epsilon > 0, \exists \delta \in (0, \rho), \text{ s.t. } |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \epsilon.$$

Then Lemma 3.13 yields (3.18). The proof is completed by z_0 being arbitrary. $\square$

Exercise 3.43. Determine whether the following functions have primitives on the indicated domains and fully justify your answer.

- (a) $f(z) = \frac{1}{z}$ on $\mathbb{C}^\bullet$;
- (b) $g(z) = \bar{z}$ on $\mathbb{C}$.

Exercise 3.44. Let $\gamma : [0, 1] \rightarrow \mathbb{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ be a piecewise smooth curve with $\gamma(0) = 1 + \mathbf{i}$ and $\gamma(1) = \mathbf{i}$.

- (a) Do z and $\frac{1}{z}$ have primitives on $\mathbb{H}$? Justify your answer.
- (b) Calculate $\int_\gamma z dz$ and $\int_\gamma \frac{1}{z} dz$.

Theorem 3.45 (Green). Let α be a counter-clockwise oriented, piecewise smooth, simple closed curve in the plane. Denote by $\mathcal{D}$ the bounded complement of α and $u, v : \mathcal{D} \rightarrow \mathbb{R}$ functions with continuous partial derivatives on $\mathcal{D}$. Then

$$\oint u(x, y) dx + v(x, y) dy = \iint \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy. \quad (3.19)$$

Lemma 3.46. If $f(z) = u(x, y) + \mathbf{i}v(x, y)$ is continuous along a piecewise smooth curve α , then $f(z)$ is integrable along α and

$$\int_\alpha f(z) dz = \int_\alpha u dx - v dy + \mathbf{i} \int_\alpha v dx + u dy. \quad (3.20)$$

Proof. This follows from $dz = dx + \mathbf{i}dy$, Definition 3.4, and Theorem B.119 that a continuous function is integrable. $\square$

Theorem 3.47 (Cauchy). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$ and f' is continuous in $\mathcal{D}$, then

$$\oint_{\partial \mathcal{D}} f(\zeta) d\zeta = 0. \quad (3.21)$$

Proof. Let $f(z) = u(x, y) + \mathbf{i}v(x, y)$ and $z = x + \mathbf{i}y$. In 1851, Riemann reasoned that

$$\begin{aligned} \oint_{\partial \mathcal{D}} f(\zeta) d\zeta &= \oint_{\partial \mathcal{D}} u dx - v dy + \mathbf{i} \oint_{\partial \mathcal{D}} v dx + u dy \\ &= \iint_{\mathcal{D}} -\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) dx dy + \mathbf{i} \iint_{\mathcal{D}} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy = 0, \end{aligned}$$

where the first step follows from Lemma 3.46, the second from Theorem 3.45, and the third from Theorem 2.56. $\square$

Lemma 3.48 (Polynomial growth). For a complex polynomial of degree $n \in \mathbb{N}^+$,

$$P(z) = \sum_{j=0}^n a_j z^j, \quad (3.22)$$

where $a_j \in \mathbb{C}$ for all $j = 0, \dots, n$ and $a_n \neq 0$, we have

$$\frac{1}{2}|a_n||z|^n \leq |P(z)| \leq \frac{3}{2}|a_n||z|^n \quad (3.23)$$

for any z with a sufficiently large modulus $|z| \geq \rho$, e.g.,

$$\rho := \max \left\{ 1, \frac{2}{|a_n|} \sum_{j=0}^{n-1} |a_j| \right\}. \quad (3.24)$$

Proof. Define $Q(z) = P(z) - a_n z^n$. Then the triangle inequality and the condition $|z| \geq \rho$ yield

$$|Q(z)| \leq \left(\sum_{j=0}^{n-1} |a_j| \right) |z|^{n-1} \leq \frac{|a_n|}{2} |z|^n.$$

Apply the triangle inequality, i.e. Corollary 1.66, again to $Q(z) + a_n z^n = P(z)$ and we have

$$|a_n||z|^n - |Q(z)| \leq |P(z)| \leq |a_n||z|^n + |Q(z)|.$$

Combining the two inequalities yields (3.23). $\square$

Theorem 3.49 (Fundamental theorem of Algebra). Any nonconstant complex polynomial (3.22) has a root.

Proof. Suppose $P(z) \neq 0$ for all $z \in \mathbb{C}$. Define $Q(z)$ by $P(z) = zQ(z) + a_0$. Then for any $z \neq 0$ we have

$$\frac{1}{z} = \frac{zQ(z) + a_0}{zP(z)} = \frac{Q(z)}{P(z)} + \frac{a_0}{zP(z)}.$$

By Corollary 3.17 and Theorem 3.47, the integral of $\frac{1}{z}$ along the circle $\alpha(t) = R \exp(\mathbf{i}t)$ with $t \in [0, 2\pi]$ and $R > 0$ yields

$$2\pi \mathbf{i} = \oint_\alpha \frac{a_0}{zP(z)} dz,$$

where we have used the facts of $P(z) \neq 0$ and Q/P being analytic. Then Lemma 3.13 and Lemma 3.48 yield

$$2\pi = \left| \oint_\alpha \frac{a_0}{zP(z)} dz \right| \leq \oint_\alpha \left| \frac{a_0}{zP(z)} \right| |dz| \leq 2\pi R \frac{2|a_0|}{R|a_n|R^n},$$

which cannot hold for any sufficiently large $R > 0$. $\square$

3.3.3 A proof by Goursat

Definition 3.50. The triangle spanned by $z_1, z_2, z_3 \in \mathbb{C}$ is the point set

$$\Delta := \left\{ \sum_{j=1}^3 t_j z_j : t_j \geq 0, \sum_{j=1}^3 t_j = 1 \right\}. \quad (3.25)$$

The triangular path $\langle z_1, z_2, z_3 \rangle$ is the closed curve

$$\langle z_1, z_2, z_3 \rangle = \alpha := \alpha_1 \oplus \alpha_2 \oplus \alpha_3, \quad (3.26)$$

where the edges constituting the triangle boundary are

$$\alpha_1(t) = z_1 + (t-0)(z_2 - z_1), \quad t \in [0, 1];$$

$$\alpha_2(t) = z_2 + (t-1)(z_3 - z_2), \quad t \in [1, 2];$$

$$\alpha_3(t) = z_3 + (t-2)(z_1 - z_3), \quad t \in [2, 3].$$

Lemma 3.51. If $\{\mathcal{P}_i\}$ is a collection of compact subsets of a metric space such that the intersection of every finite subcollection of $\{\mathcal{P}_i\}$ is nonempty, then $\cap_i \mathcal{P}_i$ is nonempty.

Proof. Denote by $\mathcal{Q}_i$ the complement of $\mathcal{P}_i$. Suppose there exists $\mathcal{P}_j$ such that no points of $\mathcal{P}_j$ belongs to every $\mathcal{P}_i$. The the sets $\{\mathcal{Q}_i\}$ is an open cover of $\mathcal{P}_j$. By Definition D.145, there exist finitely many indices $k_1, \dots, k_n$ such that $\mathcal{P}_j \subset \cup_{\ell=k_1}^{k_n} \mathcal{Q}_\ell$. But this implies $\mathcal{P}_j \cap \mathcal{P}_{k_1} \cap \mathcal{P}_{k_2} \cap \dots \cap \mathcal{P}_\ell = \emptyset$, contradicting the starting point. $\square$

Theorem 3.52 (Cantor intersection theorem). Suppose $(\mathcal{P}_k)$ is a sequence of non-empty, compact subsets of $\mathcal{P}$ satisfying $\mathcal{P}_0 \supseteq \mathcal{P}_1 \supseteq \dots \supseteq \mathcal{P}_k \supseteq \dots$. Then $\cap_k \mathcal{P}_k \neq \emptyset$.

Proof. This follows directly from Lemma 3.51. $\square$

Exercise 3.53. Prove that if $\{E_n\}$ is a sequence of closed nonempty and bounded sets in a complete metric space $\mathcal{X}$, if $E_{n+1} \subset E_n$, and if $\lim_{n \rightarrow \infty} \text{diam } E_n = 0$, cf. Definition D.22. Then $\cap_{n=1}^{\infty} E_n$ consists of exactly one point.

Exercise 3.54. Use the conclusion in Exercise 3.53 to prove the Baire category theorem D.80.

Theorem 3.55 (Cauchy integral theorem on triangular paths). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic on a domain $\mathcal{D}$ and the triangle $\Delta(z_1, z_2, z_3)$ is a subset of $\mathcal{D}$, then we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0. \quad (3.27)$$

Proof. As shown in the plot below, we first construct a sequence of triangular paths

$$\alpha^{(n)} = \left\langle z_1^{(n)}, z_2^{(n)}, z_3^{(n)} \right\rangle$$

by inductively defining

$$\alpha^{(0)} := \alpha = \langle z_1, z_2, z_3 \rangle,$$

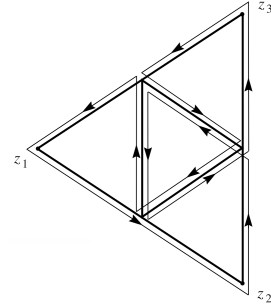
$$\alpha_1^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, z_2^{(n)}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle,$$

$$\alpha_2^{(n)} := \left\langle \frac{z_2^{(n)} + z_3^{(n)}}{2}, z_3^{(n)}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle,$$

$$\alpha_3^{(n)} := \left\langle \frac{z_3^{(n)} + z_1^{(n)}}{2}, z_1^{(n)}, \frac{z_1^{(n)} + z_2^{(n)}}{2} \right\rangle,$$

$$\alpha_4^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle.$$

and choosing $\alpha^{(n+1)} = \alpha_1^{(n)}$ or $\alpha_2^{(n)}$ or $\alpha_3^{(n)}$ or $\alpha_4^{(n)}$.



It follows that

$$\oint_{\alpha^{(n)}} = \oint_{\alpha_1^{(n)}} + \oint_{\alpha_2^{(n)}} + \oint_{\alpha_3^{(n)}} + \oint_{\alpha_4^{(n)}}$$

and we can choose $\alpha^{(n+1)}$ such that

$$\left| \oint_{\alpha^{(n)}} f \right| \leq 4 \left| \oint_{\alpha^{(n+1)}} f \right|,$$

which implies

$$\left| \oint_{\alpha} f \right| \leq 4^n \left| \oint_{\alpha^{(n)}} f \right|.$$

The closed triangles $\Delta^{(n)}$ are nested and Exercise 3.53 states that there exists a point $z_0 \in \cap_n \Delta^{(n)} \subset \mathcal{D}$. By Lemma 2.18, the condition of f being analytic at z_0 yields $f(z) = f(z_0) + f'(z_0)(z - z_0) + r(z)$ where

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.28)$$

Then we have $\oint_{\alpha^{(n)}} f(\zeta) d\zeta = \oint_{\alpha^{(n)}} r(\zeta) d\zeta$ from Theorem 3.16 and Exercise 2.22. It follows that

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right|. \quad (3.29)$$

We show $\lim_{n \rightarrow \infty} 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right| = 0$ to complete the proof. By (3.28), we have

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |z - z_0| < \delta \Rightarrow |r(z)| \leq \epsilon |z - z_0|, \quad (3.30)$$

where the condition $|z - z_0| < \delta$ holds for any sufficiently large n . In other words, we can choose n large enough such that $\Delta^{(n)} \subset \mathcal{U}_\delta(z_0)$. Denote by $\ell(\alpha)$ the arc length of α as

in Definition C.382. Then by construction of the sequence of triangles, we have

$$|z - z_0| \leq \ell(\alpha^{(n)}) = \frac{1}{2^n} \ell(\alpha),$$

which, together with (3.29), (3.30), and Lemma 3.13, yields

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \epsilon \frac{1}{2^n} \ell(\alpha) \ell(\alpha^{(n)}) = \epsilon \ell^2(\alpha).$$

To sum up, $\left| \oint_{\alpha} f(\zeta) d\zeta \right|$ can be made arbitrarily small, hence (3.27) holds. $\square$

Corollary 3.56. If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise linear simple closed curve $\beta \subset \mathcal{D}$ we have

$$\oint_{\beta} f(\zeta) d\zeta = 0. \quad (3.31)$$

Proof. This follows from Theorem 3.55 and the two ears theorem, which states that every simple polygon with more than three vertices has at least two ears. $\square$

Theorem 3.57 (Heine 1872). If $K \subset \mathbb{C}$ is compact and $f : K \rightarrow \mathbb{C}$ is continuous, then f is uniformly continuous.

Proof. This is a specialization of Theorem D.150. $\square$

Lemma 3.58. If $f : \mathcal{D} \rightarrow \mathbb{C}$ is continuous in a domain $\mathcal{D}$, then for any piecewise smooth simple curve $\alpha \subset \mathcal{D}$ and any $\epsilon > 0$, there exists a piecewise linear curve $\sigma \subset \mathcal{D}$ such that

$$\left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| < \epsilon. \quad (3.32)$$

In other words, the integral of f on a piecewise smooth curve can be approximated arbitrarily well by an integral of f on line segments.

Proof. Since $\mathcal{D}$ is open, there exists a compact subset $G \subset \mathcal{D}$ such that $\alpha \cap \partial G = \emptyset$ and $\alpha \subset G$. Define

$$\rho = \inf\{|z_1 - z_2| : z_1 \in \alpha, z_2 \in \partial G\}.$$

Clearly $\rho > 0$. We have $\forall z \in \alpha, \forall r < \rho, \mathcal{U}_r(z) \subset G$ and

$$\forall z_1, z_2 \in \alpha, |z_1 - z_2| < \rho \Rightarrow \{z_1 t + z_2(1-t) : t \in [0, 1]\} \subset G.$$

By Theorem 3.57, the continuity of f and the compactness of G yield the uniform continuity of f on G , i.e.,

$$\forall \epsilon > 0, \exists \delta_1 > 0 \text{ s.t. } \forall z_1, z_2 \in G,$$

$$|z_1 - z_2| < \delta_1 \Rightarrow |f(z_1) - f(z_2)| < \frac{\epsilon}{2\ell}, \quad (3.33)$$

where ℓ is the arc length of α .

Choose $n + 1$ points $z_1, z_2, \dots, z_{n+1} \in \alpha$ such that $\max_{j=1}^n |z_j - z_{j+1}| < \min(\rho, \delta_1)$: if α is closed then we set $z_{n+1} = z_1$; if α is not closed we set z_1 and z_{n+1} as the starting and ending points, respectively. We also require that $\alpha^{-1}(z_1), \alpha^{-1}(z_2), \dots, \alpha^{-1}(z_{n+1})$ increases monotonically. Then the piecewise linear closed curve σ formed by connecting z_j 's consecutively by linear segments is contained

in $\mathcal{D}$. Its edges σ_j 's correspond to the arcs α_j 's. For each j , Lemma 3.14 yields

$$\int_{\sigma_j} f(z_j) d\zeta = f(z_j)(z_{j+1} - z_j) = \int_{\alpha_j} f(z_j) d\zeta. \quad (3.34)$$

Consequently, we have

$$\begin{aligned} & \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| \\ & \leq \left| \int_{\alpha_j} [f(\zeta) - f(z_j)] d\zeta \right| + \left| \int_{\sigma_j} [f(\zeta) - f(z_j)] d\zeta \right| \\ & \leq \ell(\alpha_j) \sup_{\zeta \in \alpha_j} |f(\zeta) - f(z_j)| + \ell(\sigma_j) \sup_{\zeta \in \sigma_j} |f(\zeta) - f(z_j)| \\ & < \frac{\epsilon}{2\ell} [\ell(\alpha_j) + \ell(\sigma_j)] \leq \frac{\epsilon}{\ell} \ell(\alpha_j), \end{aligned}$$

where the first step follows from (3.34) and the triangle inequality, the second step from Lemma 3.13 and Definition C.382, and the third step from (3.33). Then we have

$$\begin{aligned} & \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| = \left| \sum_{j=1}^n \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| \\ & \leq \sum_{j=1}^n \left| \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| = \epsilon, \end{aligned}$$

which completes the proof. $\square$

Theorem 3.59 (Cauchy integral theorem for simply connected domains). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise smooth closed curve $\alpha \subset \mathcal{D}$ we have

$$\oint_{\alpha} f(\zeta) d\zeta = 0. \quad (3.35)$$

Proof. This follows directly from Corollary 3.56 and Lemma 3.58. $\square$

Exercise 3.60. For an analytic function $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $\forall z \in \mathbb{C}, f(z+1) = f(z)$, prove that the value of $\int_{[w, w+1]} f(\zeta) d\zeta$ is independent of the choice of $w \in \mathbb{C}$.

Exercise 3.61. For bounded simply connected domain $\mathcal{D}$,

- (a) prove $\oint_{\partial \mathcal{D}} \bar{z} dz = 2iS(\mathcal{D})$, where $S(\mathcal{D})$ is the area of $\mathcal{D}$,
- (b) show that f being analytic on $\bar{\mathcal{D}}$ implies

$$\max_{z \in \partial \mathcal{D}} |\bar{z} - f(z)| \geq 2 \frac{S(\mathcal{D})}{\ell(\partial \mathcal{D})},$$

where $\partial \mathcal{D}$ is the boundary of $\mathcal{D}$ and $\ell(\partial \mathcal{D})$ its arc length.

Definition 3.62. A *star-shaped domain* is an open set $\mathcal{D} \subset \mathbb{C}$ that satisfies

$$\exists z_* \in \mathcal{D} \text{ s.t. } \forall z \in \mathcal{D}, \{z_* + t(z - z_*) : t \in [0, 1]\} \subset \mathcal{D},$$

where z_* is called a *star center*.

Example 3.63. Each convex domain, in particular the open disk, is star-shaped. The ring sector

$$\{z_0 + \zeta \rho e^{i\varphi} : \rho \in (r, R), \varphi \in (0, \beta); \zeta, z_0 \in \mathbb{C}, |\zeta| = 1\}$$

is star-shaped if $\beta < \pi$ and $\cos \frac{\beta}{2} > \frac{r}{R}$.

Theorem 3.64 (Cauchy integral theorem for star-shaped domains). Each analytic function f defined on a star-shaped domain $\mathcal{D}$ has a primitive in $\mathcal{D}$.

Proof. This follows from Theorems 3.59 and 3.42 since a star-shaped domain is a simply connected domain. $\square$

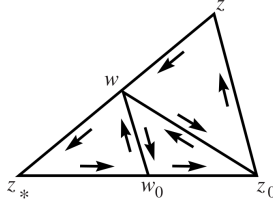
Exercise 3.65 (Alternative proof of Corollary 3.15). Prove Corollary 3.15 via Theorems 3.64 and 3.42 by choosing a suitable integral path from 1 to z .

Corollary 3.66. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be continuous in a star-shaped domain $\mathcal{D}$ with center z_* . If f is analytic at every point $z \neq z_*$, then f has a primitive in $\mathcal{D}$.

Proof. By Theorem 3.42, it suffices to show

$$\oint_{\langle z, z_0, z_* \rangle} f(\zeta) d\zeta = 0,$$

where $z \neq z_*$ and $z_0 \neq z_*$ are two arbitrary points in $\mathcal{D}$ and the triangle is entirely contained within $\mathcal{D}$.



As shown in the above plot, Theorem 3.55 yields

$$\oint_{\langle z_*, z_0, z \rangle} = \oint_{\langle z_*, w_0, w \rangle} + \oint_{\langle w_0, z_0, w \rangle} + \oint_{\langle z_0, z, w \rangle} = \oint_{\langle z_*, w_0, w \rangle}.$$

The proof is completed by passing to the limit $w \rightarrow z_*$ and $w_0 \rightarrow z_*$. $\square$

3.3.4 Multiply connected domains

Theorem 3.67 (Jordan curve theorem). The complement of a Jordan curve γ consists of two components: one is bounded and the other is unbounded; both of them are open and each has γ as its boundary.

Definition 3.68. A Jordan curve is said to be *positively oriented* or *counterclockwise oriented* iff an observer traversing in the direction of its parametrization always has the bounded complement on her left; it is *negatively oriented* or *clockwise oriented* iff the observer always has the unbounded complement on her left.

Definition 3.69. The *interior* of an oriented Jordan curve γ , denoted by $\text{int}(\gamma)$, is the complement of γ that always lies to the left of an observer who traverses the curve in the increasing direction of its parameter; the other complement of γ is called the *exterior* of γ and denoted by $\text{ext}(\gamma)$.

Lemma 3.70 (Seifert decomposition). Any oriented closed curve can be decomposed into the union of a finite number of oriented, pairwise non-crossing Jordan curves.

Proof. The local neighborhood of each self-intersection contains an even number of arc pieces. Rearrange them so that the resulting Jordan curves have no proper intersections. $\square$

Definition 3.71. A *disk with n holes* ($n \in \mathbb{N}$) is a connected open set $\mathcal{D} \subset \mathbb{C}$ that can be expressed as

$$\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j), \quad (3.36)$$

where $\text{int}(\gamma)$ is the interior of a Jordan curve γ in Definition 3.69, γ_0 is a counter-clockwise oriented Jordan curve, and $\gamma_1, \dots, \gamma_n$ are clockwise oriented Jordan curves satisfying

$$\forall j = 1, \dots, n, \text{ext}(\gamma_j) \subset \text{int}(\gamma_0), \quad (3.37a)$$

$$\forall j, k = 1, \dots, n, j \neq k \Rightarrow \text{ext}(\gamma_j) \cap \text{ext}(\gamma_k) = \emptyset. \quad (3.37b)$$

The boundary and closure of a disk with holes are denoted respectively by

$$\partial\mathcal{D} := \bigcup_{j=0}^n \gamma_j, \quad \overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}.$$

Example 3.72. It can be shown that any regular open semianalytic set with bounded boundary is a disk with holes. For example, an open disk is a disk with no holes.

Theorem 3.73 (Cauchy integral theorem for a disk with holes). A continuous function $f : \overline{\mathcal{D}} \rightarrow \mathbb{C}$ that is analytic on the disk with holes $\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j)$ satisfies

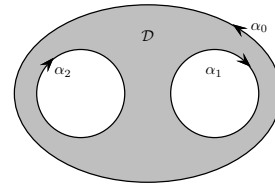
$$\int_{\partial\mathcal{D}} f(z) dz = 0, \quad (3.38)$$

or equivalently,

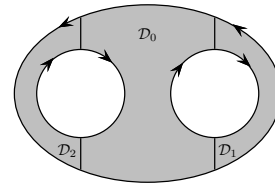
$$\oint_{\gamma_0} f(z) dz = \sum_{j=1}^n \oint_{\gamma_j^-} f(z) dz, \quad (3.39)$$

where γ_j^- is the reciprocal of γ_j as in Definition 3.30.

Proof. The domain $\mathcal{D}$ is exemplified for $n = 2$ as follows.



It follows from a simple induction that $\bigcup_{j=0}^n \gamma_j$ can be reorganized into positively-oriented Jordan curves with disjoint interiors $\mathcal{D}_0, \mathcal{D}_1, \dots, \mathcal{D}_m$ as follows.



Then (3.38) follows from $\bigcup_{j=0}^n \gamma_j = \bigcup_{j=0}^m \partial\mathcal{D}_j$ and the Cauchy integral theorem 3.59. (3.39) follows from (3.38), Definition 3.71, and moving the integrals over clockwise oriented Jordan curves to the RHS and reversing the orientation of each γ_j 's for each $j = 1, 2, \dots, n$. $\square$

3.4 Cauchy integral formulas

Lemma 3.74. For $c \in \mathbb{C}$ and $r > 0$, represent $\partial\mathcal{U}_r(c)$ as a counter-clockwise oriented circle

$$\forall t \in [0, 2\pi], \quad \alpha(t) := c + re^{it}, \quad (3.40)$$

and we have

$$\forall z \in \mathcal{U}_r(c), \quad \oint_{\alpha} \frac{d\zeta}{\zeta - z} = 2\pi i. \quad (3.41)$$

Proof. Referring to Definition 3.71, we construct a disk $\mathcal{D}$ with one hole by setting γ_0 as the circle α and γ_1 as a clockwise oriented circle of which the center is z and the radius is $\rho \leq r - |c - z|$. Theorem 3.73 yields

$$0 = \int_{\partial\mathcal{D}} = \oint_{\gamma_0} + \oint_{\gamma_1} \implies \oint_{\gamma_0} = \oint_{\gamma_1}.$$

The rest of the proof follows from Corollary 3.17. $\square$

Theorem 3.75 (Cauchy integral formula). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall \overline{\mathcal{U}_r(c)} \subset \mathcal{D}, \quad \forall z \in \mathcal{U}_r(c), \quad f(z) = \frac{1}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad (3.42)$$

where α is the oriented representation of $\partial\mathcal{U}_r(c)$ in (3.40).

Proof. The compactness of $\overline{\mathcal{U}_r(c)}$ gives

$$\exists R > r \text{ s.t. } \overline{\mathcal{U}_r(c)} \subset \mathcal{U}_R(c) \subset \mathcal{D}.$$

We can thus assume that $\mathcal{D}$ is an open disk. The function

$$g(w) := \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } w \neq z, \\ f'(z) & \text{if } w = z \end{cases}$$

is analytic in $\mathcal{D} \setminus \{z\}$ and is continuous in $\mathcal{D}$. Then Corollary 3.66 and Theorem 3.42 yield

$$\oint_{\alpha} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = 0 \implies f(z) \oint_{\alpha} \frac{1}{\zeta - z} d\zeta = \oint_{\alpha} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

The proof is completed by Lemma 3.74. $\square$

Corollary 3.76 (Cauchy integral formula for a disk with holes). A continuous function $f : \overline{\mathcal{D}} \rightarrow \mathbb{C}$ that is analytic on the disk with holes $\mathcal{D} := \cap_{j=0}^n \text{int}(\gamma_j)$ satisfies

$$\forall z \in \mathcal{D}, \quad f(z) = \frac{1}{2\pi i} \oint_{\gamma_0} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \sum_{j=1}^n \oint_{\gamma_j^-} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad (3.43)$$

where γ_j^- is the reciprocal of γ_j as in Definition 3.30.

Proof. Choose $r > 0$ sufficiently small so that $\overline{\mathcal{U}_r(z)} \subset \mathcal{D}$, denote by γ_{n+1} the clockwise oriented curve that represents $\partial\mathcal{U}_r(z)$, and define $\mathcal{D}_r := \mathcal{D} \setminus \overline{\mathcal{U}_r(z)}$. Definition 3.71 implies $\mathcal{D}_r = \cap_{i=0}^{n+1} \text{int}(\gamma_i)$. Since the function $g : \mathcal{D}_r \rightarrow \mathbb{C}$ given by $g(\zeta) := \frac{f(\zeta)}{\zeta - z}$ is analytic, Theorems 3.73 and 3.75 imply

$$\oint_{\gamma_0} g(\zeta) d\zeta = \sum_{j=1}^{n+1} \oint_{\gamma_j^-} g(\zeta) d\zeta = \sum_{j=1}^n \oint_{\gamma_j^-} g(\zeta) d\zeta + 2\pi i f(z),$$

which completes the proof. $\square$

Example 3.77. Let α denote the circle defined in (3.40) with $c = 0$ and $r = 2$. We compute

$$\oint_{\alpha} \frac{\zeta}{(9 - \zeta^2)(\zeta + i)} d\zeta = \oint_{\alpha} \frac{\frac{\zeta}{9 - \zeta^2}}{\zeta - (-i)} d\zeta = 2\pi i \left. \frac{\zeta}{9 - \zeta^2} \right|_{\zeta = -i} = \frac{\pi}{5},$$

where the second step follows from Theorem 3.75.

Corollary 3.78 (Mean value equation). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall \overline{\mathcal{U}_r(c)} \subset \mathcal{D}, \quad f(c) = \frac{1}{2\pi} \int_0^{2\pi} f(c + re^{it}) dt. \quad (3.44)$$

Proof. This follows directly from Theorem 3.75 by setting $z = c$ and $\zeta = c + re^{it}$. $\square$

Example 3.79. To compute $I = \oint_{|z|=2} \frac{dz}{z^2 + 2}$, we construct two counter-clockwise oriented circles α_1 and α_2 surrounding the two singularities $z = \pm\sqrt{2}i$. For sufficiently small radii, Theorem 3.73 yields

$$\begin{aligned} \oint_{|z|=2} f(z) dz &= \oint_{\alpha_1} f(z) dz + \oint_{\alpha_2} f(z) dz \\ &= \oint_{\alpha_1} \frac{1}{z - \sqrt{2}i} dz + \oint_{\alpha_2} \frac{1}{z - (-\sqrt{2}i)} dz \\ &= \left. \frac{2\pi i}{z + \sqrt{2}i} \right|_{z=\sqrt{2}i} + \left. \frac{2\pi i}{z - \sqrt{2}i} \right|_{z=-\sqrt{2}i} = 0. \end{aligned}$$

Exercise 3.80. Compute $I = \oint_{|z|=1} \frac{3z-1}{z(z-2)} dz$.

Exercise 3.81. For some $r > 0$, f is an analytic function on $\{z \in \mathbb{C} : |z| > r\}$. Show that $\lim_{z \rightarrow \infty} zf(z) = A$ implies

$$\forall R > r, \quad \oint_{|z|=R} f(z) dz = 2\pi i A.$$

Exercise 3.82. Refer to Notations 1 and 3.

(a) For an analytic function $f : \mathcal{U}_R \rightarrow \mathbb{C}$, prove

$$\forall r \in (0, R), \forall z \in \mathcal{U}_r, \quad f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(re^{it}) \frac{r^2 - |z|^2}{|re^{it} - z|^2} dt.$$

(b) Show that $\text{Re} \left(\frac{re^{it} + z}{re^{it} - z} \right) = \frac{r^2 - |z|^2}{|re^{it} - z|^2}$.

(c) Show that u being harmonic in an open set containing $\overline{\mathbb{E}}$ implies

$$z \in \mathbb{E}, \quad u(z) = \int_0^{2\pi} P(z, e^{it}) u(e^{it}) dt,$$

where $P(z, e^{it})$ is the Poisson kernel defined by

$$P(z, e^{it}) = \frac{1}{2\pi} \frac{1 - |z|^2}{|e^{it} - z|^2} = \frac{1}{2\pi} \text{Re} \left(\frac{e^{it} + z}{e^{it} - z} \right).$$

Hint: For (a), note that if $w = \frac{r^2}{\bar{z}}$, then the integral of $\frac{f(\zeta)}{\zeta - w}$ over the circle $|\zeta| = r$ is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired result. For (c), apply Exercise 3.41, then use (a) and (b).

Lemma 3.83. Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function that is also analytic in $\mathcal{D}$ for any fixed $t \in [a, b]$. Also suppose that the partial derivative $\frac{\partial f}{\partial z} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ is continuous. Then the function

$$g(z) := \int_a^b f(t, z) dt$$

is analytic in $\mathcal{D}$ and

$$g'(z) = \int_a^b \frac{\partial f(t, z)}{\partial z} dt. \quad (3.45)$$

Proof. For any fixed t , write $f(t, z) = p(t, x, y) + \mathbf{i}q(t, x, y)$ and $g(z) = u(x, y) + \mathbf{i}v(x, y)$. Then we have $u = \int p dt$ and $v = \int q dt$. Then the Cauchy–Riemann theorem 2.56 applied to f yields

$$\begin{aligned} \frac{\partial u}{\partial x} &= \int \frac{\partial p}{\partial x} = \int \frac{\partial q}{\partial y} = \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= \int \frac{\partial p}{\partial y} = - \int \frac{\partial q}{\partial x} = - \frac{\partial v}{\partial x}. \end{aligned}$$

Since $\frac{\partial f}{\partial \bar{z}} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ is continuous, $g(z)$ is totally differentiable and thus analytic by the above equations and the Cauchy–Riemann theorem. By Theorem 2.72, we have $\frac{\partial f}{\partial \bar{z}} = 0$. Then (3.45) follows from Theorem B.131. $\square$

Theorem 3.84 (Generalized Cauchy integral formula). Every analytic function is arbitrarily often complex differentiable. More precisely, any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\overline{\mathcal{U}_r(c)} \subset \mathcal{D}, \quad \forall z \in \mathcal{U}_r(c), \quad f^{(n)}(z) = \frac{n!}{2\pi\mathbf{i}} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta, \quad (3.46)$$

where α is the oriented representation of $\partial\mathcal{U}_r(c)$ in (3.40).

Proof. For $G(t, z) := \frac{f(\zeta(t))\zeta'(t)}{\zeta(t) - z}$, Theorem 3.75 yields

$$f(z) = \frac{1}{2\pi\mathbf{i}} \oint_{\alpha} G(t, z) dt.$$

Then Lemma 3.83 and an induction on n yield (3.46). $\square$

Corollary 3.85 (Cauchy inequality). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\overline{\mathcal{U}_r(c)} \subset \mathcal{D}, \quad \forall n \in \mathbb{N}^+, \quad |f^{(n)}(c)| \leq \frac{n!}{r^n} M(r), \quad (3.47)$$

where $M(r) := \max_{|\zeta - c| = r} |f(\zeta)|$.

Proof. For the circle α defined in (3.40), Theorem 3.84 yields

$$|f^{(n)}(c)| = \left| \frac{n!}{2\pi\mathbf{i}} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - c)^{n+1}} d\zeta \right| \leq \frac{n!}{2\pi} \frac{M(r)}{r^{n+1}} 2\pi r = \frac{n! M(r)}{r^n}.$$

$\square$

Example 3.86. For a Jordan curve α around $\mathbf{i}$, we have

$$\oint_{\alpha} \frac{\cos z}{(z - \mathbf{i})^3} dz = \frac{2\pi\mathbf{i}}{2!} (\cos z)''|_{z=\mathbf{i}} = -\pi\mathbf{i} \cos \mathbf{i} = -\pi \frac{e^{-1} + e}{2} \mathbf{i},$$

where the last step follows from Corollary 1.94.

Exercise 3.87. Compute $I = \oint_{|z|=2\pi} \frac{z^3}{(z-\pi)^3} dz$.

Exercise 3.88. Suppose $f : \mathbb{E} \rightarrow \mathbb{C}$ is analytic. Show that the diameter $d = \sup_{z, w \in \mathbb{E}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Hint: Show that $2f'(0) = \frac{1}{2\pi\mathbf{i}} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$ for any $r \in (0, 1)$.

Exercise 3.89. Let f be a bounded analytic function on the unit disk $\mathbb{E}$. Prove that

$$\sup_{z \in \mathbb{E}} (1 - |z|) |f'(z)| < +\infty.$$

Hint: For $z \in \mathbb{E}$, use the Cauchy inequality at the circle centered at z with radius $1 - |z|$.

Exercise 3.90. Let f be an analytic function on the strip $\mathcal{D} := \{z \in \mathbb{C} : |\operatorname{Im} z| < 1\}$. For a fixed $\eta \in \mathbb{R}$, there exists $A > 0$ such that

$$\forall z \in \mathcal{D}, \quad |f(z)| \leq A(1 + |z|)^{\eta}.$$

Show that for each $n \in \mathbb{N}$, there exists $A_n > 0$ such that

$$\forall x \in \mathbb{R}, \quad |f^{(n)}(x)| \leq A_n(1 + |x|)^{\eta}.$$

Hint: Use the Cauchy inequality.

Exercise 3.91. Let f be analytic in an open set containing $\overline{\mathbb{E}}$.

(a) Prove that

$$\frac{1}{2\pi\mathbf{i}} \oint_{|\zeta|=1} \frac{\overline{f(\zeta)}}{\zeta - z} d\zeta = \begin{cases} \overline{f(0)} & \text{if } |z| < 1, \\ \overline{f(0) - f(\frac{1}{\bar{z}})} & \text{if } |z| > 1. \end{cases}$$

(b) If $|z| < 1$, compute

$$\frac{1}{2\pi\mathbf{i}} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

(c) If P is a polynomial, show that

$$\oint_{|\zeta|=1} \overline{P(\zeta)} d\zeta = 2\pi\mathbf{i} \overline{P'(0)}.$$

Hint: You can verify that

- $\int_{\alpha} \overline{f(\zeta)} d\zeta = \overline{\int_{\alpha} f(\zeta) d\bar{\zeta}}$, where α is any piecewise smooth curve.
- For $|\zeta| = 1$, it holds that $\bar{\zeta} = \frac{1}{\zeta}$, $d\bar{\zeta} = -\frac{d\zeta}{\zeta^2}$ and $d|\zeta| = \frac{d\zeta}{\mathbf{i}\zeta}$.

3.5 Some consequences

3.5.1 Morera's theorem

Theorem 3.92 (Morera 1886). Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function. If

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0 \quad (3.48)$$

for every triangular path $\langle z_1, z_2, z_3 \rangle$ whose triangle is entirely contained in $\mathcal{D}$, then f is analytic.

Proof. For any given point $z_0 \in \mathcal{D}$, there is an open neighborhood $\mathcal{U}_\epsilon(z_0) \subset \Delta(z_1, z_2, z_3) \subset \mathcal{D}$. It suffices to show that f is analytic in $\mathcal{U}_\epsilon(z_0)$. By (3.48), Lemma 3.58, and the fact that every polygon has a triangulation, the integral of f along any Jordan curve in $\mathcal{U}_\epsilon(z_0)$ vanishes. Theorem 3.42 then implies that f has a primitive F and the proof is completed by applying Theorem 3.84 to F . $\square$

3.5.2 Liouville's theorem

Definition 3.93. An *entire function* is an analytic function $f : \mathbb{C} \rightarrow \mathbb{C}$.

Theorem 3.94 (Liouville 1847). Every bounded entire function is a constant.

Proof. We show $f'(z) = 0$ for all $z \in \mathbb{C}$. Since f is bounded, there exists $C > 0$ such that $|f(z)| \leq C$ for all $z \in \mathbb{C}$. Then Corollary 3.85 implies that

$$\forall z \in \mathbb{C}, \forall r > 0, \quad |f'(z)| \leq \frac{C}{r}.$$

Passing to the limit $r \rightarrow +\infty$ completes the proof. $\square$

Example 3.95. $f(z) = \cos(z)$ is not bounded since

$$\lim_{x \rightarrow +\infty} \cos(ix) = \lim_{x \rightarrow +\infty} \frac{e^x + e^{-x}}{2} = +\infty.$$

Hence Theorem 3.92 does not apply to $\cos(z)$.

Exercise 3.96 (The generalized Liouville). Let f be an entire function. For a given integer $m \geq 1$, there exists $C > 0$ such that

$$\forall z \in \mathbb{C}, \quad |f(z)| \leq C(1 + |z|)^m.$$

Show that f is a polynomial of degree at most m .

Hint: Use the Cauchy inequality to show $f^{(m+1)} \equiv 0$.

Exercise 3.97. We say $f : \mathbb{C} \rightarrow \mathbb{C}$ is *doubly periodic function* if there exist $\tau \in \mathbb{C} \setminus \mathbb{R}$ such that

$$\forall z \in \mathbb{C}, \quad f(z+1) = f(z+\tau) = f(z).$$

Show that every entire doubly periodic function is a constant.

Exercise 3.98. Let f be a nonconstant entire function. Show that $f(\mathbb{C})$ is dense in $\mathbb{C}$, that is, for any $\zeta \in \mathbb{C}$, there exists a sequence $\{z_n\}$ such that $f(z_n) \rightarrow \zeta$.

Hint: Suppose there exist $\zeta \in \mathbb{C}$ and $r > 0$ such that $\mathcal{U}_r(\zeta) \cap f(\mathbb{C}) = \emptyset$. Then apply Liouville's theorem to $\frac{1}{f(z)-\zeta}$.

Exercise 3.99 (Alternative proof of the fundamental theorem of algebra). Prove Theorem 3.49 by applying Liouville's theorem.

3.5.3 The stability of complex calculus on power series

Lemma 3.100. Suppose a sequence $(f_n : \mathcal{D} \rightarrow \mathbb{C})_{n \in \mathbb{N}}$ of continuous functions with $\mathcal{D}$ open converges locally uniformly to f . Then we have

$$\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz, \quad (3.49)$$

where $\gamma : [0, 1] \rightarrow \mathcal{D}$ is any piecewise smooth curve.

Exercise 3.101. Prove Lemma 3.100.

Theorem 3.102 (Weierstrass 1841). Let $\mathcal{D} \subset \mathbb{C}$ be open and $(f_n : \mathcal{D} \rightarrow \mathbb{C})$ be a sequence of analytic functions. If (f_n) converges locally uniformly to f in the sense of Definition D.168, then f is analytic and the sequence of derivatives (f'_n) converges locally uniformly to f' .

Proof. Let $\gamma := \langle z_1, z_2, z_3 \rangle$ be an arbitrary triangular path whose triangle is entirely contained in $\mathcal{D}$. We have

$$\oint_{\gamma} f(\zeta) d\zeta = \oint_{\gamma} \lim_{n \rightarrow \infty} f_n(\zeta) d\zeta = \lim_{n \rightarrow \infty} \oint_{\gamma} f_n(\zeta) d\zeta = 0,$$

where the first equality follows from the local uniform convergence of (f_n) to f , the second from Lemma 3.100, and the last from Cauchy's Theorem 3.47. Then f being analytic follows from Morera's theorem 3.92.

(f'_n) converges locally uniformly to f' because

$$\begin{aligned} \forall c \in \mathcal{D}, \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c) \subset \mathcal{D}, \\ \lim_{n \rightarrow \infty} f'_n(z) &= \lim_{n \rightarrow \infty} \left(\frac{1}{2\pi i} \oint_{\alpha} \frac{f_n(\zeta)}{\zeta - z} d\zeta \right)' \\ &= \lim_{n \rightarrow \infty} \frac{1}{2\pi i} \oint_{\alpha} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta = \frac{1}{2\pi i} \oint_{\alpha} \lim_{n \rightarrow \infty} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta \\ &= \frac{1}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta = f'(z), \end{aligned}$$

where α is the circle in (3.40), the first step follows from the Cauchy integral formula (Theorem 3.75), the second from the generalized Cauchy integral formula (Theorem 3.84), the third from Lemma 3.100, the fourth from the locally uniform convergence of (f_n) to f , and the last from the Cauchy integral formula. $\square$

Exercise 3.103. With the assumptions of Theorem 3.102, show that for each $k \in \mathbb{N}$, $(f_n^{(k)})$ converges locally uniformly to $f^{(k)}$.

Theorem 3.104 (Weierstrass 1841). If a series $\sum_{n=0}^{\infty} f_n$ of analytic functions $\mathcal{D} \mapsto \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open converges normally to f in the sense of Definition 1.74, then f is analytic and the series $\sum_{n=0}^{\infty} f'_n$ converges normally to f' .

Proof. By Theorem 1.75, the normal convergence of $\sum_{n=0}^{\infty} f_n$ implies the local uniform convergence of the sequence $(\sum_{i=0}^n f_i)_{n=0}^{\infty}$ to f . By Theorem 3.102, f is analytic. For any given $c \in \mathcal{D}$ we choose $\epsilon > 0$ small enough such that the closed disk $\overline{\mathcal{U}_{2\epsilon}(c)}$ is entirely contained in $\mathcal{D}$. Then the generalized Cauchy integral formula yields

$$\forall z \in \mathcal{U}_{\epsilon}(c), \quad |f'_n(z)| = \left| \frac{1}{2\pi i} \oint_{|\zeta-c|=2\epsilon} \frac{f_n(\zeta)}{(\zeta-z)^2} d\zeta \right| \leq \frac{2M_n}{\epsilon},$$

where $\frac{1}{|\zeta-z|^2} \leq \frac{1}{\epsilon^2}$ holds for all $z \in \mathcal{U}_{\epsilon}(c)$ and the inequality follows from Lemma 3.13. The rest follows from Definition 1.74. $\square$

Corollary 3.105. The derivative of the function $f : \mathbb{C} \rightarrow \mathbb{C}$

$$f(z) = \sum_{n=0}^{\infty} a_n(z-c)^n \quad (3.50)$$

can be obtained termwise in its convergence disk $\mathcal{U}_r(c)$:

$$\forall z \in \mathcal{U}_r(c), \quad f'(z) = \sum_{n=1}^{\infty} n a_n(z-c)^{n-1}. \quad (3.51)$$

Proof. This follows from Theorems 1.77 and 3.104. $\square$

Exercise 3.106. The k -th derivative of the function f in (3.50) can be obtained termwise in its convergence disk $\mathcal{U}_r(c)$:

$$\forall z \in \mathcal{U}_r(c), \quad f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n(z-c)^{n-k}.$$

Exercise 3.107. Show that if f is analytic in $\mathbb{E}$ and continuous on $\overline{\mathbb{E}}$, then $\oint_{|\zeta|=1} f(\zeta) d\zeta = 0$.

Hint: Consider a sequence of functions $(f_r)_{0 < r < 1}$, where

$$f_r : \overline{\mathcal{U}_{\frac{1}{r}}(0)} \rightarrow \mathbb{C}, \quad f_r(z) = f(rz).$$

Exercise 3.108. Use the proof of Exercise 3.107 to show that the result in Exercise 3.82(c) still holds if u is harmonic in $\mathbb{E}$ and continuous on $\overline{\mathbb{E}}$.

Definition 3.109. The p -series or the *hyperharmonic series* is defined for any $p > 0$ as

$$S_p := \sum_{n=1}^{\infty} \frac{1}{n^p}. \quad (3.52)$$

It is called an *overharmonic series* if $p > 1$.

Theorem 3.110. The p -series diverges for $p \leq 1$ and converges for $p > 1$.

Proof. Use the integral test in Theorem B.50. $\square$

Definition 3.111. *Riemann zeta function* is defined as

$$\zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad (3.53)$$

where $s \in \mathbb{C}$ and $\operatorname{Re} s > 1$.

Corollary 3.112. The series in (3.53) converges normally and the Riemann zeta function is analytic in the half-plane

$$\mathcal{D} = \{s \in \mathbb{C} : \operatorname{Re} s > 1\}. \quad (3.54)$$

Proof. By Corollary 2.25, we have

$$|n^s| = |\exp(s \operatorname{Log} n)| = |n^{\operatorname{Re} s}|.$$

Put $M_n := \left(\frac{1}{n}\right)^{\operatorname{Re} s}$ and the proof is completed by Definition 1.74, Theorem 3.104, and Theorems 3.110. $\square$

Exercise 3.113. *Bessel function of order m* is defined as

$$\mathcal{J}_m(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(m+n)!} \left(\frac{z}{2}\right)^{2n+m},$$

where $m \in \mathbb{N}$. By computing the convergence radius of the above power series, show that $\mathcal{J}_m$ is an entire function.

Exercise 3.114. The *hypergeometric series* is defined as

$$F(a, b, c; z) := \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!},$$

where $a, b, c \in \mathbb{C}$ are fixed, $-c \notin \mathbb{N}$, and $(z)_n$ is the *Pochhammer symbol* defined as

$$\forall z \in \mathbb{C}, \quad (z)_n := \begin{cases} 1 & \text{if } n = 0, \\ z(z+1) \cdots (z+n-1) & \text{if } n \in \mathbb{N}^+. \end{cases}$$

Show that $F(z)$ converges for $|z| < 1$ and satisfies the differential equation

$$z(1-z)F''(z) + (c - (a+b+1)z)F'(z) - abF(z) = 0.$$

Chapter 4

Characterizing Analytic Functions

4.1 Power series

Theorem 4.1 (Power series expansion, Cauchy 1831). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall c \in \mathcal{D}, \forall \mathcal{U}_R(c) \subseteq \mathcal{D}, \forall z \in \mathcal{U}_R(c), f(z) = \sum_{n=0}^{\infty} a_n(z-c)^n, \quad (4.1)$$

where $\mathcal{U}_R(c)$ is the open disk with radius R and center c in Notation 1 and each a_n is uniquely determined as

$$a_n := \frac{f^{(n)}(c)}{n!} = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta, \quad (4.2)$$

whose value does not depend on the choice of $\rho \in (0, R)$.

Proof. $|z-c| < |\zeta-c|$ holds for all $z \in \mathcal{U}_\rho(c)$. Then the geometric series yields

$$\frac{1}{\zeta-z} = \frac{1}{(\zeta-c)(1-\frac{z-c}{\zeta-c})} = \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}}.$$

The Cauchy integral formula (Theorem 3.75) yields

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta \\ &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} f(\zeta) \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} (z-c)^n \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} a_n(z-c)^n, \end{aligned}$$

where the third step follows from Theorems 1.77, 1.75 and D.178, and the last step from (4.2) and Theorem 3.84. Finally, the uniqueness of a_n follows from the uniqueness of the above derivation. $\square$

Exercise 4.2 (Gutzmer's inequality). With the assumptions of Theorem 4.1, show that for any $r \in (0, R)$,

$$\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq M(r)^2, \quad (4.3)$$

where $M(r)$ is defined in Corollary 3.85.

Exercise 4.3 (Alternative proof of Corollary 3.85). Prove (3.47) by using (4.3). Furthermore, show that the equality in (3.47) holds for some $n_0 \in \mathbb{N}$ if and only if $f(z) = a_{n_0}(z-c)^{n_0}$.

Example 4.4. Theorem 4.1 applied to $\text{Log}(z)$ on $\mathbb{C}_-$ in (2.8) yields a power series expansion,

$$\text{Log}(z) = \text{Log}(c) + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{c^n n} (z-c)^n, \quad (4.4)$$

where we have used the fact $(\text{Log} z)' = \frac{1}{z}$ (why?). By Theorem 1.79, the convergence radius is $|c|$. Interestingly, this example shows that the convergence radius may be bigger than the distance of the series center to the boundary of the domain of definition.

One needs to be careful about the convergence of a Taylor series. If $c \in \mathbb{C}_-$ satisfies $\text{Re } c < 0$, then $\mathbb{C}_- \cap \mathcal{U}_r(c)$ might consist of two disjoint separable connected components, where the power series converge to different functions,

$$f(z) = \begin{cases} \text{Log}(z) & \text{if } \text{Im } c > 0; \\ \text{Log}(z) + 2\pi i & \text{if } \text{Im } c < 0. \end{cases}$$

Example 4.5. $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges for $|z| = 1$, $\sum_{n=1}^{\infty} z^n$ diverges for $|z| = 1$, and the series

$$p(z) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n} \quad (4.5)$$

diverges for $z = -1$ and converges for all other z satisfying $|z| \leq 1$. Why? The answers lie in Theorem 3.110 and the fact that $p(z) = \text{Log}(1+z)$.

Example 4.6. Consider the geometric series

$$g(z) = \frac{1}{1-z^2} = \sum_{n=0}^{\infty} z^{2n}.$$

The divergence of $\sum_{n=0}^{\infty} z^{2n}$ for $|z| \geq 1, z \in \mathbb{R}$ is easily understood: $g(z)$ has two singularities at $z = \pm 1$. However the divergence of the geometric series

$$f(z) = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}.$$

for $|z| \geq 1, z \in \mathbb{R}$ is perplexing. After all, $f(z)$ is always bounded for any given $z \in \mathbb{R}$. This puzzle is solved in the context of complex analysis by observing that $f(z)$ has two singularities at $z = \pm i$. Indeed, $f(z) = g(iz)$ and the two functions are really the same complex function with different coordinates.

Exercise 4.7 (Double series theorem of Weierstrass). Let the power series

$$f_j(z) = \sum_{k=0}^{\infty} c_{jk}(z-c)^k \quad (4.6)$$

be convergent in the disk $\mathcal{U}_r(c)$ with $r > 0$. If the series $F := \sum_{j=0}^{\infty} f_j$ converges normally in $\mathcal{U}_r(c)$, show that F is analytic in $\mathcal{U}_r(c)$ and can be represented as

$$F(z) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{\infty} c_{jk} \right) (z-c)^k. \quad (4.7)$$

Example 4.8. The following $\mathcal{C}^\infty$ function due to Cauchy

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}), & x \neq 0; \\ 0, & x = 0 \end{cases} \quad (4.8)$$

satisfies $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$, but it does not have a Taylor expansion at any neighborhood at 0.

4.1.1 Fundamental equivalence on open sets

Theorem 4.9 (Equivalence of analytic functions on open sets). The following assertions are equivalent for a function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open.

- (A) f is analytic at any point $z \in \mathcal{D}$.
- (B) f is totally differentiable in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$) and satisfies the Cauchy–Riemann equations (2.34).
- (C) f is continuous and for each triangle path $\langle z_1, z_2, z_3 \rangle$ whose convex hull is contained in $\mathcal{D}$ we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0.$$

- (D) f locally admits primitives, i.e., for each point $c \in \mathcal{D}$ there exists an open disk $\mathcal{U}(c) \subset \mathcal{D}$ such that $f|_{\mathcal{U}(c)}$ has a primitive.
- (E) f is continuous, and for each disk $\mathcal{U}_\rho(c)$ satisfying $\overline{\mathcal{U}_\rho(c)} \subset \mathcal{D}$ we have

$$\forall z \in \mathcal{U}_\rho(c), \quad f(z) = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta.$$

- (F) f is locally representable as a convergent power series.
- (G) f is representable as a power series in each open disk fully contained in $\mathcal{D}$.

Proof. We list the main points of the proof as follows.

- $(A) \Leftrightarrow (B)$: Cauchy–Riemann theorem 2.56;
- $(A) \Rightarrow (C)$: Cauchy theorem 3.55;
- $(C) \Rightarrow (A)$: Morera’s theorem 3.92;
- $(C) \Rightarrow (D)$: Theorem 3.42 and Lemma 3.58;
- $(D) \Rightarrow (C)$: Theorem 3.42;

- $(A) \Rightarrow (E)$: Cauchy integral formula 3.75;
- $(E) \Rightarrow (F)$: Taylor theorem 4.1;
- $(F) \Rightarrow (A)$: Corollary 3.105;
- $(A) \Rightarrow (G)$: Taylor theorem 4.1;
- $(G) \Rightarrow (F)$: A universal statement implies a existential statement. $\square$

Exercise 4.10. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_r(0)$. Show that if f is nonconstant and $\sum_{n=2}^{\infty} n|a_n|r^{n-1} \leq |a_1|$, then f is injective.

Exercise 4.11. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_R(0)$. Define $A(r) = \max_{|z|=r} \operatorname{Re} f(z)$, Show that for each $r \in (0, R)$ and $n \in \mathbb{N}^+$,

- (a) $a_n r^n = \frac{1}{\pi} \int_0^{2\pi} [\operatorname{Re} f(re^{it})] e^{-int} dt$;
- (b) $|a_n| r^n \leq 2A(r) - 2\operatorname{Re} f(0)$.

4.1.2 Elementary domains

Definition 4.12. A domain $\mathcal{D} \subset \mathbb{C}$ is called an *elementary domain* iff any analytic function defined on $\mathcal{D}$ has a global primitive in $\mathcal{D}$.

Lemma 4.13. Let $\mathcal{D}$ and Ω be two elementary domains. If $\mathcal{D} \cap \Omega$ is non-empty and connected, then $\mathcal{D} \cup \Omega$ is also an elementary domain.

Proof. Let $f : \mathcal{D} \cup \Omega \rightarrow \mathbb{C}$ be analytic. We need to show that f has a primitive. Since $\mathcal{D}$ and Ω are both elementary domains, f has a primitive $F_1 : \mathcal{D} \rightarrow \mathbb{C}$ on $\mathcal{D}$ and a primitive $F_2 : \Omega \rightarrow \mathbb{C}$ on Ω . The difference $F_1 - F_2$ must be locally constant in $\mathcal{D} \cap \Omega$, and therefore constant since $\mathcal{D} \cap \Omega$ is connected. By addition of a constant if necessary, we can assume WLOG that $F_1|_{\mathcal{D} \cap \Omega} = F_2|_{\mathcal{D} \cap \Omega}$. Hence the functions F_1 and F_2 now glue to a single function $F : \mathcal{D} \cup \Omega \rightarrow \mathbb{C}$. $\square$

Corollary 4.14. Slit annuli are elementary domains.

Proof. By Example 3.63, the ring sector is a star-shaped domain in the sense of Definition 3.62 and thus an elementary domain. By Lemma 4.13, the union of two ring sectors are also an elementary domain if their intersection is non-empty and connected. Hence a ring sector can be extended to a slit annulus while retaining the property of elementary domains. $\square$

Corollary 4.15. Let $\mathcal{D}_1 \subset \mathcal{D}_2 \subset \dots$ be an increasing sequence of elementary domains. Then their union $\cup_{n=1}^{\infty} \mathcal{D}_n$ is also an elementary domain.

Proof. This follows directly from Lemma 4.13. $\square$

Lemma 4.16. Suppose $\mathcal{D} \subset \mathbb{C}$ is an elementary domain and $\varphi : \mathcal{D} \rightarrow \mathcal{D}^*$ a globally conformal map in the sense of Definition 2.38. Then $\mathcal{D}^*$ is also an elementary domain.

Proof. It suffices to show that any analytic function $f_* : \mathcal{D}^* \rightarrow \mathbb{C}$ has a primitive F_* satisfying $F'_* = f_*$. The condition of φ being globally conformal, Definition 2.38, Theorem 2.39, Theorem 2.74, and Theorem 3.84 imply that φ , φ^{-1} , and φ' are all analytic. Hence $(f_* \circ \varphi)\varphi' : \mathcal{D} \rightarrow \mathbb{C}$

is analytic. Since $\mathcal{D}$ is an elementary domain, there exists $F : \mathcal{D} \rightarrow \mathbb{C}$ such that $F' = (f_* \circ \varphi)\varphi'$. Hence we can construct $F_* : \mathcal{D}^* \rightarrow \mathbb{C}$ by $F_* = F \circ \varphi^{-1}$ such that

$$(F \circ \varphi^{-1})' = (F' \circ \varphi^{-1})\frac{1}{\varphi'} = f_*. \quad \square$$

Exercise 4.17. Show that the sickle-shaped domain

$$\mathcal{D} = \mathcal{U}_1(0) \setminus \overline{\mathcal{U}_{\frac{1}{2}}\left(\frac{1}{2}\right)} = \left\{z \in \mathbb{C} : |z| < 1, \left|z - \frac{1}{2}\right| > \frac{1}{2}\right\}$$

is an elementary domain.

Hint: Consider the conformal map $z \mapsto \frac{1}{1-z}$. What is the image of $\mathcal{D}$ under this map?

4.2 Zeros or roots

Definition 4.18. A point $a \in \Omega$ is called a *zero* or a *root* of an analytic function $f : \Omega \rightarrow \mathbb{C}$ iff $f(a) = 0$. A zero point a of f is of *multiplicity* m iff $\exists m \in \mathbb{N}^+$ such that

$$f^{(m)}(a) \neq 0 \text{ and } \forall j = 0, 1, \dots, m-1, f^{(j)}(a) = 0.$$

In particular, a zero of multiplicity 1 is a *simple zero* or a *simple root*.

Example 4.19. The zeros of $f(z) = \sin z - 1$ form the set $\{\frac{\pi}{2} + 2k\pi : k \in \mathbb{Z}\}$. Each zero is of multiplicity 2 and thus a critical point of f .

Lemma 4.20. Let Ω be a domain and $a \in \Omega$ a zero point of an analytic function $f : \Omega \rightarrow \mathbb{C}$. Then the multiplicity of f at a is m if and only if $f(z) = (z-a)^m \varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$.

Proof. This follows directly from Theorem 4.1. $\square$

Exercise 4.21. For any analytic function $f : \mathcal{U}_R(0) \rightarrow \mathbb{C}$ with $R > 0$, show $f \equiv 0$ if there exist constants $\rho \in (0, 1)$ and $C > 0$ such that $|f(\frac{1}{n})| \leq C\rho^n$ holds for sufficiently large $n \in \mathbb{N}$.

4.2.1 Roots of a nonzero analytic function are discrete

Definition 4.22. A subset $\mathcal{M} \subseteq \mathcal{X}$ in a metric space $(\mathcal{X}, d)$ is *discrete* iff

$$\forall x \in \mathcal{X}, \exists \epsilon > 0 \text{ s.t. } \forall y \in \mathcal{M} \setminus \{x\}, d(x, y) > \epsilon. \quad (4.9)$$

Such a set consists of *isolated points*.

Lemma 4.23. A subset $\mathcal{M} \subseteq \mathcal{X}$ is *discrete* in $\mathcal{X}$ if and only if there is no limit point of $\mathcal{M}$ in $\mathcal{X}$.

Proof. By Definition 2.16, the existence of a limit point x of $\mathcal{M}$ in $\mathcal{X}$ implies

$$\exists x \in \mathcal{X} \text{ s.t. } \forall \epsilon > 0, \exists y \in \mathcal{M} \text{ s.t. } d(x, y) \in (0, \epsilon).$$

The negation of the above statement yields (4.9). $\square$

Example 4.24. The set $\mathcal{M} := \{\frac{1}{n} : n \in \mathbb{N}^+\}$ is not discrete in $\Omega = \mathbb{C}$, but is indeed discrete in $\Omega = \mathbb{C} \setminus \{0\}$.

Example 4.25. The zeros of $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0, \end{cases} \quad (4.10)$$

are $0, \pm \frac{1}{n\pi}$ where $n \in \mathbb{N}^+$ and 0 is a limit point of the zeros.

Example 4.26. Combine (4.8) and (4.10) and we have a C^∞ function whose zeros have a limit point in $\mathbb{R}$,

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}) \sin \frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0. \end{cases} \quad (4.11)$$

Theorem 4.27. The set of zeros of a nonzero analytic function $f : \Omega \rightarrow \mathbb{C}$ with a domain $\Omega \subset \mathbb{C}$ is discrete in Ω .

Proof. Let a be a zero of f with multiplicity m . Lemma 4.20 yields $f(z) = (z-a)^m \varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$. The continuity of φ gives

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(a), \varphi(z) \neq 0.$$

Then the proof is completed by Definition 4.22. $\square$

4.2.2 Local mapping behaviors

Theorem 4.28. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function on an elementary domain, and $f(z) \neq 0$ for all $z \in \mathcal{D}$. There exists an analytic function $h : \mathcal{D} \rightarrow \mathbb{C}$ such that

$$f(z) = \exp(h(z)), \quad (4.12)$$

where the function h is called an *analytic branch of the logarithm* of f .

Proof. By Theorem 3.84, f being analytic implies that f' is also analytic. Hence $\frac{f'}{f}$ is also analytic and by Definition 4.12 it has a primitive, say F . Then the function $G(z) := \frac{\exp(F(z))}{f(z)}$ satisfies

$$\forall z \in \mathcal{D}, G'(z) = \frac{1}{f^2} \left[\frac{f'}{f} \exp(F) f - \exp(F) f' \right] = 0.$$

Hence for all $z \in \mathcal{D}$ we have $\exp(F(z)) = C f(z)$ with $C \neq 0$. Set $C = \exp(c)$ and the following function satisfies (4.12):

$$h(z) = F(z) - c. \quad \square$$

Corollary 4.29. Under the assumption of Theorem 4.28, there exists for any $n \in \mathbb{N}$ an analytic function $H : \mathcal{D} \rightarrow \mathbb{C}$ such that $H^n = f$.

Proof. Set $H(z) = \exp(\frac{1}{n} h(z))$ and use Theorem 4.28. $\square$

Theorem 4.30 (Local behavior of an analytic function at a root). For a nonconstant analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ where $\mathcal{D}$ is a domain, $0 \in \mathcal{D}$, and $f(0) = 0$, there exists $r > 0$ and a globally conformal map $g : \mathcal{U}_r(0) \rightarrow \mathbb{C}$ such that

$$\forall z \in \mathcal{U}_r(0), f(z) = [g(z)]^m \quad (4.13)$$

where m is the multiplicity of the root of f at 0.

Proof. Set $a = 0$ in Lemma 4.20 and we have

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(0), \quad f(z) = z^m \varphi(z), \quad (4.14)$$

where φ is analytic in $\mathcal{U}_R(0)$ and $\varphi(0) \neq 0$. The continuity of φ implies the existence of $\bar{r} \in (0, R]$ such that $\varphi(z) \neq 0$ for all $z \in \mathcal{U}_{\bar{r}}(0)$. By Corollary 4.29, there exists an analytic function h such that $h^m = \varphi$. Then (4.13) follows from

$$g(z) := zh(z).$$

Clearly, $g : \mathcal{U}_{\bar{r}}(0) \rightarrow \mathbb{C}$ is analytic. For g to be globally conformal in the sense of Definition 2.38, Theorem 2.39 implies that it suffices to show $g'(0) \neq 0$, because then Theorem 4.27 and the inverse function theorem would imply the existence of $\tilde{r} > 0$ such that g is a homeomorphism in $\mathcal{U}_{\tilde{r}}(0)$ and then we can set $r = \min(\bar{r}, \tilde{r})$. Theorem 4.1 gives $\varphi(z) = \sum_{j=0}^{\infty} a_{m+j} z^j$ and thus

$$f(z) = z^m \sum_{j=0}^{\infty} a_{m+j} z^j, \quad (4.15)$$

where $a_m \neq 0$ follows from Definition 4.18. On the one hand, (4.15) yields $f^{(m)}(0) = m!a_m$. On the other hand, (4.13) gives $f^{(m)}(0) = m!(g'(0))^m$. Then $0 \neq a_m = (g'(0))^m$ implies $g'(0) \neq 0$. $\square$

Exercise 4.31. Let l_1, l_2 be two rays that both start from the origin and have an angle $\theta \in (0, \frac{2\pi}{m})$. Find the angle between the image curves $f(l_1)$ and $f(l_2)$ at 0 by assuming the conditions in Theorem 4.30.

Corollary 4.32. A $\mathcal{C}^1$ function $f : \mathcal{D} \rightarrow \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{R}^2$ open is globally conformal if and only if f is bijective and analytic when $\mathcal{D}, \Omega$ are interpreted as open subsets of $\mathbb{C}$.

Proof. The necessity follows directly from Definition 2.38 and Theorem 2.39. For the sufficiency, Theorem 4.30 implies that the algebraic multiplicity m in (4.13) be 1; otherwise f would not be injective, letting alone being bijective. Since g is globally conformal, $g'(0) \neq 0$, which gives $f'(0) \neq 0$. $\square$

Corollary 4.33. If $\varphi : \mathcal{D} \rightarrow \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{C}$ open is biholomorphic, i.e., bijective and analytic, then so is its inverse function $\varphi^{-1} : \Omega \rightarrow \mathcal{D}$.

Proof. The follows from Corollaries 2.75 and 4.32. $\square$

Exercise 4.34. For a nonconstant analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ where $\mathcal{D}$ is a domain, $0 \in \mathcal{D}$, $f(0) = f'(0) = 0$, and $f''(0) \neq 0$, construct two curves Γ_1 and Γ_2 so that

- (a) both Γ_1 and Γ_2 pass through 0,
- (b) Γ_1 and Γ_2 are orthogonal at 0,
- (c) $f(\Gamma_1)$ is a subset of $\mathbb{R}$ and has a minimum at 0,
- (d) $f(\Gamma_2)$ is a subset of $\mathbb{R}$ and has a maximum at 0.

4.2.3 The identity theorem

Theorem 4.35 (Identity). For two analytic functions $f, g : \mathcal{D} \rightarrow \mathbb{C}$ defined on a nonempty domain $\mathcal{D}$, the following statements are equivalent:

- (a) $f = g$.
- (b) The coincidence set $\{z \in \mathcal{D} : f(z) = g(z)\}$ has a limit point in $\mathcal{D}$.
- (c) $\exists a \in \mathcal{D}$ s.t. $\forall n \in \mathbb{N}, f^{(n)}(a) = g^{(n)}(a)$.

Proof. The equivalence of (a) and (b) follows from applying Theorem 4.27 to $f - g$ and that of (a) and (c) from applying Theorems 4.27 and 4.1. $\square$

Lemma 4.36. If the product of two analytic functions on a domain $\mathcal{D}$ vanishes identically, then one of the two functions vanishes identically on $\mathcal{D}$.

Proof. Suppose $f(z)g(z) = 0$ for all $z \in \mathcal{D}$. We show $f \neq 0$ implies $g = 0$. $f \neq 0$ gives $\exists a \in \mathcal{D}$ s.t. $f(a) \neq 0$. Then the continuity of f yields

$$\exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(a), \quad f(z) \neq 0.$$

Hence $fg = 0$ and Theorem 4.27 imply $g(z) = 0$ for all $z \in \mathcal{U}_r(a)$. The proof is completed by Theorem 4.35. $\square$

Exercise 4.37. Determine whether there exist analytic functions $f, g : \mathcal{U}_R(0) \rightarrow \mathbb{C}$ satisfying

- (a) $f(\frac{1}{n}) = f(-\frac{1}{n}) = \frac{1}{n^2}, n \geq 1$;
- (b) $g(\frac{1}{2n-1}) = g(\frac{1}{2n}) = \frac{1}{2^n}, n \geq 1$.

Exercise 4.38. Suppose an analytic function f on a domain $\mathcal{D}$ satisfies

$$\forall z \in \mathcal{D}, \quad \exists n \in \mathbb{N} \text{ s.t. } f^{(n)}(z) = 0.$$

Show that f is a polynomial.

Hint: Write $\mathcal{D} = \cup_{n=0}^{\infty} g_n^{-1}(\{0\})$, where $g_n(z) = f^{(n)}(z)$.

4.2.4 The open mapping theorem

Definition 4.39. A function is *open* iff it maps open sets to open sets.

Example 4.40. The function $f(x) = \sin x$ is $\mathcal{C}^\infty$ and $\mathbb{R}$ is open, but $\sin(\mathbb{R}) = [-1, 1]$ is not open.

Theorem 4.41 (Open mapping). Any nonconstant analytic function on a domain is open.

Proof. The function $p(z) = z^n$ maps an open disk of radius r onto to an open disk of radius r^n : $re^{i\varphi} \mapsto r^n e^{in\varphi}$. At each point z of an open set Ω , there exists an open disk $\mathcal{U}_r(z)$ at z such that $\mathcal{U}_r(z) \subset \Omega$. Then Ω can be considered as the union of open disks and Definition 4.39 implies that p is open. The rest of the proof follows from Theorem 4.30 and the fact that the function g is a globally conformal map and thus a homeomorphism. $\square$

Corollary 4.42. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant function, then f is a constant function.

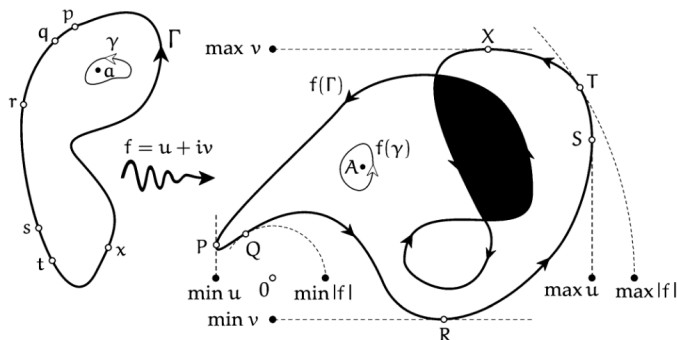
Proof. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant, $f(\mathcal{D})$ cannot be an open set in $\mathbb{C}$, and thus Definition 4.39 and $\mathcal{D}$ being open imply that f is not an open map. The rest of the proof follows directly from the contraposition of Theorem 4.41. $\square$

Exercise 4.43. Show that if $f : \mathcal{D} \rightarrow \mathcal{D}$ is analytic on a domain $\mathcal{D}$ and satisfies $f \circ f = f$, then f is either constant or the identity map.

4.2.5 Principles of maximum and minimum moduli

Theorem 4.44 (Maximum modulus). Suppose the modulus of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ on a domain $\mathcal{D}$ reaches its maximum on $\mathcal{D}$. Then f is a constant.

Proof. Suppose $|f|$ reaches its maximum, say, at $a \in \mathcal{D}$. Then $f(\mathcal{U})$ is not open for any open ball $\mathcal{U}$ centered at a . Hence f is not open and the proof is then completed by the contraposition of Theorem 4.41. $\square$



Corollary 4.45. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function. For a compact set $K \subset \mathcal{D}$, the restriction $f|_K$ has a maximal modulus on K . Furthermore, the maximal modulus value occurs on the boundary of K .

Proof. This follows directly from Theorem 4.44. $\square$

Exercise 4.46. Let $w_1, \dots, w_n$ be points on the unit circle. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j is at least 1, i.e., $\prod_{j=1}^n |z - w_j| \geq 1$. Conclude that there exists a point w on the unit circle such that $\prod_{j=1}^n |w - w_j| = 1$.

Exercise 4.47. Show that if the real part of an entire function f is bounded, then f is constant.

Theorem 4.48 (Minimum modulus). For a nonconstant analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ on a domain $\mathcal{D}$, a (local) minimum modulus of f at $a \in \mathcal{D}$ implies that a is a root of f , i.e., $f(a) = 0$.

Proof. Suppose $f(a) \neq 0$. Then $|f(a)| > 0$. Since $|f|$ reaches its local minimum at a , $f(\mathcal{U})$ is not open for any open ball $\mathcal{U}$ centered at a . Hence $f(\mathcal{D})$ is not open. But the contraposition of Theorem 4.41 implies that f is constant, which contradicts the given condition. $\square$

Exercise 4.49. Prove the fundamental theorem of algebra by Theorem 4.48.

Exercise 4.50. Let f be nonconstant and analytic in an open set containing $\mathbb{E}$. Show that if $|f(z)| = 1$ for all $|z| = 1$, then $\mathbb{E} \subseteq f(\mathbb{E})$.

Hint: One must show that $f(z) = w_0$ has a root for every $w_0 \in \mathbb{E}$. Then it suffices to consider the case of $w_0 = 0$. Why?

Exercise 4.51. Show that there is no analytic function f in $\mathbb{E}$ that extends continuously to $\partial\mathbb{E}$ and satisfies $f(z) = \frac{1}{z}$ for all $z \in \partial\mathbb{E}$.

4.2.6 The Schwarz lemma

Lemma 4.52 (Schwarz 1869). An analytic function $f : \mathbb{E} \rightarrow \mathbb{E}$ with $f(0) = 0$ must satisfy $|f'(0)| \leq 1$ and

$$\forall z \in \mathbb{E}, \quad |f(z)| \leq |z|. \quad (4.16)$$

Furthermore, if any of the two equalities holds, i.e., if $|f'(0)| = 1$ or $|f(a)| = |a|$ for some $a \neq 0$, then f must be a rotation, i.e., $f(z) = \zeta z$ for some $\zeta \in \mathbb{C}$ with $|\zeta| = 1$.

Proof. By Theorem 4.1, we can write $f = \sum_{j=0}^{\infty} a_j z^j$. Then $f(0) = 0$ implies $a_0 = 0$. The function $g : \mathbb{E} \rightarrow \mathbb{C}$ given by

$$g(z) := \sum_{j=1}^{\infty} a_j z^{j-1}$$

satisfies $f(z) = zg(z)$ and $f'(0) = g(0)$ and is analytic. Then the condition $f(z) \in \mathbb{E}$ implies $|f(z)| \leq 1$, which gives $|g(z)| \leq \frac{1}{|z|}$. Furthermore, Corollary 4.45 gives

$$\forall r \in (0, 1), \quad \forall z \in \overline{\mathcal{U}_r(0)}, \quad |g(z)| \leq \frac{1}{r}.$$

Pass to the limit $r \rightarrow 1$ and we have $|g(z)| \leq 1$, which implies $|f'(0)| \leq 1$ and (4.16).

Suppose $|f'(0)| = 1$ or $|f(a)| = |a|$ for some $z = a \neq 0$. Then $|g(0)| = 1$ or $|g(a)| = 1$ and thus $g(z)$ attains its maximum modulus on $\mathbb{E}$. By Theorem 4.44, $g(z)$ is a constant on $\mathbb{E}$: $g(z) \equiv \zeta$, which completes the proof. $\square$

Exercise 4.53 (Alternative proof of Liouville's theorem). Prove Theorem 3.94 by Lemma 4.52.

Exercise 4.54. For biholomorphic functions $f, g : \mathbb{E} \rightarrow \mathbb{E}$ with $f(0) = g(0)$ and $f'(0) = g'(0)$, show that $f = g$.

Exercise 4.55 (Study 1913). For a biholomorphic function $f : \mathbb{E} \rightarrow \Omega$, show that if Ω is convex, then for any $r \in (0, 1)$, the image $\Omega_r = f(\mathcal{U}_r(0))$ is also convex.

Hint: For two points $|a| \leq |b| < r$ and $t \in [0, 1]$, consider the map $g_t(z) = (1-t)f(\frac{az}{b}) + tf(z)$.

4.2.7 $\text{Aut}(\mathbb{E})$: the automorphism group of $\mathbb{E}$

Definition 4.56. The group of automorphisms of a domain $\mathcal{D} \subseteq \mathbb{C}$, denoted by $\text{Aut}(\mathcal{D})$, is the group of all globally conformal maps (or equivalently all biholomorphic functions) $\mathcal{D} \rightarrow \mathcal{D}$ with function composition.

Corollary 4.57. A biholomorphic function $\varphi_0 : \mathbb{E} \rightarrow \mathbb{E}$ with $\varphi_0(0) = 0$ must be a pure rotation, i.e.,

$$\exists \zeta \in \mathbb{C} \text{ with } |\zeta| = 1 \text{ s.t. } \varphi_0(z) = \zeta z. \quad (4.17)$$

Proof. By Corollary 4.33, the inverse φ_0^{-1} exists and is also analytic. Applying Lemma 4.52 to both φ_0 and φ_0^{-1} gives

$$|\varphi_0(z)| \leq |z| = |\varphi_0^{-1}(\varphi_0(z))| \leq |\varphi_0(z)|.$$

The rest follows from the last sentence of Lemma 4.52. $\square$

Lemma 4.58. For any fixed $a \in \mathbb{E}$, the map $\varphi_a : \mathbb{E} \rightarrow \mathbb{E}$ given by

$$\varphi_a(z) := \frac{z - a}{\bar{a}z - 1} \quad (4.18)$$

is a bijective map of the unit disk onto itself which satisfies

- (a) $\varphi_a(a) = 0$,
- (b) $\varphi_a(0) = a$,
- (c) $\varphi_a^{-1} = \varphi_a$.

In particular, φ_a is in both directions analytic.

Proof. First, φ_a in (4.18) is well defined as the denominator has no zeros in $\mathbb{E}$. Second, φ_a is injective because

$$z_1 \neq z_2 \implies \varphi_a(z_1) - \varphi_a(z_2) = \frac{(z_2 - z_1)(1 - |a|^2)}{(\bar{a}z_1 - 1)(\bar{a}z_2 - 1)} \neq 0.$$

Third, φ_a is surjective because

$$|\varphi_a(z)| < 1 \iff |z - a|^2 < |\bar{a}z - 1|^2 \iff (1 - |a|^2)(1 - |z|^2) > 0.$$

Finally, (a), (b), and (c) follow directly from (4.18). $\square$

Exercise 4.59. Consider an analytic function $f : \mathbb{E} \rightarrow \mathbb{E}$ that is not the identity function.

- (a) Show that f has at most one fixed point.
- (b) Give an example of f having no fixed points.

Exercise 4.60. For an analytic function $f : \mathbb{E} \rightarrow \mathbb{E}$, show that if there exists a point $a \in \mathbb{E} \setminus \{0\}$ such that $f(0) = a$ and $f(a) = 0$, then $f = \varphi_a$, where φ_a is as defined in (4.18).

Theorem 4.61. Any $f \in \text{Aut}(\mathbb{E})$ is of the form

$$f(z) = e^{i\theta} \frac{z - a}{\bar{a}z - 1}, \quad (4.19)$$

where $\theta \in [0, 2\pi)$ and $a \in \mathbb{E}$ depend only on f .

Proof. Since f is bijective, it maps one and only one point $a \in \mathbb{E}$ to 0, i.e., $a = f^{-1}(0)$. By Lemma 4.58(b), the map $g := f \circ \varphi_a$ is biholomorphic and satisfies $g(0) = 0$. The rest follows from Corollary 4.57 and Lemma 4.58(c). $\square$

Theorem 4.62 (Schwarz–Pick). Any nonconstant analytic function $f : \mathbb{E} \rightarrow \mathbb{E}$ satisfies

$$\forall z \in \mathbb{E}, \quad \frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}, \quad (4.20)$$

where the equality holds for some $z \in \mathbb{E}$ if and only if $f \in \text{Aut}(\mathbb{E})$.

Exercise 4.63. Prove Theorem 4.62.

4.2.8 Brouwer's fixed point theorem

Definition 4.64. A *fixed point* of a map $f : X \rightarrow Y$ is a point $p \in X$ that satisfies $f(p) = p$.

Definition 4.65. A subspace X of a topological space Y is a *retract* of Y if there is a continuous function $r : Y \rightarrow X$ with $r(x) = x$ for all $x \in X$; such a function is called a *retraction*.

Definition 4.66. The *inclusion* $i_A : A \hookrightarrow X$ of a topological space X and a subset A of X is

$$\forall x \in A, \quad i_A(x) = x. \quad (4.21)$$

Lemma 4.67. A continuous map $r : Y \rightarrow X$ is a retraction if and only if $r \circ i = 1_X$ where $i : X \hookrightarrow Y$ is an inclusion.

Proof. For the sufficiency, $r \circ i = 1_X$ implies

$$\forall x \in X, \quad r(x) = r \circ i(x) = 1_X(x) = x.$$

Hence, r is a retraction. As for the necessity, $r \circ i$ maps X to X and $\forall x \in X$, $r \circ i(x) = r(x) = x$, hence $r \circ i = 1_X$. $\square$

Lemma 4.68. For $n \geq 0$, the n -sphere $\mathbb{S}^n$ cannot be a retract of the $(n + 1)$ -ball $\mathbb{D}^{n+1}$.

Proof. See [Rotman, 1988, p. 3]. $\square$

Theorem 4.69 (Brouwer's fixed point). A continuous function $f : \mathbb{D}^n \rightarrow \mathbb{D}^n$ must have a fixed point $\mathbf{x}^* = f(\mathbf{x}^*)$.

Proof. Suppose that there exists $f : \mathbb{D}^n \rightarrow \mathbb{D}^n$ such that $f(\mathbf{x}) \neq \mathbf{x}$ for all $\mathbf{x} \in \mathbb{D}^n$. Then we can determine a ray from $f(\mathbf{x})$ to $\mathbf{x}$, which intersects $\mathbb{S}^{n-1}$ at $\mathbf{y}$. Define $g : \mathbb{D}^n \rightarrow \mathbb{S}^{n-1}$ by $g(\mathbf{x}) = \mathbf{y}$ and this construction implies that g is a retraction, but this contradicts Lemma 4.68. $\square$

Exercise 4.70. The ray in the proof of Theorem 4.69 starts from $f(\mathbf{x})$. Can we start the ray from $\mathbf{x}$ and still get g as a retraction? Why?

Example 4.71. Let $f : \mathbb{D}^n \rightarrow \mathbb{R}^n$ be a homeomorphism between $\mathbb{D}^n$ and $\mathbb{R}^n$. For any $\mathbf{v} \neq \mathbf{0}$, the translation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $T(\mathbf{y}) = \mathbf{y} + \mathbf{v}$ has no fixed points, neither does the map $f^{-1} \circ T \circ f : \mathbb{D}^n \rightarrow \mathbb{D}^n$. For $n = 2$, a homeomorphism between $\mathbb{E}$ and $\mathbb{C}$ is $f(z) = \frac{z}{1 - |z|}$ since its inverse $w \mapsto \frac{w}{1 + |w|}$ is also continuous.

4.3 Analytic continuation

Definition 4.72. Let $\mathcal{G}$ and $\mathcal{D}$ be two subsets of $\mathbb{C}$. An analytic function $F : \mathcal{G} \rightarrow \mathbb{C}$ is said to be an *analytic continuation* of another analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ iff $\mathcal{G} \supset \mathcal{D}$ and $F(z) = f(z)$ for all $z \in \mathcal{D}$.

Example 4.73. Corollary 3.66 is an example of analytic continuation for f from $\mathcal{D} \setminus \{z_*\}$ to $\mathcal{D}$.

Lemma 4.74. Let $\mathcal{D} \subseteq \mathbb{C}$ be a domain and $L \subset \mathbb{C}$ be a straight line. If a continuous function $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic at all points $z \in \mathcal{D} \setminus L$, then f is analytic on $\mathcal{D}$.

Proof. We use Morera's theorem 3.92 to prove this. WLOG, we can assume that L is the real axis $\mathbb{R}$; otherwise we consider the function $f \circ g$ where $g(z) = e^{i\theta}z + c$ is a Euclidean isometry for any fixed θ and c given *a priori*. There are four possibilities with respect to the intersection of $\mathbb{R}$ to an arbitrary triangle $\Delta(z_1, z_2, z_3)$ in Definition 3.50.

- (a) No intersections: (3.48) follows from the Cauchy integral theorem 3.55.

- (b) A single intersection: (3.48) follows from Corollary 3.66 and Theorem 4.9.
- (c) An edge, say $\overline{z_1 z_2}$, as a subset of $\mathbb{R}$: (3.48) follows from setting $\overline{w_1 w_2} := \overline{z_1 z_2} + i\epsilon$ for $\epsilon \in \mathbb{R}^+$ and arguing that

$$\begin{aligned} & \lim_{w_1 w_2 \rightarrow \overline{z_1 z_2}} \left(\int_{w_1 w_2} f(z) dz - \int_{\overline{z_1 z_2}} f(z) dz \right) \\ &= \lim_{\epsilon \rightarrow 0} \int_{\overline{z_1 z_2}} (f(z + i\epsilon) - f(z)) dz \\ &= \int_{\overline{z_1 z_2}} \lim_{\epsilon \rightarrow 0} (f(z + i\epsilon) - f(z)) dz = 0, \end{aligned}$$

where the second step follows from Lemma 3.100 and the last from Heine's theorem 3.57 and Lemma 3.13.

- (d) All other cases: we decompose $\Delta(z_1, z_2, z_3)$ into a finite number of triangles and apply (b) and (c). $\square$

Theorem 4.75 (Schwarz reflection principle 1867). Let $\mathcal{D}$ be a domain symmetric with respect to the real axis. Define three subsets of $\mathcal{D}$ as

$$\mathcal{D}_+ := \{z \in \mathcal{D} : \operatorname{Im} z > 0\}, \quad (4.22a)$$

$$\mathcal{D}_- := \{z \in \mathcal{D} : \operatorname{Im} z < 0\}, \quad (4.22b)$$

$$\mathcal{D}_0 := \mathcal{D} \cap \mathbb{R}. \quad (4.22c)$$

If f is continuous on $\mathcal{D}_+ \cup \mathcal{D}_0$, f is analytic on $\mathcal{D}_+$, and $f(\mathcal{D}_0) \subset \mathbb{R}$, then f can be extended to an analytic function $F : \mathcal{D} \rightarrow \mathbb{C}$ given by

$$F(z) := \begin{cases} f(z) & \text{if } z \in \mathcal{D}_+ \cup \mathcal{D}_0, \\ \overline{f(\overline{z})} & \text{if } z \in \mathcal{D}_-. \end{cases} \quad (4.23)$$

Proof. By $f(z)$ being analytic on $\mathcal{D}_+$, we have, $\forall z \in \mathcal{D}_-$,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{z+h-z} &= \lim_{h \rightarrow 0} \frac{\overline{f(\overline{z+h})} - \overline{f(\overline{z})}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\overline{f(\overline{z} + \overline{h})} - \overline{f(\overline{z})}}{h} = \lim_{h \rightarrow 0} \frac{\overline{f(\overline{z} + \overline{h})} - \overline{f(\overline{z})}}{\overline{h}} = \overline{f'(\overline{z})}, \end{aligned}$$

where the last step follows from $\overline{h} \rightarrow 0 \Leftrightarrow h \rightarrow 0$ for $h \in \mathbb{C}$ and the third equality follows from $\overline{\left(\frac{u}{v}\right)} = \frac{\overline{u}}{\overline{v}}$, which can be easily shown by Corollary 1.62. Then the proof is completed by Lemma 4.74. $\square$

Corollary 4.76. If $\mathcal{D} \subseteq \mathbb{C}$ is a domain with $\mathcal{D} \cap \mathbb{R} \neq \emptyset$ and if $f, g : \mathcal{D} \rightarrow \mathbb{C}$ are analytic functions with $f|_{\mathcal{D} \cap \mathbb{R}} = g|_{\mathcal{D} \cap \mathbb{R}}$, then we have $f(z) = g(z)$ for all $z \in \mathcal{D}$.

Proof. This follows directly from the identity theorem 4.35. $\square$

Corollary 4.77. Let $r > 0$ and $\mathcal{D}$ be a domain symmetric with respect to the circle $\partial\mathcal{U}_r(0) = \{z \in \mathbb{C} : |z| = r\}$, that is, $z \in \mathcal{D}$ if and only if $\frac{r^2}{\overline{z}} \in \mathcal{D}$. Define three subsets of $\mathcal{D}$ as

$$\mathcal{D}_+ := \mathcal{D} \cap \mathcal{U}_r(0), \quad (4.24a)$$

$$\mathcal{D}_- := \{z \in \mathcal{D} : |z| > r\}, \quad (4.24b)$$

$$\mathcal{D}_0 := \mathcal{D} \cap \partial\mathcal{U}_r(0). \quad (4.24c)$$

If f is continuous on $\mathcal{D}_+ \cup \mathcal{D}_0$, f is analytic on $\mathcal{D}_+$, and there exists $\rho > 0$ such that for all $z \in \mathcal{D}_0$, $|f(z)| = \rho$, then f can be extended to an analytic function $F : \mathcal{D} \rightarrow \mathbb{C}$ given by

$$F(z) := \begin{cases} f(z) & \text{if } z \in \mathcal{D}_+ \cup \mathcal{D}_0, \\ \rho^2 / \overline{f(\frac{r^2}{\overline{z}})} & \text{if } z \in \mathcal{D}_-. \end{cases} \quad (4.25)$$

Exercise 4.78. Prove Corollary 4.77 by similar arguments as in the proof of Theorem 4.75.

Exercise 4.79. Show that an analytic function $f : \mathbb{E} \rightarrow \mathbb{C}$ that is continuous on $\overline{\mathbb{E}}$ must be a constant if

$$\begin{cases} f(z) \neq 0 & \text{if } z \in \mathbb{E}, \\ |f(z)| = 1 & \text{if } z \in \partial\mathbb{E}. \end{cases}$$

Definition 4.80. A power series in two variables centered at a point (x_0, y_0) is a power series of the form

$$f(x, y) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_{n,m} (x - x_0)^n (y - y_0)^m. \quad (4.26)$$

Definition 4.81. A function $f : \mathcal{D} \rightarrow \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ open is *real analytic* iff for each $(x_0, y_0) \in \mathcal{D}$ there exists $r > 0$ such that for any $(x, y) \in \mathcal{U}_r(x_0, y_0)$ there exists a power series in two variables that equals $f(x, y)$, i.e., (4.26) holds.

Exercise 4.82. Show that the real part and the imaginary part of an analytic function is real analytic in the sense of Definition 4.81.

Lemma 4.83. Let $\mathcal{I} \subset \mathbb{R}$ be a non-degenerate interval, i.e., an interval that contains more than one point. A function $f : \mathcal{I} \rightarrow \mathbb{R}$ has an analytic continuation $F : \mathcal{D} \rightarrow \mathbb{C}$ on a domain $\mathcal{D} \subset \mathbb{C}$ if and only if f is real analytic in the sense of Definition B.102.

Proof. For necessity, F being analytic, Theorem 4.1, and $\mathcal{I} \subset \mathcal{D}$ yield

$$\forall c \in \mathcal{I}, \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c) \subset \mathcal{D}, F(z) = \sum_{n=0}^{\infty} a_n (z - c)^n.$$

Then Definition 4.72 gives $F|_{\mathcal{I}} = f$, i.e.,

$$\forall x \in \mathcal{I} \cap \mathcal{U}_r(c), f(x) = \sum_{n=0}^{\infty} a_n (x - c)^n.$$

Hence f is real analytic in the sense of Definition B.102.

As for the sufficiency, f being real analytic gives

$$\forall c \in \mathcal{I}, \exists r_c > 0 \text{ s.t. } \forall x \in \mathcal{I} \cap \mathcal{U}_{r_c}(c), f(x) = \sum_{n=0}^{\infty} a_n (x - c)^n.$$

Define a series $F_c(z) := \sum_{n=0}^{\infty} a_n (z - c)^n$. Abel's Theorem 1.77 dictates that $F_c(z)$ converges to some complex number for any given $z \in \mathcal{U}_{r_c}(c)$. By Theorem 4.9 (A) $\Leftrightarrow$ (F), $F_c(z) : \mathcal{U}_{r_c}(c) \rightarrow \mathbb{C}$ is analytic. For each $c \in \mathcal{I}$, the above construction gives an analytic function F_c .

Starting from $F|_{c_0} = F_{c_0}$ for any $c_0 \in \mathcal{I}$, we choose c_1 sufficiently close to c_0 so that $I_{0,1} := \mathcal{U}_{r_{c_1}}(c_1) \cap \mathcal{U}_{r_{c_0}}(c_0) \neq \emptyset$. The two analytic functions F_{c_0} and F_{c_1} agree with each other

on the infinite set $\mathcal{I} \cap I_{0,1}$. By the identity theorem 4.35 and Definition 4.72, the function $F_{\{0,1\}} : \mathcal{U}_{r_{c_1}}(c_1) \cup \mathcal{U}_{r_{c_0}}(c_0) \rightarrow \mathbb{C}$ is an analytic continuation of F_{c_0} . Then the proof is completed by repeating the above analytic continuation so that $\mathcal{I}$ is covered by $\mathcal{D} := \cup_{i \in \mathbb{N}} \mathcal{U}_{r(c_i)}(c_i)$ and F is given by $F := F_{\mathbb{N}}$. $\square$

Corollary 4.84. Let $\mathcal{D} \subset \mathbb{C}$ be a domain, $\Omega \subset \mathcal{D}$ a subset with at least one limit point in $\mathcal{D}$, and $f : \Omega \rightarrow \mathbb{C}$ an analytic function. If there exists a function $F : \mathcal{D} \rightarrow \mathbb{C}$ as the analytic continuation of f , then F is unique.

Proof. This follows from the identity theorem 4.35. $\square$

4.4 Isolated singularities

Example 4.85. Functions like $\frac{\sin z}{z}$, $\frac{1}{z}$, and $\exp(\frac{1}{z})$ are not defined at $z = 0$, but are all analytic in some punctured disk $\dot{\mathcal{U}}_r(0)$, cf. (1.22).

Definition 4.86. An (*isolated*) *singularity* of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is a point $c \notin \mathcal{D}$ such that $\mathcal{D}$ contains a punctured disk at c , i.e.,

$$\exists r > 0 \text{ s.t. } \dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}.$$

Example 4.87. The singularity $z = 0$ of the analytic function $f(z) = \frac{1}{\sin \frac{1}{z}}$ is not isolated because it is the limit point of the sequence $(\frac{1}{n\pi})_{n \in \mathbb{N}^+}$, of isolated singularities of f .

4.4.1 Classifying singularities in $\mathbb{C}$

Definition 4.88. A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open is *removable* iff f can be analytically extended to $\mathcal{D} \cup \{c\}$, i.e. iff there exists an analytic continuation $F : \mathcal{D} \cup \{c\} \rightarrow \mathbb{C}$ with $F|_{\mathcal{D}} = f$.

Theorem 4.89 (Riemann removability condition 1851). A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is removable if and only if there exists $r > 0$ such that $\dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}$ and f is bounded over $\dot{\mathcal{U}}_r(c)$.

Proof. The necessity is obvious and we only prove the sufficiency. WLOG, we assume $c = 0$. Define a function $h : \mathcal{U}_r(0) \rightarrow \mathbb{C}$ by

$$h(z) = \begin{cases} z^2 f(z), & z \neq 0; \\ 0, & z = 0. \end{cases}$$

h is analytic at 0 because $f(z)$ is bounded at $z = 0$ and

$$\lim_{z \rightarrow 0} \frac{h(z) - h(0)}{z - 0} = \lim_{z \rightarrow 0} z f(z) = 0.$$

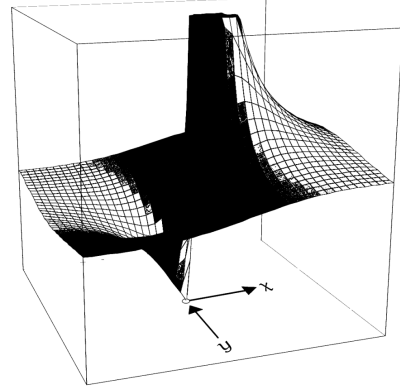
Thus h is analytic in $\mathcal{U}_r(0)$ and Theorem 4.1 yields

$$h(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

Then $h(0) = h'(0) = 0$ implies $a_0 = a_1 = 0$. For all $z \neq 0$, the power series $a_2 + a_3 z + a_4 z^2 + \cdots$ defines an analytic function F that satisfies $F|_{\dot{\mathcal{U}}_r(0)} = f$ and is also analytic at $z = 0$. The proof is then completed by Definition 4.88. $\square$

Example 4.90. Since $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$, Theorem 4.89 implies that the singularity $z = 0$ of $f(z) = \frac{\sin z}{z}$ is removable.

Example 4.91. The singularity $z = 0$ is not removable for $f(z) = \frac{1}{z}$.



Example 4.92. The singularity $z = 0$ is not removable for $g(z) = \exp(\frac{1}{z})$. Since $|g(z)| = \exp(\frac{\cos \theta}{r})$, the behavior of g at 0 depends upon the direction in which we approach 0: $\cos \theta < 0$ implies $|g(z)| \rightarrow 0$; $\cos \theta = 0$ implies $|g(z)| \rightarrow 1$; and $\cos \theta > 0$ implies $|g(z)| \rightarrow +\infty$ as a rate faster than any polynomial! See the plot above.

Exercise 4.93. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that the following statements are equivalent:

- (a) c is a removable singularity of f .
- (b) There exists $r > 0$ such that $\dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}$ and f is bounded over $\dot{\mathcal{U}}_r(c)$.
- (c) The limit $\lim_{z \rightarrow c} f(z)$ exists.
- (d) $\lim_{z \rightarrow c} (z - c)f(z) = 0$.

Exercise 4.94. Is 0 a removable singularity of $g(z) = \sin \frac{1}{z}$? Why?

Exercise 4.95. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that c is removable singularity of f if there exist positive constants r , A , and ϵ such that

$$\forall z \in \dot{\mathcal{U}}_r(c), \quad |f(z)| \leq A|z - c|^{-1+\epsilon}.$$

Definition 4.96. A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is said to be *non-essential* iff there exists an integer $m \in \mathbb{Z}$ such that c is a removable singularity of

$$g(z) := (z - c)^m f(z). \quad (4.27)$$

In particular, a *pole* is a non-essential singularity that is not removable.

Definition 4.97. A singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is said to be *essential* if it is not non-essential.

Lemma 4.98. Let $c \notin \mathcal{D}$ be a non-essential singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. If f does

not vanish identically in some neighborhood of c , then there exists a minimal integer $k \in \mathbb{Z}$ such that the function

$$h(z) = (z - c)^k f(z) \quad (4.28)$$

has a removable singularity at c and $h(c) \neq 0$.

Proof. If c is a non-essential singularity of f , then by Definitions 4.96 and 4.88 the function $g(z)$ in (4.27) can be expanded as a power series at c . Furthermore, f does not vanish identically in some neighborhood of c , hence

$$\begin{aligned} \exists n \in \mathbb{N} \text{ s.t. } a_n \neq 0 \text{ and} \\ g(z) = a_n(z - c)^n + a_{n+1}(z - c)^{n+1} + \cdots \end{aligned} \quad (4.29)$$

Multiply both sides with $(z - c)^{-n}$, refer to (4.27), and we see that the function $h(z)$ in (4.28) has a removable singularity at c with $k = m - n$. In addition, $h(c) \neq 0$ follows from $a_n \neq 0$. Furthermore, k is the minimal integer with this property. Otherwise the power series

$$(z - c)^{k-1} f(z) = \frac{a_n}{z - c} + a_{n+1} + a_{n+2}(z - c) + \cdots$$

would have a removable singularity. But this is false because Theorem 4.89 dictates that the singularity c of $\frac{a_n}{z - c}$ is not removable. $\square$

Definition 4.99. The *order of a non-essential singularity* c of an analytic function f , written $\text{ord}(f; c)$, is the maximum number $m \in \mathbb{Z}$ such that $\varphi(z) := (z - c)^{-m} f(z)$ has a removable singularity at c and $\varphi(c) \neq 0$. In particular,

$$\text{ord}(f; c) = +\infty \iff \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c), f(z) = 0.$$

Example 4.100. The analytic function $f : \mathbb{C} \setminus \{1\} \rightarrow \mathbb{C}$,

$$f(z) = (z - 1)^5 + 2(z - 1)^6 = (z - 1)^5(1 + 2(z - 1)),$$

has a non-essential singularity at $z = 1$ of order 5. By Definition 4.18 and Lemma 4.20, $z = 1$ is a zero point of f with multiplicity 5. This suggests that the order of a non-essential singularity reduces to the multiplicity of a zero point for $\text{ord}(f; c) > 0$.

Lemma 4.101. A non-essential singularity c of an analytic function f is removable if and only if $\text{ord}(f; c) \geq 0$.

Proof. We show that $m := \text{ord}(f; c) \geq 0$ implies removability and $m < 0$ implies non-removability. Since c is a non-essential singularity of order m , Definition 4.99 and Theorem 4.89 yield

$$\exists C_m > 0, \exists r > 0, \text{ s.t. } \forall z \in \dot{\mathcal{U}}_r(c), |f(z)(z - c)^{-m}| \in (0, C_m).$$

When $m = 0$, the above equation reduces to the condition of f being bounded in $\dot{\mathcal{U}}_r(c)$, and then Theorem 4.89 implies that c is removable. For any given $m > 0$, c is also removable by Theorem 4.89 because

$$\begin{aligned} \exists C = C_m, \exists r = \min(1, r_m), \text{ s.t. } \forall z \in \dot{\mathcal{U}}_r(c), \\ |f(z)| = |f(z)(z - c)^{-m}| \cdot |z - c|^m < C_m = C, \end{aligned}$$

where the inequality follows from $|z - c| < 1$ and $m > 0$.

For any given $m < 0$, let $h(z) = (z - c)^{-m} f(z)$. Then Definition 4.99 implies that $h(c) \neq 0$. The continuity of h at c then yields

$$\exists \rho > 0 \text{ s.t. } \forall z \in \dot{\mathcal{U}}_\rho(c), 0 \neq |h(z)| \geq h_m := \inf_{z \in \dot{\mathcal{U}}_\rho(c)} |h(z)|.$$

It follows that $f(z)$ is not bounded at c because

$$\begin{aligned} \forall C > 0, \exists r = \min\left(\rho, \frac{h_m^{-\frac{1}{m}}}{C^{-\frac{1}{m}}}\right) \text{ s.t. } \forall z \in \mathcal{U}_r(c), \\ |f(z)| = \frac{|h(z)|}{|z - c|^{-m}} \geq \frac{h_m}{|z - c|^{-m}} > \frac{h_m}{r^{-m}} \geq C. \end{aligned} \quad (4.30)$$

Then Theorem 4.89 completes the proof. $\square$

Definition 4.102. The *order of a pole* c is the negative of its order as a non-essential singularity. In particular, a pole of order 1, 2, 3 is called a *simple*, *double*, *triple pole*, respectively.

Corollary 4.103. A non-essential singularity c of an analytic function f is a pole if and only if its order is negative.

Proof. This follows from Definitions 4.96 and 4.99 and Lemma 4.101. $\square$

Exercise 4.104. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that the following statements are equivalent:

- (a) c is a pole of f of order $k \in \mathbb{N}$.
- (b) There exist $r > 0$ and an analytic function $h : \mathcal{U}_r(c) \rightarrow \mathbb{C}$ with $h(c) \neq 0$ such that $\mathcal{U}_r(c) \subseteq \mathcal{D}$ and

$$\forall z \in \dot{\mathcal{U}}_r(c), \quad f(z) = \frac{h(z)}{(z - c)^k}.$$

- (c) There exist $r > 0$ and an analytic function $g : \mathcal{U}_r(c) \rightarrow \mathbb{C}$ not vanishing in $\dot{\mathcal{U}}_r(c)$ such that g has a zero of multiplicity k at c and $f = \frac{1}{g}$ over $\dot{\mathcal{U}}_r(c)$.
- (d) There exist $r > 0$ and constants $M_1, M_2 > 0$ such that

$$\forall z \in \dot{\mathcal{U}}_r(c), \quad M_1|z - c|^{-k} \leq |f(z)| \leq M_2|z - c|^{-k}.$$

Example 4.105. The analytic function $f : \mathbb{C}^\bullet \rightarrow \mathbb{C}$,

$$f(z) = \frac{1}{z^2} + \frac{1}{z} = \frac{1 + z}{z^2}$$

has a double pole at $z = 0$.

Corollary 4.106. At a pole $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, we have

$$\lim_{z \rightarrow c; z \in \mathcal{D}} |f(z)| = \infty, \quad (4.31)$$

i.e., $\forall C > 0, \exists r > 0 \text{ s.t. } \forall z \in \mathcal{D}, |z - c| < r \implies |f(z)| \geq C$.

Proof. By Corollary 4.103, a pole is a non-essential singularity with a negative order. The proof is then completed by repeating the arguments in proving (4.30). $\square$

Lemma 4.107. Suppose $c \notin \mathcal{D}$ is a non-essential singularity of analytic functions $f, g : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Then c is also a non-essential singularity of the functions

$$f \pm g, f \cdot g, \frac{f}{g} \text{ satisfying } \forall z \in \mathcal{D}, g(z) \neq 0,$$

and we have

$$\text{ord}(f \pm g; c) \geq \min\{\text{ord}(f; c), \text{ord}(g; c)\}, \quad (4.32a)$$

$$\text{ord}(f \cdot g; c) = \text{ord}(f; c) + \text{ord}(g; c), \quad (4.32b)$$

$$\text{ord}\left(\frac{f}{g}; c\right) = \text{ord}(f; c) - \text{ord}(g; c). \quad (4.32c)$$

Proof. By Definition 4.99, we write $f(z) = (z-c)^m h(z)$ with $h(c) \neq 0$ and $g(z) = (z-c)^n \ell(z)$ with $\ell(c) \neq 0$. The rest of the proof follows trivially. $\square$

Theorem 4.108 (Casorati–Weierstrass). For any neighborhood $\dot{\mathcal{U}}_r(c)$ of an essential singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, the image $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is dense in $\mathbb{C}$ in the sense of Definition D.19.

Proof. By Definition D.19, A being dense in $\mathbb{C}$ means

$$\forall x \in \mathbb{C}, \forall \epsilon > 0, A \cap \mathcal{U}_\epsilon(x) \neq \emptyset. \quad (4.33)$$

Suppose there exists $r > 0$ such that $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is not dense in $\mathbb{C}$. Negating (4.33) with $A = f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ gives

$$\exists b \in \mathbb{C} \exists \epsilon > 0 \text{ s.t. } f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D}) \cap \mathcal{U}_\epsilon(b) = \emptyset.$$

The analytic function $g(z) := \frac{1}{f(z)-b}$ is bounded in $\dot{\mathcal{U}}_\delta(c)$ for some $\delta \in (0, r)$. Theorem 4.89 then implies that $z = c$ is a removable singularity of $g(z)$, which contradicts the fact that c is an essential singularity of f :

- if $g(c) \neq 0$, $f(c) = \frac{1}{g(c)} + b$ is bounded and thus c is a removable singularity of f ;
- if $g(c) = 0$, c is a root of g with multiplicity m . Thus $\lim_{z \rightarrow c} f(z) = \lim_{z \rightarrow c} \frac{1}{g(z)} + b = \infty$. By Definitions 4.96 and 4.102, c is a pole of f of order m . $\square$

Theorem 4.109 (Classifying isolated singularities by their mapping behaviors). A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open must be one of the three types as follows.

- c is removable $\Leftrightarrow f$ is bounded in $\dot{\mathcal{U}}_r(c)$ for some $r > 0$;
- c is a pole $\Leftrightarrow \lim_{z \rightarrow c; z \in \mathcal{D}} |f(z)| = \infty$;
- c is essential $\Leftrightarrow \forall r > 0$, $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is dense in $\mathbb{C}$.

Proof. By Definitions 4.96 and 4.97, a singularity must be one of the three types. We need to prove the three equivalences. (a) is just Theorem 4.89. If $\lim_{z \rightarrow c; z \in \mathcal{D}} |f(z)| = \infty$, then c is neither removable nor essential, hence it must be a pole. Then (b) holds from Corollary 4.106. If $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is dense in $\mathbb{C}$, then by Theorem 4.89 c is not removable. Also, $\lim_{z \rightarrow c; z \in \mathcal{D}} |f(z)| = \infty$ does not hold; otherwise for any $c \in \mathbb{C}$ there exists $r_c > 0$ small enough so that the distance from c to $f(\dot{\mathcal{U}}_{r_c}(c) \cap \mathcal{D})$ is arbitrarily large. The proof is completed by Theorem 4.108. $\square$

Exercise 4.110. Find all isolated singularities of

$$f(z) = \frac{(z-1)^2(z+3)}{1 - \sin \frac{\pi z}{2}}$$

and determine for each one its type.

4.4.2 Laurent decomposition

Notation 5. For $r \in [0, +\infty)$ and $R \in (r, +\infty]$, the *annulus centered at c with radii r, R* is

$$\mathcal{A}_c(r, R) := \{z \in \mathbb{C} : r < |z - c| < R\}. \quad (4.34)$$

In particular, $\mathcal{A}_c(r, +\infty) = \{z \in \mathbb{C} : |z - c| > r\}$.

Theorem 4.111 (Cauchy integral formula for annuli). Any analytic function $f : \mathcal{A}_c(r, R) \rightarrow \mathbb{C}$ satisfies

$$\forall \rho_1, \rho_2 \in (r, R) \text{ with } \rho_1 < \rho_2, \forall z \in \mathcal{A}_c(\rho_1, \rho_2),$$

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - c| = \rho_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{|\zeta - c| = \rho_1} \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (4.35)$$

Proof. The conclusion follows as a special case of Corollary 3.76 with $n = 1$. $\square$

Exercise 4.112. Give an alternative proof of Theorem 4.111 by applying the Riemann removability theorem 4.89 to the function $g_z : \mathcal{A}_c(r, R) \setminus \{z\} \rightarrow \mathbb{C}$ given by

$$g_z(\zeta) := \frac{f(\zeta) - f(z)}{\zeta - z}.$$

You are not supposed to use Corollary 3.76.

Theorem 4.113 (Laurent decomposition). A continuous function $f : \mathcal{A}_c(r, R) \rightarrow \mathbb{C}$ that is analytic on $\mathcal{A}_c(r, R)$ admits a decomposition as

$$f(z) = g(z) + h(z), \quad (4.36)$$

where $g : \mathcal{U}_R(c) \rightarrow \mathbb{C}$ and $h : \mathcal{A}_c(r, +\infty) \rightarrow \mathbb{C}$ are analytic. Moreover, the decomposition (4.36) is unique if $h(\infty) = 0$.

Proof. The decomposition follows from Theorem 4.111 and

$$g(z) := +\frac{1}{2\pi i} \oint_{|\zeta - c| = R} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

$$h(z) := -\frac{1}{2\pi i} \oint_{|\zeta - c| = r} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

where $h(\infty) = 0$. Both g and h are clearly analytic.

Suppose $f(z) = \tilde{h}(z) + \tilde{g}(z)$ is a different decomposition with $\tilde{h}(\infty) = 0$. Then we have $\tilde{h}(z) + \tilde{g}(z) = h(z) + g(z)$ and the functions $H := h - \tilde{h}$ and $G := \tilde{g} - g$ satisfies $G(z) = H(z)$ for all $z \in \mathcal{A}_c(r, R)$. Since H is analytic on $\mathcal{A}_c(r, +\infty)$, G is analytic on $\mathcal{U}_R(c)$, $r < R$, and $\mathcal{A}_c(r, R) = \mathcal{U}_R(c) \cap \mathcal{A}_c(r, +\infty)$, the identity theorem 4.35 and Definition 4.72 imply that G and H glue to a single entire function F satisfying $\lim_{z \rightarrow \infty} F(z) = H(\infty) = h(\infty) - \tilde{h}(\infty) = 0$. Liouville's theorem 3.94 dictates that both $G(z)$ and $H(z)$ be the constant zero, which gives $g = \tilde{g}$ and $h = \tilde{h}$. $\square$

Definition 4.114. In the Laurent decomposition (4.36), the analytic functions $g : \mathcal{U}_R(c) \rightarrow \mathbb{C}$ and $h : \mathcal{A}_c(r, +\infty) \rightarrow \mathbb{C}$ are respectively called the *analytic/regular part* and the *principal part*.

4.4.3 Laurent series

Definition 4.115. A *doubly infinite series* is an infinite series of the form

$$\sum_{j=-\infty}^{+\infty} a_j = \sum_{j=0}^{+\infty} a_j + \sum_{j=1}^{+\infty} a_{-j}, \quad (4.37)$$

it *converges* iff the series $\sum_{j=0}^{+\infty} a_j$ and $\sum_{j=1}^{+\infty} a_{-j}$ both converge independently.

Definition 4.116. A *Laurent series* on $\mathcal{U}_r(c)$ is a doubly infinite series of the form $\sum_{j=-\infty}^{+\infty} a_j(z-c)^j$.

Lemma 4.117. If a Laurent series $\sum_{j=-\infty}^{+\infty} a_j(z-c)^j$ converges at some $z = z_0$, then there exist unique $r, R \in [0, \infty]$ such that the series converges normally in its *annulus of convergence* $\mathcal{A}_c(r, R)$ and diverges for all $z \in \mathcal{A}_c(R, +\infty)$ and all $z \in \mathcal{A}_c(0, r)$. In addition, for r_1 and r_2 satisfying $r < r_1 < r_2 < R$, the Laurent series converges uniformly and absolutely on $\overline{\mathcal{A}_c(r_1, r_2)}$.

Proof. WLOG, we set $c = 0$ and assume the point of convergence $z_0 \neq 0$. By Definition 4.115, the convergence of $\sum_{j=-\infty}^{+\infty} a_j(z-c)^j$ at z_0 implies those of

$$p_+(z) = \sum_{j=0}^{+\infty} a_j z^j, \quad p_-\left(\frac{1}{z}\right) = \sum_{j=1}^{+\infty} a_{-j} \left(\frac{1}{z}\right)^j. \quad (4.38)$$

By the Abel theorem 1.77, there exists $r_2 > |z_0|$ such that $p_+(z)$ converges normally for all $z \in \mathcal{U}_{r_2}(c)$ and there exists $r > \left|\frac{1}{z_0}\right|$ such that $p_-\left(\frac{1}{z}\right)$ converges normally for all z satisfying $\left|\frac{1}{z}\right| < r$. Set $r_1 = \frac{1}{r}$ and we have $r_1 < r_2$ and the first conclusion, which, together with Theorem 1.75 and $\overline{\mathcal{A}_c(r_1, r_2)}$ being compact implies the second conclusion. $\square$

Example 4.118. Apply Lemma 4.117 to Laurent series of

$$f(z) := \frac{2}{z^2 - 4z + 3} = \frac{1}{1-z} + \frac{1}{z-3}$$

on three annuli $\mathcal{A}_0(0, 1)$, $\mathcal{A}_0(1, 3)$, and $\mathcal{A}_0(3, +\infty)$.

$z \in \mathcal{A}_0(0, 1)$ implies $r = 0$ and $R = 1$ because

$$\begin{aligned} \frac{1}{1-z} &= \sum_{n=0}^{\infty} z^n \text{ and } \frac{1}{3-z} = \frac{1}{3} \cdot \frac{1}{1-z/3} = \sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} \\ \implies g(z) = f(z) &= \sum_{n=0}^{\infty} \left(1 - \frac{1}{3^{n+1}}\right) z^n \text{ and } h(z) = 0. \end{aligned}$$

$z \in \mathcal{A}_0(1, 3)$ implies $r = 1$ and $R = 3$ because

$$\begin{aligned} \begin{cases} -h(z) = \frac{1}{z-1} = \frac{1}{z} \cdot \frac{1}{1-1/z} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}, \\ -g(z) = \frac{1}{3-z} = \sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} \end{cases} \\ \implies f(z) = g(z) + h(z) = -\sum_{n=1}^{\infty} \frac{1}{z^n} - \sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}}. \end{aligned}$$

$z \in \mathcal{A}_0(3, +\infty)$ implies $r = 3$ and $R = +\infty$ because

$$\begin{aligned} \frac{1}{z-1} &= \sum_{n=1}^{\infty} \frac{1}{z^n} \text{ and } \frac{1}{3-z} = \frac{1}{z} \cdot \frac{1}{1-3/z} = \sum_{n=0}^{\infty} \frac{3^n}{z^{n+1}} \\ \implies h(z) = f(z) &= \sum_{n=1}^{\infty} (3^{n-1} - 1) \frac{1}{z^n} \text{ and } g(z) = 0. \end{aligned}$$

Exercise 4.119. Find all possible Laurent series of

$$f(z) := \frac{1}{z(z-1)(z-2)}.$$

Theorem 4.120 (Laurent series). An analytic function $f : \mathcal{A}_c(r, R) \rightarrow \mathbb{C}$ can be represented by a Laurent series:

$$\forall z \in \mathcal{A}_c(r, R), \quad f(z) = \sum_{j=-\infty}^{+\infty} a_j(z-c)^j, \quad (4.39)$$

which converges absolutely and uniformly on each compact subset of $\mathcal{A}_c(r, R)$ with each coefficient uniquely given by

$$\forall j \in \mathbb{Z}, \quad a_j = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{j+1}} d\zeta, \quad (4.40)$$

which does not depend on $\rho \in (r, R)$ and satisfies

$$|a_j| \leq \frac{1}{\rho^j} \sup\{|f(\zeta)| : |\zeta-c| = \rho\}. \quad (4.41)$$

Proof. For any $\forall z \in \mathcal{A}_c(r, R)$, there exists r_1 and r_2 satisfying $r < r_1 < r_2 < R$ such that $z \in \mathcal{A}_c(r_1, r_2)$. By Theorems 4.111 and 4.113, the following decomposition is unique:

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta-c|=r_2} \frac{f(\zeta)}{\zeta-z} d\zeta - \frac{1}{2\pi i} \oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{\zeta-z} d\zeta. \quad (4.42)$$

Repeat the proof of Taylor expansions in Theorem 4.1 for the first RHS integral in (4.42) and we have

$$\oint_{|\zeta-c|=r_2} \frac{f(\zeta)}{\zeta-z} d\zeta = \sum_{j=0}^{+\infty} \left(\oint_{|\zeta-c|=r_2} \frac{f(\zeta)}{(\zeta-c)^{j+1}} d\zeta \right) (z-c)^j.$$

For the second RHS integral in (4.42), we have

$$\begin{aligned} \oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{\zeta-z} d\zeta &= - \oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{1 - \frac{\zeta-c}{z-c}} \cdot \frac{1}{z-c} d\zeta \\ &= - \oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{z-c} \sum_{j=0}^{+\infty} \frac{(\zeta-c)^j}{(z-c)^j} d\zeta \\ &= - \sum_{j=-\infty}^{-1} \left[\oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{(\zeta-c)^{j+1}} d\zeta \right] (z-c)^j. \end{aligned}$$

Consequently, the a_j 's for non-negative and negative j 's have the same form except the domain integration:

$$\forall j \in \mathbb{N}, \quad a_j = \frac{1}{2\pi i} \oint_{|\zeta-c|=r_2} \frac{f(\zeta)}{(\zeta-c)^{j+1}} d\zeta, \quad (4.43a)$$

$$\forall j \in -\mathbb{N}^+, \quad a_j = \frac{1}{2\pi i} \oint_{|\zeta-c|=r_1} \frac{f(\zeta)}{(\zeta-c)^{j+1}} d\zeta. \quad (4.43b)$$

The Cauchy integral theorem implies that we can set r_1 and r_2 in (4.43) to any $\rho \in (r, R)$ without changing the value of a_j 's. The uniqueness of a_j 's in (4.40) then follows from the uniqueness of (4.43) and Theorem 4.1. Finally, (4.41) follows from a variable substitution and Lemma 3.13. $\square$

Example 4.121. The Laurent series of the function $f(z) = \exp(-\frac{1}{z^2})$ at the essential singularity $z = 0$ is obtained from the power series of $\exp(w)$ with $w = -\frac{1}{z^2}$:

$$f(z) = 1 - \frac{1}{z^2} + \frac{1}{2!} \frac{1}{z^4} - \frac{1}{3!} \frac{1}{z^6} + \dots = 1 + h(z),$$

where $h(z)$ is the principal part of $f(z)$.

Exercise 4.122. Let $f : \mathcal{A}_c(r, R) \rightarrow \mathbb{C}$ be an analytic function with the Laurent expansion (4.39). Prove that

$$\forall \rho \in (r, R), \quad \sum_{j=-\infty}^{+\infty} |a_j|^2 \rho^{2j} \leq [M(\rho)]^2,$$

where $M(\rho)$ is defined in Corollary 3.85. In particular, for any $j \in \mathbb{Z}$, $|a_j| \leq \rho^{-j} M(\rho)$; the equality holds for some $j = j_0$ and $\rho = \rho_0$ if and only if $f(z) = a_{j_0}(z - c)^{j_0}$.

Lemma 4.123. The Laurent expansion of an analytic function $f : \mathcal{A}_0(r, R) \rightarrow \mathbb{C}$ only contains even (or odd) powers of z if f is respectively even (or odd).

Exercise 4.124. Prove Lemma 4.123.

Corollary 4.125. For a singularity c of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, there exists $R > 0$ such that f has a unique Laurent expansion at c in $\mathcal{A}_c(0, R)$, i.e.,

$$\forall z \in \mathcal{A}_c(0, R), \quad f(z) = \sum_{j=-\infty}^{+\infty} a_j(z - c)^j. \quad (4.44)$$

Proof. This follows directly from Theorem 4.120 and the fact that a punctured disk is an annulus with zero inner radius and finite outer radius. $\square$

Theorem 4.126 (Classifying isolated singularities by Laurent series). According to (4.44), a singularity c of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is

(a) removable iff

$$\forall j < 0, \quad a_j = 0, \quad (4.45)$$

(b) a pole of order $k \in \mathbb{N}^+$ iff

$$a_{-k} \neq 0 \quad \text{and} \quad \forall n < -k, \quad a_n = 0, \quad (4.46)$$

(c) essential iff neither (4.45) nor (4.46) holds.

Proof. By (4.45), the function f can be expressed as a power series, thus it is analytic at some punctured neighborhood of c and thus bounded at this punctured neighborhood. Then it follows from Theorem 4.89 that (4.45) is a necessary and sufficient condition of c being a removable singularity. The condition (4.46) is equivalent to (4.31), then it follows from Theorem 4.109 (b) that (4.46) is a necessary and sufficient condition of c being a pole. The rest of the proof follows from Definitions 4.96 and 4.97. $\square$

Exercise 4.127. Give an alternative proof of (d) $\Rightarrow$ (a) in Exercise 4.93 by applying Exercise 4.122.

Definition 4.128. A function $f : \mathcal{S}_{a,b} \rightarrow \mathbb{C}$ with

$$\mathcal{S}_{a,b} := \{z = x + yi \in \mathbb{C} : y \in (a, b)\} \quad (4.47)$$

and $-\infty \leq a < b \leq +\infty$ is *1-periodic* iff $f(z + 1) = f(z)$ holds for any $z \in \mathcal{S}_{a,b}$.

Corollary 4.129 (Fourier series of a 1-periodic analytic function). Any 1-periodic analytic function $f : \mathcal{S}_{a,b} \rightarrow \mathbb{C}$ as in Definition 4.128 can be represented as a uniquely normally convergent *complex Fourier series*

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n e^{2\pi i n z}, \quad (4.48)$$

where $z = x + iy$ and the *Fourier coefficients* are given by

$$a_n = \int_0^1 f(z) e^{-2\pi i n z} dx. \quad (4.49)$$

Proof. Write $r := e^{-2\pi b}$ and $R := e^{-2\pi a}$, construct a conformal map $q : \mathcal{S}_{a,b} \rightarrow \mathcal{A}_0(r, R)$ by $q(z) := e^{2\pi i z}$, define an analytic function $g : \mathcal{A}_0(r, R) \rightarrow \mathbb{C}$ by $f = g \circ q$, and we can represent f by a Laurent series

$$f(z) = g(q(z)) = \sum_{n=-\infty}^{+\infty} a_n q^n.$$

By Theorem 4.120, the coefficient a_n is given by

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \oint_{|\zeta|=\rho} \frac{g(\zeta)}{\zeta^{n+1}} d\zeta = \int_0^1 \frac{g(\rho e^{2\pi i x})}{\rho^n e^{2\pi i n x}} dx \\ &= \int_0^1 f(z) e^{-2\pi i n z} dx, \end{aligned}$$

where the last step follows from $\rho = e^{-2\pi y}$. $\square$

Exercise 4.130 (Laurent series as an expansion in an orthonormal basis). Denote by $\mathcal{H}$ the space of all analytic functions on $\mathcal{A}_0(r, R)$ where $r < 1 < R$ and define a sesquilinear map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$,

$$\langle f, g \rangle := \frac{1}{2\pi i} \oint_{|\zeta|=1} f(\zeta) \overline{g(\zeta)} \frac{1}{\zeta} d\zeta. \quad (4.50)$$

Show that

- (a) $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{H}$ in the sense of Definition C.170;
- (b) the set $\{z^j : j \in \mathbb{Z}\}$ is an orthonormal basis of $\mathcal{H}$;
- (c) the Laurent series of any $f \in \mathcal{H}$ is exactly the expansion of f in terms of the orthonormal basis $\{z^j : j \in \mathbb{Z}\}$. In other words, if $f(z) = \sum_{j=-\infty}^{+\infty} a_j z^j$ is the Laurent expansion of f , then

$$\forall j \in \mathbb{Z}, \quad a_j = \langle f, z^j \rangle = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta^{j+1}} d\zeta.$$

4.4.4 The residue theorem

Definition 4.131. The *residue of an analytic function* $f : \mathcal{U}_R(c) \rightarrow \mathbb{C}$ at an isolated singularity c , written $\text{Res}(f; c)$ or $\text{Res}_{z=c} f(z)$, is the coefficient a_{-1} of the Laurent series

$$f(z) = a_{-1}(z - c)^{-1} + \sum_{n=0}^{+\infty} a_n(z - c)^n + \sum_{n=-2}^{-\infty} a_n(z - c)^n. \quad (4.51)$$

Theorem 4.132 (Residue). At a non-essential singularity c of an analytic function $f : \Omega \rightarrow \mathbb{C}$ with Ω open, we have

$$\oint_{|\zeta-c|=\rho} f(\zeta) d\zeta = 2\pi i \operatorname{Res}(f; c), \quad (4.52)$$

where $\rho > 0$ can be arbitrary provided that $\dot{\mathcal{U}}_\rho(c) \subset \Omega$.

Proof. By Definition 4.88 and Ω being open, there exists $R > \rho$ such that $\dot{\mathcal{U}}_R(c) \subset \Omega$. Since $\dot{\mathcal{U}}_R(c)$ equals the annulus $\mathcal{A}_c(0, R)$, we can construct a disk with one hole $\mathcal{A}_c(\epsilon, \rho)$ for any arbitrarily small $\epsilon > 0$. Then the Cauchy integral theorem (3.39) gives

$$\oint_{|\zeta-c|=\rho} f(\zeta) d\zeta = \oint_{|\zeta-c|=\epsilon} f(\zeta) d\zeta. \quad (4.53)$$

Theorems 4.126 and 4.120 yield the Laurent expansion

$$f(\zeta) = g(\zeta) + \sum_{n=-m}^{-1} a_n (\zeta - c)^n, \quad (4.54)$$

where g is the regular part of f at c . If c is removable, we have $m = 0$ and the second term in (4.54) drops out; otherwise c is a pole and the pole order m is a positive integer. The Cauchy integral theorem 3.59 gives

$$\oint_{|\zeta-c|=\epsilon} g(\zeta) d\zeta = 0. \quad (4.55)$$

Apply the Cauchy integral formula (3.42) to the constant function $\zeta \mapsto a_{-1}$ and we have

$$\oint_{|\zeta-c|=\epsilon} \frac{a_{-1}}{\zeta - c} d\zeta = 2\pi i a_{-1}. \quad (4.56)$$

For any $j = 2, \dots, m$, the function $G(\zeta) := \frac{a_{-j}}{(1-j)(\zeta-c)^{j-1}}$ satisfies $G'(\zeta) = \frac{a_{-j}}{(\zeta-c)^j}$ and is continuous on the circle $|\zeta - c| = \epsilon$. Hence we have

$$\forall n = -m, \dots, -2, \oint_{|\zeta-c|=\epsilon} a_n (\zeta - c)^n d\zeta = 0. \quad (4.57)$$

The proof is completed by plugging (4.54) into the RHS of (4.53) and applying (4.55), (4.56), and (4.57). $\square$

Lemma 4.133. The residue of a pole c of order m of an analytic function f is

$$\operatorname{Res}_{z=c} f(z) = \frac{\varphi^{(m-1)}(c)}{(m-1)!}, \quad (4.58)$$

where $\varphi(z) := (z - c)^m f(z)$ is analytic in $\mathcal{U}_r$ for some $r > 0$.

Proof. By Definitions 4.102 and 4.99, there exists $r > 0$ such that $\varphi : \mathcal{U}_r(c) \rightarrow \mathbb{C}$ is analytic. Then (4.58) follows from the generalized Cauchy formula (Theorem 3.84) and the residue theorem 4.132 applied to φ . $\square$

Corollary 4.134. The residue of a simple pole c of f

$$\operatorname{Res}_{z=c} f(z) = \lim_{z \rightarrow c} (z - c) f(z). \quad (4.59)$$

Proof. Set $m = 1$ in Lemma 4.133 and we get (4.59). $\square$

Corollary 4.135. The residue of a double pole c of an analytic function f is $\operatorname{Res}_{z=c} f(z) = \varphi'(c)$ where $\varphi(z) = (z - c)^2 f(z)$.

Proof. This follows directly from Lemma 4.133. $\square$

Corollary 4.136. Suppose c is a simple pole of $f(z) = \frac{\varphi(z)}{\phi(z)}$ where φ and ϕ are analytic with $\varphi(c) \neq 0$, $\phi(c) = 0$, and $\phi'(c) \neq 0$. Then we have $\operatorname{Res}_{z=c} f(z) = \frac{\varphi(c)}{\phi'(c)}$.

Proof. Since c is a simple pole, Corollary 4.134 yields

$$\operatorname{Res}_{z=c} f(z) = \lim_{z \rightarrow c} \frac{\varphi(z)}{\phi(z)} (z - c) = \lim_{z \rightarrow c} \frac{\varphi(z)}{\frac{\phi(z) - \phi(c)}{z - c}} = \frac{\varphi(c)}{\phi'(c)}. \quad \square$$

Lemma 4.137. Any analytic function $g : \mathcal{D} \rightarrow \mathbb{C}$ with domain $\mathcal{D}$ and any analytic function $f : \mathcal{D} \setminus \{c\} \rightarrow \mathbb{C}$ with a non-essential singularity $c \in \mathcal{D}$ satisfy

$$\operatorname{Res} \left(g \frac{f'}{f}; c \right) = g(c) \operatorname{ord}(f; c). \quad (4.60)$$

Proof. By Definition 4.99, the function $\varphi(z) := (z - c)^m f(z)$ is analytic in $\mathcal{U}_r(c)$ for some $r > 0$ where $m := -\operatorname{ord}(f; c)$.

$$f'(z) = \frac{(z - c)\varphi'(z) - m\varphi(z)}{(z - c)^{m+1}} \implies g \frac{f'}{f} = g \frac{\varphi'}{\varphi} - \frac{mg}{z - c}$$

leads to (4.60) via $\alpha := \partial \mathcal{U}_r(c)$ and

$$\operatorname{Res} \left(g \frac{f'}{f}; c \right) = \frac{1}{2\pi i} \oint_\alpha g \frac{f'}{f} d\zeta = \frac{-m}{2\pi i} \oint_\alpha \frac{g(\zeta)}{\zeta - c} d\zeta = g(c) \operatorname{ord}(f; c),$$

where the first step follows from Theorem 4.132, the second from the Cauchy integral theorem 3.59 and $g \frac{\varphi'}{\varphi}$ being analytic in $\mathcal{U}_R(c)$ for some $R > r$, and the last from the Cauchy integral formula (Theorem 3.75). $\square$

Example 4.138. $f(z) = \frac{\exp(iz)}{z^2 + 1}$ has a simple pole at $z = i$. Corollary 4.134 or 4.136 yields $\operatorname{Res}(f; i) = -\frac{1}{2e}$.

Example 4.139. $f(z) = \frac{\pi \cos(\pi z)}{\sin(\pi z)}$ has a simple pole at each $n \in \mathbb{Z}$. Corollary 4.136 yields $\operatorname{Res}(f; n) = 1$.

Example 4.140. $f(z) = \frac{1}{(z^2 + 1)^3}$ has a triple pole at $z = i$. Lemma 4.133 yields $\operatorname{Res}(f; i) = -\frac{3i}{16}$.

Theorem 4.141 (Residue: a disk with holes). For a disk with n holes $\mathcal{D}$, a finite discrete set $Z \subset \mathcal{D}$, and a continuous function $f : \overline{\mathcal{D}} \setminus Z \rightarrow \mathbb{C}$ that is analytic in $\mathcal{D} \setminus Z$, we have

$$\int_{\partial \mathcal{D}} f(z) dz = 2\pi i \sum_{j=1}^m \operatorname{Res}(f; z_j) \quad (4.61)$$

where each point $z_j \in Z$ is a non-essential singularity of f .

Proof. Define clockwise oriented Jordan curves as follows,

$$\forall j = 1, 2, \dots, m, \quad \alpha_{n+j} := \partial \mathcal{U}_{r_j}(z_j),$$

where the r_j 's are small enough such that $\alpha_0, \alpha_1, \dots, \alpha_{n+m}$ form a new disk with $m+n$ holes. Apply the Cauchy integral theorem 3.73 to $\cap_{j=0}^{n+m} \operatorname{int}(\alpha_j)$ and we have

$$\begin{aligned} \int_{\cup_{j=0}^{n+m} \alpha_j} f(z) dz &= 0 \iff \left(\int_{\partial \mathcal{D}} + \int_{\cup_{j=n+1}^{n+m} \alpha_j} \right) f(z) dz = 0 \\ &\iff \int_{\partial \mathcal{D}} f(z) dz = \sum_{j=n+1}^{n+m} \oint_{\alpha_j^+} f(z) dz = 2\pi i \sum_{j=1}^m \operatorname{Res}(f; z_j), \end{aligned}$$

where α_j^+ 's in the last line are counter-clockwise oriented and the last equality follows from Theorem 4.132. $\square$

4.4.5 Calculating $I_1 := \int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$ where $R := \frac{P(x,y)}{Q(x,y)}$ is a rational function

Roadmap 4.142. The key to computing the above I_1 is to introduce the change of variable $z = e^{i\theta}$, and then substitute

$$\cos \theta = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{z - z^{-1}}{2i}, \quad d\theta = \frac{dz}{iz}$$

into I_1 to obtain

$$I_1 = \oint_{|z|=1} R\left(\frac{z + z^{-1}}{2}, \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}. \quad (4.62)$$

Lemma 4.143. For all $p \in \mathbb{C}$ with $|p| \in [0, 1)$, we have

$$I_1 = \int_0^{2\pi} \frac{d\theta}{1 - 2p \cos \theta + p^2} = \frac{2\pi}{1 - p^2}. \quad (4.63)$$

Proof. If $p = 0$, (4.63) clearly holds. Otherwise (4.62) yields $I_1 = \oint_{|z|=1} f(z) dz$ with $f(z) = \frac{-i}{(z-p)(1-pz)}$. Since $f(z)$ has a simple pole $z = p$ inside the unit disk, Corollary 4.134 yields

$$\text{Res}(f; p) = \left. \frac{-i}{1 - pz} \right|_{z=p} = \frac{-i}{1 - p^2}.$$

The rest follows from Theorem 4.141. $\square$

Lemma 4.144. For all $a > b > 0$, we have

$$I_1 = \int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{2\pi}{b^2} \left(a - \sqrt{a^2 - b^2} \right). \quad (4.64)$$

Proof. By (4.62), we have $I_1 = \frac{i}{2b} \oint_{|z|=1} f(z) dz$ with

$$f(z) = \frac{(z^2 - 1)^2}{z^2(z - u)(z - v)},$$

where $u = \frac{-a + \sqrt{a^2 - b^2}}{b}$ and $v = \frac{-a - \sqrt{a^2 - b^2}}{b}$. Since $|v| > 1 > |u|$, in the unit disk $f(z)$ has a simple pole at $z = u$ and another double pole at $z = 0$. Corollaries 4.134 and 4.135 give

$$\begin{aligned} \text{Res}(f; 0) &= \left[\frac{(z^2 - 1)^2}{z^2 + \frac{2a}{b}z + 1} \right]'_{z=0} = -\frac{2a}{b}. \\ \text{Res}(f; u) &= \left. \frac{(z^2 - 1)^2}{z^2(z - v)} \right|_{z=u} = \frac{2\sqrt{a^2 - b^2}}{b}. \end{aligned}$$

The rest follows from Theorem 4.141. $\square$

Lemma 4.145. For all $m \in \mathbb{N}^+$, we have

$$I_1 = \int_0^\pi \frac{\cos mx}{5 - 4 \cos x} dx = \frac{\pi}{3 \cdot 2^m}. \quad (4.65)$$

Exercise 4.146. Prove Lemma 4.145.

4.4.6 Calculating $I_2 = \int_{-\infty}^{+\infty} P(x)/Q(x) dx$

Roadmap 4.147. The integral limit in I_2 spans the real axis and thus does not form a loop. The main idea is to enclose an interval inside the loop, let (part of) the loop boundary go to infinity, and then apply the residue theorem. Usually we do this by defining an integral on the arc with an arbitrarily large radius and require it to vanish when the radius goes to infinity. This strategy for calculating I_2 requires Lemma 4.149.

Definition 4.148. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is integrable on every finite interval $[-r, r]$, the *Cauchy principal value of an improper integral of the form* $\int_{-\infty}^{+\infty} f(x) dx$ is

$$\text{P.V.} \int_{-\infty}^{+\infty} f(x) dx := \lim_{r \rightarrow +\infty} \int_{-r}^r f(x) dx, \quad (4.66)$$

provided the limit exists. When there is no danger of ambiguity, we omit the notation “P.V.” and simply write $\int_{-\infty}^{+\infty} f(x) dx$ to denote the principal value.

Lemma 4.149. Suppose, for any sufficiently large $R > 0$, $f(z)$ is continuous on the arc

$$S_R(\theta_1, \theta_2) := \{Re^{i\theta} : -\pi < \theta_1 \leq \theta \leq \theta_2 < \pi\}, \quad (4.67)$$

where $\lim_{R \rightarrow +\infty} \int_{S_R} z f(z) dz = \lambda$ holds regardless of θ . Then

$$\lim_{R \rightarrow +\infty} \int_{S_R(\theta_1, \theta_2)} f(z) dz = i(\theta_2 - \theta_1)\lambda. \quad (4.68)$$

Proof. $(\text{Log } z)' = \frac{1}{z}$, Lemma 3.14, and Theorem 1.111 yield

$$i(\theta_2 - \theta_1)\lambda = \int_{S_R(\theta_1, \theta_2)} \frac{\lambda}{z} dz.$$

The condition $\lim_{R \rightarrow +\infty} \int_{S_R} z f(z) dz = \lambda$ implies

$$\begin{aligned} \forall \epsilon > 0, \exists R_0 > 0 \text{ s.t. } \forall R > R_0, \forall z \in S_R(\theta_1, \theta_2) \\ |zf(z) - \lambda| < \frac{\epsilon}{\theta_2 - \theta_1}. \end{aligned}$$

Then we have

$$\forall \epsilon > 0, \left| \int_{S_R(\theta_1, \theta_2)} \frac{zf(z) - \lambda}{z} dz \right| < \frac{\epsilon}{\theta_2 - \theta_1} \frac{R(\theta_2 - \theta_1)}{R} = \epsilon,$$

which completes the proof. $\square$

Theorem 4.150. Suppose $P(z) := \sum_{j=0}^m c_j z^j$ with $c_m \neq 0$ and $Q(z) := \sum_{j=0}^n b_j z^j$ with $b_n \neq 0$ satisfy

- (i) P and Q have no common zeros,
- (ii) $n \geq m + 2$,
- (iii) $\forall z \in \mathbb{R}, Q(z) \neq 0$.

Then for $f(z) := \frac{P(z)}{Q(z)}$ we have

$$\int_{-\infty}^{+\infty} f(x) dx = 2\pi i \sum_{j=1}^k \text{Res}(f; z_j), \quad (4.69)$$

where $z_1, \dots, z_k$ are poles of f in the upper half plane

$$\mathbb{H} := \{z \in \mathbb{C} : \text{Im}(z) > 0\}. \quad (4.70)$$

Proof. By (ii) and (iii), the integral in (4.69) exists and equals to its Cauchy principal value. Construct a counter-clockwise Jordan curve α by concatenating the interval $[-R, R]$ and $S_R = \{Re^{i\theta} : \theta \in (0, \pi)\}$. Since the degree of $Q(z)$ is finite, for a sufficiently large R all singularities of f in $\mathbb{H}$ are inside $\text{int}(\alpha)$. The residue theorem 4.141 yields

$$2\pi i \sum_{j=1}^k \text{Res}(f; z_j) = \int_{\alpha} f(z) dz = \int_{-R}^{+R} f(x) dx + \int_{S_R} f(z) dz.$$

Since $n - m - 1 \geq 1$, we have

$$\lim_{z \rightarrow \infty} |zf(z)| = \lim_{z \rightarrow \infty} \left| \frac{1}{z^{(n-m-1)}} \right| \left| \frac{c_m + \cdots + \frac{c_0}{z^m}}{b_n + \cdots + \frac{b_0}{z^n}} \right| = 0.$$

Lemma 4.149 then yields $\lim_{R \rightarrow +\infty} \int_{S_R} f(z) dz = 0$, which completes the proof. $\square$

Corollary 4.151. For all $a > 0$, we have

$$I_2 = \int_0^{+\infty} \frac{dx}{x^4 + a^4} = \frac{\pi}{2\sqrt{2}a^3}. \quad (4.71)$$

Proof. $f(z) := \frac{1}{z^4 + a^4}$ has simple poles $a_k := a \exp(i\frac{\pi+2k\pi}{4})$ for $k = 0, 1, 2, 3$. Then Corollary 4.136 gives

$$(*) : \text{Res}(f; a_k) = \frac{1}{4z^3} \Big|_{z=a_k} = \frac{1}{4a_k^3} = \frac{a_k}{4a_k^4} = -\frac{a_k}{4a^4}.$$

Since only a_0 and a_1 are in $\mathbb{H}$, Theorem 4.150 yields

$$(\square) : I_2 = \frac{1}{2} \int_{-\infty}^{+\infty} f(x) dx = \pi i \text{Res}(f; a_0) + \pi i \text{Res}(f; a_1).$$

Then (4.71) follows from substituting $(*)$ into $(\square)$. $\square$

Exercise 4.152. Prove $\int_{-\infty}^{+\infty} \frac{x^4}{(2+3x^2)^4} dx = \frac{\pi}{288\sqrt{6}}$.

Exercise 4.153. Prove $\int_{-\infty}^{+\infty} \frac{dx}{(1+x^2)^{n+1}} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi$.

4.4.7 Calculating $I_3 = \int_{-\infty}^{+\infty} e^{imx} P(x)/Q(x) dx$

Lemma 4.154 (Jordan's inequality).

$$\forall x \in \left[0, \frac{\pi}{2}\right], \quad \frac{2}{\pi} x \leq \sin x \leq x.$$

Exercise 4.155. Prove Lemma 4.154.

Lemma 4.156. Suppose, for any sufficiently large $R > 0$, $g(z)$ is continuous on the half circle

$$\Gamma_R := S_R(0, \pi) = \{Re^{i\theta} : 0 \leq \theta \leq \pi\},$$

where $\lim_{R \rightarrow +\infty; z \in \Gamma_R} g(z) = 0$ holds regardless of θ . Then

$$\forall m > 0, \quad \lim_{R \rightarrow +\infty} \int_{\Gamma_R} g(z) e^{imz} dz = 0. \quad (4.72)$$

Proof. For any $m > 0$, $\lim_{R \rightarrow +\infty; z \in \Gamma_R} g(z) = 0$ gives

$$\forall \epsilon > 0, \exists R_0 > 0 \text{ s.t. } \forall R > R_0, \forall z \in \Gamma_R, |g(z)| < \frac{m\epsilon}{\pi},$$

which implies

$$\begin{aligned} \left| \int_{\Gamma_R} g(z) e^{imz} dz \right| &= \left| \int_0^\pi g(Re^{i\theta}) e^{imR \exp(i\theta)} Re^{i\theta} i d\theta \right| \\ &\leq R \frac{m\epsilon}{\pi} \left| \int_0^\pi e^{-mR \sin \theta} d\theta \right| = 2R \frac{m\epsilon}{\pi} \left| \int_0^{\pi/2} e^{-mR \sin \theta} d\theta \right| \\ &\leq 2R \frac{m\epsilon}{\pi} \left| \int_0^{\pi/2} e^{-\frac{2mR\theta}{\pi}} d\theta \right| = \epsilon (1 - e^{-mR}) < \epsilon, \end{aligned}$$

where the fourth step follows from Lemma 4.154. $\square$

Theorem 4.157. Suppose $P(z) := \sum_{j=0}^m c_j z^j$ with $c_m \neq 0$ and $Q(z) := \sum_{j=0}^n b_j z^j$ with $b_n \neq 0$ satisfy

- (i) P and Q have no common zeros,
- (ii) $n \geq m + 1$,
- (iii) $\forall z \in \mathbb{R}, Q(z) \neq 0$.

Then, for $f(z) = \frac{P(z)}{Q(z)} e^{ikz}$ with $k > 0$, we have

$$\int_{-\infty}^{+\infty} f(x) dx = 2\pi i \sum_{j=1}^{\ell} \text{Res}(f; z_j), \quad (4.73)$$

where $z_1, \dots, z_\ell$ are poles of f in the upper half plane $\mathbb{H}$.

Proof. The proof is similar to that of Theorem 4.150. $\square$

Corollary 4.158. For all $m > 0$, we have

$$I_3 = \int_{-\infty}^{+\infty} \frac{\cos(mx)}{1+x^2} dx = \pi e^{-m}. \quad (4.74)$$

Proof. $f(z) := \frac{e^{imz}}{1+z^2}$ has only one simple pole i in $\mathbb{H}$. Then Corollary 4.136 gives $\text{Res}(f; i) = \frac{e^{imz}}{2z} \Big|_{z=i} = \frac{e^{-m}}{2i}$ and Theorem 4.157 completes the proof. $\square$

Exercise 4.159. Show that

$$I_3 = \int_{-\infty}^{+\infty} \frac{x \cos x}{x^2 - 2x + 10} dx = \frac{\pi}{3} e^{-3} (\cos 1 - 3 \sin 1).$$

Lemma 4.160. Suppose, for sufficiently small $r > 0$, $f(z)$ is continuous on the arc $S_r(\theta_1, \theta_2)$ in (4.67). Then

$$\lim_{r \rightarrow 0; z \in S_r} z f(z) = \lambda \implies \lim_{r \rightarrow 0} \int_{S_r(\theta_1, \theta_2)} f(z) dz = i(\theta_2 - \theta_1) \lambda. \quad (4.75)$$

Proof. The proof is similar to that of Lemma 4.149. $\square$

Corollary 4.161. $\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi$.

Proof. Construct a positively oriented Jordan curve

$$\alpha := [-R, -r] \cup S_r(0, \pi) \cup [r, R] \cup S_R(0, \pi). \quad (4.76)$$

The function $g(z) := \frac{e^{iz}}{z}$ is analytic in $\text{int}(\alpha)$ and satisfies

$$\lim_{R \rightarrow +\infty; z \in S_R} z g(z) = 0, \quad \lim_{r \rightarrow 0; z \in S_r} z g(z) = 1. \quad (4.77)$$

Cauchy's integral theorem yields

$$0 = \oint_{\alpha} g(z) dz = \mathbf{i} \left(\int_{-R}^{-r} + \int_r^R \right) \frac{\sin x}{x} dx + \left(\int_{S_R} - \int_{S_r} \right) g(z) dz,$$

where the negative sign of the integral on S_r follows from the definition of $S_r(0, \pi)$ in (4.67) and the last equality from (4.76) and $\left(\int_{-R}^{-r} + \int_r^R \right) \frac{\cos x}{x} = 0$. By (4.77) and Lemmas 4.149 and 4.160, we have

$$\lim_{R \rightarrow +\infty} \int_{S_R} g(z) dz = 0, \quad \lim_{r \rightarrow 0} \int_{S_r} g(z) dz = \pi \mathbf{i}.$$

Then the proof is completed by the existence of the integral $\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx$ and its Cauchy principal value. $\square$

Example 4.162. Let $c > 0$ and $x > 1$ be real numbers. We compute the integral $I = \frac{1}{2\pi \mathbf{i}} \int_{c-\mathbf{i}\infty}^{c+\mathbf{i}\infty} \frac{x^s}{s^2} ds$.

Construct a counter-clockwise Jordan curve

$$\alpha = [c - \mathbf{i}R, c + \mathbf{i}R] \cup S_{R,c} \left(\frac{\pi}{2}, \frac{3\pi}{2} \right),$$

where

$$S_{R,c}(\theta_1, \theta_2) = c + S_R(\theta_1, \theta_2) = \{c + Re^{\mathbf{i}\theta} : \theta_1 \leq \theta \leq \theta_2\}.$$

When R is sufficiently large, α encloses 0 and the residue theorem 4.132 yields

$$\begin{aligned} \frac{1}{2\pi \mathbf{i}} \oint_{\alpha} \frac{x^s}{s^2} ds &= \frac{1}{2\pi \mathbf{i}} \left(\int_{c-\mathbf{i}R}^{c+\mathbf{i}R} \frac{x^s}{s^2} ds + \int_{S_{R,c}} \frac{x^s}{s^2} ds \right) \\ &= \text{Res}_{s=0} \frac{x^s}{s^2} = (x^s)'|_{s=0} = \log x. \end{aligned}$$

For any $s \in S_{R,c}(\frac{\pi}{2}, \frac{3\pi}{2})$, we have $|s| \geq R - c$ and $\text{Re } s \leq c$, which imply

$$\left| \frac{x^s}{s^2} \right| = \frac{x^{\text{Re } s}}{|s|^2} \leq \frac{x^c}{(R - c)^2}.$$

Then Lemma 3.13 gives

$$\left| \int_{S_{R,c}} \frac{x^s}{s^2} ds \right| \leq \frac{x^c}{(R - c)^2} \ell(S_{R,c}) = \frac{\pi x^c R}{(R - c)^2},$$

that is, $\lim_{R \rightarrow +\infty} \int_{S_{R,c}} \frac{x^s}{s^2} ds = 0$. Thus by Definition 4.148,

$$I = \frac{1}{2\pi \mathbf{i}} \lim_{R \rightarrow +\infty} \int_{c-\mathbf{i}R}^{c+\mathbf{i}R} \frac{x^s}{s^2} ds = \log x.$$

Exercise 4.163. Let $c > 0$ and $x \in (0, 1]$ be real numbers. Show that $I = \frac{1}{2\pi \mathbf{i}} \int_{c-\mathbf{i}\infty}^{c+\mathbf{i}\infty} \frac{x^s}{s^2} ds = 0$.

4.5 Meromorphic functions

4.5.1 The extended complex plane $\overline{\mathbb{C}}$

Definition 4.164. The *extended complex plane* is

$$\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}, \quad (4.78)$$

where $\infty = \frac{1}{0}$, $\frac{1}{\infty} = 0$.

Definition 4.165. A set $\mathcal{D} \subseteq \overline{\mathbb{C}}$ is *open in $\overline{\mathbb{C}}$* iff $\mathcal{D} \cap \mathbb{C}$ is open in $\mathbb{C}$ and

$$\infty \in \mathcal{D} \implies \exists R > 0 \text{ s.t. } \{z \in \mathbb{C} : |z| > R\} \subset \mathcal{D}. \quad (4.79)$$

Lemma 4.166. A set $\mathcal{D} \subseteq \overline{\mathbb{C}}$ is open if and only if the set $\{z \in \overline{\mathbb{C}} : z^{-1} \in \mathcal{D}\}$ is open in $\overline{\mathbb{C}}$.

Exercise 4.167. Prove Lemma 4.166.

Definition 4.168. An open set $\mathcal{D} \subseteq \overline{\mathbb{C}}$ is a *domain* in $\overline{\mathbb{C}}$ iff the intersection $\mathcal{D} \cap \mathbb{C}$ is a domain in $\mathbb{C}$.

Definition 4.169 (Meromorphic functions in $\mathbb{C}$). A function $f : \mathcal{D} \rightarrow \overline{\mathbb{C}}$ with $\mathcal{D} \subseteq \mathbb{C}$ open is *meromorphic* iff

- (a) the set $S(f) = f^{-1}(\{\infty\})$ is discrete in $\mathcal{D}$,
- (b) $f_0 : \mathcal{D} \setminus S(f) \rightarrow \mathbb{C}$, the restriction of f , is analytic,
- (c) all points in $S(f)$ are poles of f_0 .

Definition 4.170 (Meromorphic functions in $\overline{\mathbb{C}}$). A function $f : \mathcal{D} \rightarrow \overline{\mathbb{C}}$ with $\mathcal{D} \subseteq \overline{\mathbb{C}}$ open is *meromorphic* iff

- (a) $f : \mathcal{D} \cap \mathbb{C} \rightarrow \overline{\mathbb{C}}$ is meromorphic,
- (b) $\hat{f}(z) := f(1/z)$ is meromorphic in the open set

$$\hat{\mathcal{D}} := \{z \in \mathbb{C} : z^{-1} \in \mathcal{D}\}. \quad (4.80)$$

Example 4.171. For polynomials P and Q , the rational function $R(z) = \frac{P(z)}{Q(z)}$ is meromorphic both in $\mathbb{C}$ and $\overline{\mathbb{C}}$.

Example 4.172. The function $f(z) = \cot(\pi z) = \frac{\cos(\pi z)}{\sin(\pi z)}$ is meromorphic in $\mathbb{C}$. Indeed, $\sin(\pi z) = 0$ is equivalent to $z \in \mathbb{Z}$. Since $\cos(\pi z) = \pm 1$ for any $z \in \mathbb{Z}$, the set of singularities of f is $\mathbb{Z}$, which is discrete in $\mathbb{C}$. Conditions (b) and (c) in Definition 4.169 are also readily verified.

Example 4.173 ($f(z) = \cot(\pi z)$ is not meromorphic in $\overline{\mathbb{C}}$). The singularity set of $\hat{f}(z) = f(1/z) = \cos(\frac{\pi}{z})/\sin(\frac{\pi}{z})$ is

$$S(\hat{f}) = \{0\} \cup \left\{ \frac{1}{n} : n \in \mathbb{Z} \setminus \{0\} \right\}.$$

Then $\mathcal{D} = \overline{\mathbb{C}}$ in Definition 4.170 implies $\hat{\mathcal{D}} = \mathbb{C}$, in which $\hat{f}$ is not meromorphic in the sense of Definition 4.169 because condition (a) does not hold: $S(\hat{f})$ is not discrete in $\mathbb{C}$.

4.5.2 Algebraic structures

Notation 6. The set of analytic functions on an open set $\Omega \subset \mathbb{C}$, the set of analytic functions on a domain $\mathcal{D} \subset \mathbb{C}$, and the set of meromorphic functions on a domain $\mathcal{D} \subset \overline{\mathbb{C}}$ are denoted by $H(\Omega)$, $\mathcal{O}(\mathcal{D})$, and $\mathcal{M}(\mathcal{D})$, respectively.

Lemma 4.174. $H(\Omega)$ is a commutative ring with unity.

Proof. By Definition G.32, $H(\Omega)$ is clearly a ring with unity since the constant function $\mathbf{1}$ is the multiplication identity. The commutativity of the functions follows from that of complex numbers. $\square$

Exercise 4.175. If an open set Ω is not connected, then $H(\Omega)$ might have divisors of zero. Give one such example.

Theorem 4.176. $H(\mathcal{D})$ is an integral domain if and only if Ω is a domain, in which case $H(\mathcal{D}) = \mathcal{O}(\mathcal{D})$.

Proof. For a nonempty open set $\mathcal{D} \subseteq \mathbb{C}$, Lemma 4.174 states that $H(\mathcal{D})$ is a commutative ring with unity, where the unity $\mathbf{1}$ is different from the additive identity $\mathbf{0}$.

The sufficiency follows from Definition G.40 and the fact of $H(\mathcal{D})$ having no divisors of zero, cf. Definition G.34 and Lemma 4.36.

For necessity, the only difference between an integral domain and a commutative ring with unity $\mathbf{1} \neq \mathbf{0}$ is that the integral domain contains no divisors of zero. Suppose $\mathcal{D}$ in Notation 6 is not connected. Then Exercise 4.175 leads to a contradiction, which completes the proof. $\square$

Example 4.177. The unit disk $\mathbb{E}$ is a domain, but the analytic function $f(z) = z$ is zero at $z = 0$ and thus does not have a multiplicative inverse in $\mathcal{O}(\mathbb{E})$.

Theorem 4.178. $\mathcal{M}(\mathcal{D})$ is a field.

Proof. Let $f, g \in \mathcal{M}(\mathcal{D})$ be two meromorphic functions with S, T as their pole sets, respectively. The function $f + g$ is analytic in the domain $\mathcal{D} \setminus (S \cup T)$. Since the addition never creates essential singularities and may reduce poles to removable singularities, $S \cup T$ contains only non-essential singularities. Then Definitions 4.86 and 4.169 yield $f + g \in \mathcal{M}(\mathcal{D})$. Similar arguments yield $fg \in \mathcal{M}(\mathcal{D})$. Hence $\mathcal{M}(\mathcal{D})$ is closed w.r.t. $+$ and $\cdot$.

It is trivial to verify the commutativity, the associativity, and the distributivity in Definition C.1. The identity holds from the functions of constant zero and constant one as the additive and multiplicative identities, respectively.

By Definition 4.169, $f \in \mathcal{M}(\mathcal{D})$ implies f being analytic on $\mathcal{D} \setminus S(f)$. By Theorem 4.27, the zero set of f is discrete and Definition 4.169 implies $\frac{1}{f} \in \mathcal{M}(\mathcal{D})$. Hence the invertibility holds and the proof is completed. $\square$

Corollary 4.179. $\mathcal{O}(\mathcal{D})$ is a subring of $\mathcal{M}(\mathcal{D})$.

Proof. This follows directly from Lemma G.45 and Theorems G.47, 4.176 and 4.178. $\square$

4.5.3 Classifying the singularity at ∞

Definition 4.180. For an analytic function $f : \mathcal{D} \setminus \{\infty\} \rightarrow \mathbb{C}$ where $\mathcal{D} \subseteq \overline{\mathbb{C}}$ is an open set that contains ∞ , the singularity of f at ∞ is said to be

- *removable*,
- *a pole of order $m \in \mathbb{N}$* ,
- *essential*,

iff the function $\hat{f} : \hat{\mathcal{D}} \setminus \{0\} \rightarrow \mathbb{C}$ has a corresponding singularity at 0, where $\hat{\mathcal{D}}$ is defined in (4.80).

Example 4.181. Consider a polynomial $p(z) = \sum_{j=0}^n a_j z^j$ with $n \in \mathbb{N}^+$ and $a_n \neq 0$. This entire function $p : \mathbb{C} \rightarrow \mathbb{C}$ has a pole of order n at ∞ because the analytic function $\hat{p} : \mathbb{C}^\bullet \rightarrow \mathbb{C}$,

$$\hat{p}(z) = p(z^{-1}) = \frac{a_n}{z^n} + \frac{a_{n-1}}{z^{n-1}} + \cdots + a_0, \quad (4.81)$$

has a pole of order n at 0.

Example 4.182. The entire function

$$f(z) = \exp(z) = \sum_{n=0}^{+\infty} \frac{z^n}{n!}$$

has an essential singularity at ∞ because Theorem 4.126 implies that the analytic function $\hat{f} : \mathbb{C}^\bullet \rightarrow \mathbb{C}$,

$$\hat{f}(z) = f(z^{-1}) = \sum_{n=0}^{+\infty} \frac{1}{n! z^n}, \quad (4.82)$$

has an essential singularity at 0.

Definition 4.183. An *entire rational function* is an entire function with a non-essential singularity at ∞ while an *entire transcendental function* is an entire function with an essential singularity at ∞ .

Theorem 4.184. An entire function $f : \mathbb{C} \rightarrow \mathbb{C}$ has a removable singularity at ∞ if and only if it is a constant; f has a pole at ∞ if and only if it is a nonconstant polynomial.

Proof. The sufficiency follows directly from Definition 4.180. For the necessity, f being entire and Theorem 4.1 imply

$$\forall z \in \mathbb{C}, f(z) = \sum_{j=0}^{+\infty} a_j z^j, \quad (4.83)$$

which gives

$$\forall z \in \mathbb{C}^\bullet, \hat{f}(z) = f(1/z) = \sum_{j=0}^{+\infty} a_j z^{-j}. \quad (4.84)$$

By the uniqueness of Laurent series in Corollary 4.125, the series in (4.84) is also the Laurent expansion of $\hat{f}$ at $z = 0$. If the singularity of f at ∞ is a pole, Definition 4.180 and Theorem 4.126 yield

$$\exists k \in \mathbb{N}^+, \text{ s.t. } \forall j > k, a_j = 0.$$

Then (4.83) dictates that $f(z)$ be a polynomial in z . Similarly, if $\hat{f}(z)$ has a removable singularity at 0, (4.84) and Definition 4.180 yield $\forall j > 0, a_j = 0$, i.e., f is a constant. $\square$

Theorem 4.185. A function $f : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ is meromorphic if and only if it is rational, i.e. of the form $f(z) = \frac{P(z)}{Q(z)}$ where P and Q are complex polynomials.

Proof. WLOG, assume P and Q have no common zeros. Hence the zeros of Q are poles of f . If the degree of P is greater than that of Q , ∞ is a pole of f ; otherwise ∞ is a removable singularity of f . Then the sufficiency follows from Definition 4.170.

As for necessity, f being meromorphic in $\overline{\mathbb{C}}$ and Definitions 4.169 and 4.170 imply that $\hat{f}$ has either a removable or a pole at 0. In both cases, $\hat{f}$ is analytic in $\mathcal{U}_r(0)$ for some $r > 0$. By Theorem 4.126 and the identity theorem 4.35, f can have only a finite number of poles in $\mathbb{C}$, say $z_1, z_2, \dots, z_n$. Let $m_1, m_2, \dots, m_n$ be orders of these poles. The function

$$F(z) := (z - z_1)^{m_1} \cdots (z - z_n)^{m_n} f(z)$$

has only removable singularities in $\mathbb{C}$ and is thus entire. Also, f is rational if and only if F is. F cannot have an essential singularity at ∞ : the Laurent series of f at ∞ has only finitely many positive powers, so does that of F at ∞ . Then Theorem 4.184 dictates that F be either a constant or a polynomial, which completes the proof. $\square$

Exercise 4.186. Give an alternative proof of the necessity part of Theorem 4.185 using Laurent expansions and Liouville's theorem 3.94.

Corollary 4.187. Any analytic function $f : \overline{\mathbb{C}} \rightarrow \mathbb{C}$ is constant.

Proof. f can be considered as a meromorphic function $\overline{\mathbb{C}} \mapsto \overline{\mathbb{C}}$ and by Theorem 4.185 it must be rational. Since ∞ is not in the codomain, f has no poles in $\mathbb{C}$, hence f must be a polynomial. Since the singularity of $f|_{\mathbb{C}}$ at ∞ is not a pole, the polynomial must be constant. $\square$

Exercise 4.188. Let f be a monic polynomial of degree d , and $M(f) := \max_{z \in \partial \mathbb{D}} |f(z)|$. Show that

- (a) $\forall |z| > 1, |f(z)| \leq |z|^d M(f)$.
- (b) $M(f) \geq 1$; the equality holds if and only if $f(z) = z^d$.

4.5.4 $\text{Aut}(\mathbb{C})$: the automorphism group of $\mathbb{C}$

Theorem 4.189. An entire function $f : \mathbb{C} \rightarrow \mathbb{C}$ is globally conformal if and only if

$$\exists a \in \mathbb{C}^\bullet, \exists b \in \mathbb{C} \text{ s.t. } \forall z \in \mathbb{C}, f(z) = az + b. \quad (4.85)$$

Proof. It is trivial to verify the sufficiency. As for the necessity, f being entire and bijective yields $\lim_{z \rightarrow \infty} f(z) = \infty$, which implies that the singularity of f at ∞ is a pole. By Theorem 4.184, f must be a nonconstant polynomial. Furthermore, f must be a polynomial of degree 1; otherwise the fundamental theorem of abstract algebra would lead to a contradiction to f being a bijection. $\square$

Corollary 4.190. A globally conformal map $f : \mathbb{C} \rightarrow \mathbb{C}$ must be of the form $f(z) = az + b$, where $a \neq 0$ and $b \in \mathbb{C}$ are constants depending on f . In other words, we have

$$\text{Aut}(\mathbb{C}) = \{z \mapsto az + b : a \in \mathbb{C}^\bullet, b \in \mathbb{C}\}. \quad (4.86)$$

Proof. This follows directly from Theorem 4.189. $\square$

Exercise 4.191. Show that the singularity of f is a pole in the proof of Theorem 4.189 by applying the Casorati-Weierstrass theorem 4.108 and the open mapping theorem 4.41 to $f(\frac{1}{z})$.

Exercise 4.192. Find all entire functions f satisfying $f(f(z)) = z$ for all $z \in \mathbb{C}$.

4.6 Möbius transformations

Example 4.193. An event in this universe can be marked by a 4-vector (t, x, y, z) in the 4-dimensional space-time. The spatial components (x, y, z) have no absolute significance in that rotating the coordinate axes gives different coordinates $(\tilde{x}, \tilde{y}, \tilde{z})$ for the same point. Nonetheless, these two sets of coordinates satisfy

$$x^2 + y^2 + z^2 = \tilde{x}^2 + \tilde{y}^2 + \tilde{z}^2.$$

In comparison, we often take it for granted that the time component t does have an absolute significance. Confirmed by innumerable experiments, Einstein's theory of relativity tells us this notion is wrong. If two momentarily coincident observers are in relative motion, they will disagree about the times at which events occur, in which case they will also disagree about the value of $x^2 + y^2 + z^2$; this is the famous *Lorentz contraction*. As a key question, is there *any* aspect of space-time that has absolute significance and on which two observers in relation motion must agree? The answer from Einstein is yes. By choosing a scaling with the speed of light being one, he discovered the invariant

$$t^2 - x^2 + y^2 + z^2 = \tilde{t}^2 - \tilde{x}^2 - \tilde{y}^2 - \tilde{z}^2.$$

The *Lorentz transformation* is a linear transformation of space-time that maps one observer's description (t, x, y, z) to another. In other words, the Lorentz transformation preserves the above invariant on which both observers agree.

As a miraculous discovery, each Lorentz transformation induces a mapping of the complex plane, which is a Möbius transformation! Conversely, every Möbius transformation yields a unique Lorentz transformation of space-time.

See the first chapter of the book by Penrose and Rindler [1984] for more details.

Definition 4.194. A *Möbius transformation* or a *linear fractional transformation* is a meromorphic function $f : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ of the form

$$f(z) = \frac{az + b}{cz + d} \quad (4.87)$$

where the constants $a, b, c, d \in \mathbb{C}$ are called the *coefficients of the Möbius transformation*.

Definition 4.195. A Möbius transformation in (4.87) is *singular* if its coefficients satisfy $ad - bc = 0$; otherwise it is *non-singular*. By default, a Möbius transformation refers to a non-singular one unless explicitly stated otherwise.

Lemma 4.196. A singular Möbius transformation is the constant map $M(z) = \frac{a}{c}$.

Proof. This follows from Definitions 4.194 and 4.195. $\square$

Definition 4.197. A Möbius transformation in (4.87) is *normalized* if its coefficients a, b, c, d satisfy $ad - bc = 1$.

Exercise 4.198. Explain how to normalize any nonsingular Möbius transformation M .

4.6.1 The Möbius group $\mathfrak{M}$

Theorem 4.199. A Möbius transformation is a continuous bijection of which the inverse is also a Möbius transformation.

Proof. The inverse of a Möbius transformation f in (4.87) is

$$g(z) = \frac{-dz + b}{cz - a} \quad (4.88)$$

because $f(g(z)) = z$ and $g(f(z)) = z$ hold for any $z \in \overline{\mathbb{C}}$. $\square$

Lemma 4.200. If f, g are two Möbius transformations, then so is $g \circ f$.

Proof. Define $g = \frac{Az+B}{Cz+D}$. Then for f in (4.87) we have

$$g(f(z)) = \frac{(Aa+Bc)z + (Ab+Bd)}{(Ca+Dc)z + (Cb+Dd)}, \quad (4.89)$$

where

$$\begin{aligned} & (Aa+Bc)(bC+Dd) - (Ab+Bd)(aC+cD) \\ &= (AD-BC)(ad-bc) \neq 0. \end{aligned} \quad \square$$

Theorem 4.201. The set of all nonsingular Möbius transformations, denoted by $\mathfrak{M}$, is a group with its binary operation as function composition.

Proof. The closure follows from Lemma 4.200, the associativity is obvious, the identity is given by setting $b = c = 0$ and $a = d \neq 0$ in (4.87), and the inverse element follows from Theorem 4.199. $\square$

4.6.2 $\text{Aut}(\overline{\mathbb{C}})$: the automorphism group of $\overline{\mathbb{C}}$

Theorem 4.202. A meromorphic function $f : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ is globally conformal if and only if it is a Möbius transformation.

Proof. By Theorem 4.185, the meromorphic function f must be rational, i.e., $f(z) = \frac{P(z)}{Q(z)}$ where P and Q are complex polynomials. Since f is injective, the equation $f(z) = w$, i.e., $P(z) - wQ(z) = 0$, can have only one root, hence the maximum degree is 1 for both P and Q .

For the sufficiency, $f(z) = \frac{az+b}{cz+d}$ implies $f'(z) = \frac{ad-bc}{(cz+d)^2}$. Hence $ad-bc \neq 0$ gives $f'(z) \neq 0$. By Definition 4.180, the singularity of f at ∞ is removable because $\hat{f}(z) = f(1/z) = \frac{a+bz}{c+dz}$ has a removable singularity at 0. Hence f is analytic in $\overline{\mathbb{C}}$ and the proof is completed by Theorem 4.199. $\square$

Theorem 4.203. $\text{Aut}(\overline{\mathbb{C}}) = \mathfrak{M}$.

Proof. This follows from Theorems 4.201 and 4.202. $\square$

4.6.3 The homomorphism of $\text{GL}_2(\mathbb{C})$ onto $\mathfrak{M}$

Definition 4.204. The coefficient matrix of a Möbius transformation f in (4.87) is

$$[f] := \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{C}^{2 \times 2}. \quad (4.90)$$

Example 4.205. The coefficient matrix of the identity Möbius transformation $f(z) = z$ is the identity matrix.

Exercise 4.206. Prove (i) $[f^{-1}(z)] = (bc-ad)[f]^{-1}$ where f is a Möbius transformation in (4.87) and (ii) $[f_2][f_1] = [f_2 \circ f_1]$ for two nonsingular Möbius transformations $f_1(z), f_2(z)$.

Definition 4.207. The general linear group of degree n over a field $\mathcal{F}$, written $\text{GL}_n(\mathcal{F})$, is the set of n -by- n invertible matrices with its binary operation as matrix multiplication. The special linear group, written $\text{SL}_n(\mathcal{F})$, is the subgroup of $\text{GL}_n(\mathcal{F})$ consisting of matrices with determinant 1.

Lemma 4.208. The map $\psi : \text{GL}_2(\mathbb{C}) \rightarrow \mathfrak{M}$ defined by

$$\psi([f]) := f \quad \text{or} \quad \psi\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \left(z \mapsto \frac{az+b}{cz+d}\right) \quad (4.91)$$

is a group homomorphism of $\text{GL}_2(\mathbb{C})$ onto $\mathfrak{M}$ with

$$\ker \psi = \{cI : c \in \mathbb{C}\}, \quad (4.92)$$

where I is the 2-by-2 identity matrix.

Proof. ψ is onto because of (4.91), Definitions 4.194 and 4.207, and the fact that the determinant of a 2-by-2 matrix being nonzero implies invertibility. Then (4.89) yields

$$\psi([g][f]) = g \circ f,$$

therefore ψ is indeed a homomorphism onto $\mathfrak{M}$. The identity of $\mathfrak{M}$ is clearly the identity map with $c = b = 0, a = d$. By Definition G.17, the kernel of ψ is $\{cI : c \in \mathbb{C}\}$. $\square$

Exercise 4.209. Prove Lemma 4.200 using Lemma 4.208. *Hint:* Use $\det([g][f]) = \det[g]\det[f]$.

4.6.4 The Riemann sphere

Definition 4.210. The Riemann sphere is

$$S^2 := \{(w, t) \in \mathbb{C} \times \mathbb{R} : |w|^2 + t^2 = 1\}. \quad (4.93)$$

Definition 4.211. The stereographic projection is a map $\sigma : S^2 \rightarrow \overline{\mathbb{C}}$ given by

$$\sigma(w, t) = \begin{cases} \frac{w}{1-t} & \text{if } (w, t) \neq N; \\ \infty & \text{if } (w, t) = N; \end{cases} \quad (4.94)$$

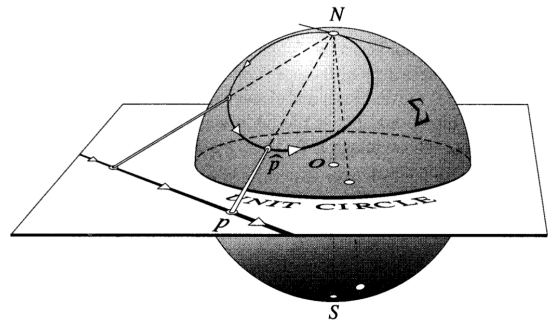
where $N := (0, 1)$ is the north pole of the Riemann sphere.

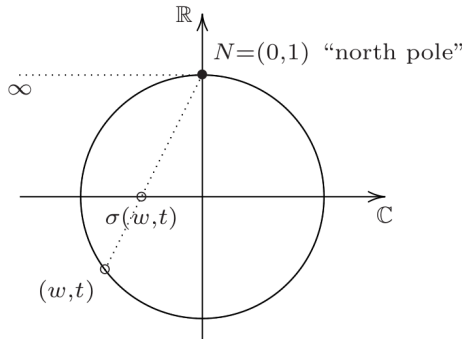
Lemma 4.212. The stereographic projection is a homeomorphism between the Riemann sphere and the extended complex plane.

Proof. It can be readily verified that σ is continuous and bijective. Furthermore, its inverse $\sigma^{-1} : \mathbb{C} \rightarrow S^2$ is given by

$$\sigma^{-1}(z) = \begin{cases} \left(\frac{2z}{1+|z|^2}, -\frac{1-|z|^2}{1+|z|^2}\right) & \text{if } z \neq \infty; \\ N & \text{if } z = \infty \end{cases} \quad (4.95)$$

and is continuous. $\square$





Definition 4.213. Two points $u, v \in \mathcal{S}^2$ are *antipodal* iff they are diametrically opposite each other on the Riemann sphere, i.e., $u + v = \mathbf{0}$.

Lemma 4.214. Two points $u, v \in \mathcal{S}^2$ are antipodal if and only if their images $p := \sigma(u)$ and $q := \sigma(v)$ under the stereographic projection are related as $p = -\frac{1}{\bar{q}}$.

Proof. By Definitions 4.211 we have

$$p = \sigma(w, t) = \frac{w}{1-t}, \quad q = \sigma(-w, -t) = \frac{-w}{1+t}$$

and the necessity follows from (4.93). $\square$

Exercise 4.215. Prove the sufficiency part of Lemma 4.214.

4.6.5 Isometries of spherical geometry

Definition 4.216. The *homogeneous coordinates*, or *projective coordinates*, of a complex number $z \in \mathbb{C}$ is an ordered pair $(z_1, z_2) \in \mathbb{C}^2$ satisfying $z = \frac{z_1}{z_2}$.

Lemma 4.217. Two vectors in $\mathbb{C}^2$ are orthogonal if and only if they are the homogeneous coordinates of antipodal points on the Riemann sphere.

Proof. Let $p = (z_1, z_2)^\top$ and $q = (w_1, w_2)^\top$ be the homogeneous coordinates of z_p and w_q , respectively. Then Definitions C.194 and C.174 yield

$$0 = \langle p, q \rangle = z_1 \bar{w}_1 + z_2 \bar{w}_2 \iff \frac{z_1}{z_2} \frac{\bar{w}_1}{\bar{w}_2} = -1 \iff z_p \bar{w}_q = -1.$$

The proof is completed by Lemma 4.214. $\square$

Theorem 4.218 (Gauss 1819). All rotations of the Riemann sphere form a group

$$\text{PSU}_2 := \left\{ f(z) = \frac{az+b}{-\bar{b}z+\bar{a}} : a, b \in \mathbb{C}; |a|^2 + |b|^2 = 1 \right\}, \quad (4.96)$$

which is a subgroup of the Möbius group $\mathfrak{M}$.

Proof. A rigid spatial rotation of the sphere must preserve distances by keeping antipodal points exactly opposite each other, which, by Lemma 4.217, is equivalent to preserving vector orthogonality in $\mathbb{C}^2$. Hence the matrix U of the rigid rotation must be unitary and in $\text{SL}_2(\mathbb{C})$. Definition C.215 states $U^*U = I$, which gives

$$U = \begin{bmatrix} a & b \\ -\bar{b} & \bar{a} \end{bmatrix} \text{ where } a, b \in \mathbb{C} \text{ and } |a|^2 + |b|^2 = 1.$$

Then (4.96) follows from Lemma 4.208. The proof is completed by checking the conditions in Definition G.16. $\square$

Lemma 4.219. The fixed point of a normalized Möbius transformation $M(z)$ can be expressed as

$$\xi_{\pm} = \frac{(a-d) \pm \sqrt{(a+d)^2 - 4}}{2c}.$$

Proof. Solve the equation $z = M(z) = \frac{az+b}{cz+d}$. $\square$

Exercise 4.220. For a Möbius transformation $M(z)$ with two fixed points $p \neq q$, show that if $M(z_0) = \infty$ and $M(\infty) = w_0$, then

$$(a) \quad z_0 + w_0 = p + q;$$

$$(b) \quad M(z) = \frac{w_0 z - p q}{z - z_0}.$$

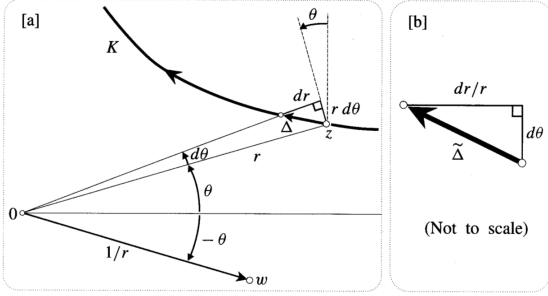
Lemma 4.221. $z = \frac{z_1}{z_2} \in \mathbb{C}$ is a fixed point of a Möbius transformation M if and only if $[z_1, z_2]^\top$ is an eigenvector of the coefficient matrix $[M]$.

Exercise 4.222. Prove Lemma 4.221.

Chapter 5

Argument Principles: from Winding Numbers to Topological Degrees

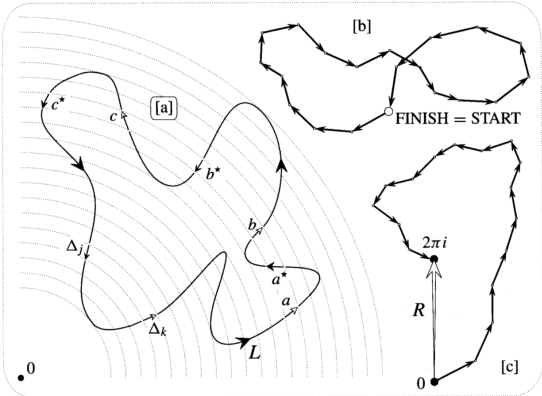
Example 5.1 (Geometry of line integrals of complex inversion). Consider $\int_{\beta} \frac{1}{z} dz$ where β is an arbitrary closed curve.



From the plot above, we have

$$\Delta = rd\theta e^{i(\theta + \frac{\pi}{2})} + dr e^{i(\theta + \pi)}, \quad w = \frac{1}{r} e^{-i\theta},$$

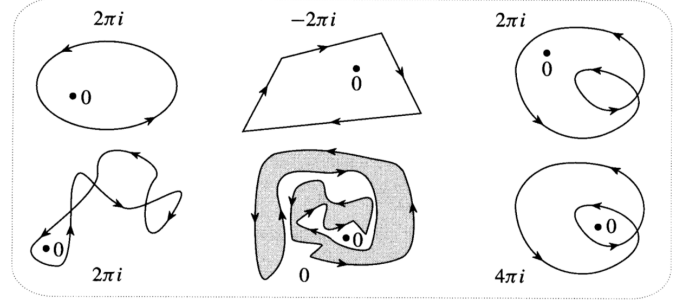
where (r, θ) are the polar coordinates of a point on the curve. Therefore we have $w\Delta = -\frac{dr}{r} + i d\theta$. As shown in subplot [a] below, for a simple closed curve β , each Δ_j can be paired up with another Δ_k such that the values $\frac{dr}{r}$ of $w_j\Delta_j$ and $w_k\Delta_k$ have the same magnitude but the different sign. Therefore, the real part of the integral $\int_{\beta} \frac{1}{z} dz$ vanishes.



The starting point and the ending point of a closed curve have the same value of r . In contrast, it is a different story for θ . If β does not enclose the origin, then the starting point and the ending point have the same value of θ and the contour integral reduces to 0, as shown in subplot [b]. However, if β does enclose the origin, we have $\int_{\beta} \frac{1}{z} dz = 2\pi i$,

with 2π being the difference of the polar coordinate θ for the ending and starting points.

Finally, for an integral on an arbitrary closed curve, the Seifert decomposition (Lemma 3.70) implies that the integral can be expressed as the sum of integrals on a number of simple closed curves, as illustrated below.



5.1 Winding numbers

Definition 5.2. The *winding number* of an oriented closed curve $\Gamma \subset \mathbb{C}$ around a point $z \in \mathbb{C} \setminus \Gamma$ is

$$w(\Gamma, z) := \frac{1}{2\pi i} \int_{\Gamma} \frac{d\gamma}{\gamma - z}, \quad (5.1)$$

where γ is a parametrization of Γ .

Lemma 5.3. Any winding number in (5.1) is an integer.

Proof. Define $\phi(t) := \exp \left(\int_0^t \frac{\gamma'(s)}{\gamma(s) - z} ds \right)$ where $t \in [0, 1]$ and Γ is parameterized as $\gamma : [0, 1] \rightarrow \mathbb{C}$. We have

$$\begin{aligned} \phi'(t) &= \frac{\gamma'(t)\phi(t)}{\gamma(t) - z} \Rightarrow \left[\frac{\phi(t)}{\gamma(t) - z} \right]' = \frac{\phi'(\gamma - z) - \phi\gamma'}{(\gamma - z)^2} = 0 \\ \Rightarrow \frac{\phi(1)}{\gamma(1) - z} &= \frac{\phi(0)}{\gamma(0) - z} \Rightarrow \phi(1) = \frac{\phi(0)(\gamma(1) - z)}{\gamma(0) - z} = 1, \end{aligned}$$

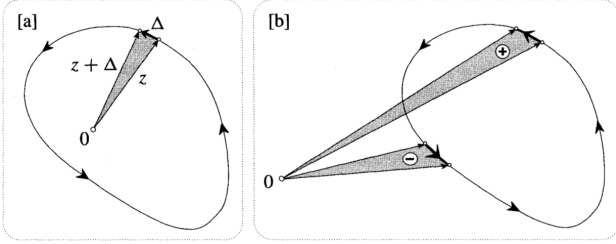
where the last equality follows from $\phi(0) = 1$ and the closeness $\gamma(0) = \gamma(1)$. The rest of the proof follows from

$$\phi(1) = \exp(w(\Gamma, z)2\pi i) = 1 \Rightarrow w(\Gamma, z) \in \mathbb{Z}. \quad \square$$

Lemma 5.4. The winding number (5.1) is a constant in each connected component of $\mathbb{C} \setminus \Gamma$. In particular, we have $w(\Gamma, z) = 0$ as $z \rightarrow \infty$, i.e., the constant is 0 in the unbounded component.

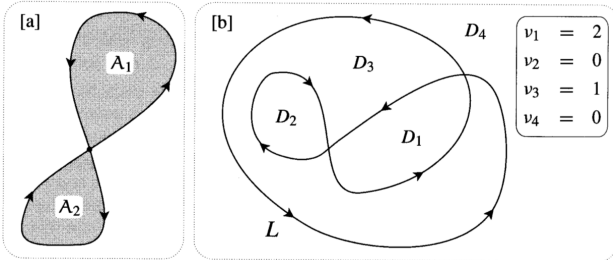
Proof. $w(\Gamma, z)$ is analytic with respect to z , and thus w is continuous in terms of z . Any two points in a component of $\mathbb{C} \setminus \Gamma$ can be connected by a path inside the component, hence Lemma 5.3 dictates that $w(\Gamma, z)$ be a constant for that component. As $z \rightarrow \infty$, the modulus of the integral in (5.1) can be made arbitrary small and thus by Lemma 5.3 it must be zero. $\square$

Example 5.5 (Geometry of the line integral of complex conjugate). Consider $\int_{\alpha} \bar{z} dz$ for a simple closed curve α .



Since $\frac{1}{2} \text{Im}(\bar{z}\Delta)$ is the area of the triangle shown in subplot [a], $\frac{1}{2} \text{Im}(\int_{\alpha} \bar{z} dz)$ equals the signed area of the simple closed region inside α . Furthermore, $\text{Arg}(\bar{z}) = \text{Arg}(z^{-1})$. By arguments similar to those in Example 5.1, we deduce that $\text{Re}(\int_{\alpha} \bar{z} dz) = 0$. These arguments also hold when the origin is outside of α , as shown in subplot [b]. Therefore we conclude that $\int_{\alpha} \bar{z} dz$ equals $2i$ times the signed area of the simple closed region bounded by α .

This result easily generalizes to arbitrary closed curves. As shown in subplot [a] below, it is possible that $\int_{\alpha} \bar{z} dz$ vanishes if $\mathcal{A}_1 = -\mathcal{A}_2$.



In more general cases such as that in subplot [b], we have

$$\int_{\alpha} \bar{z} dz = 2i \sum_j \nu_j \mathcal{A}_j, \quad (5.2)$$

where $\mathcal{A}_j$ is the signed area of the j th complement D_j of α and ν_j is the winding number of L around a point in D_j , cf. Lemma 5.4.

Lemma 5.6. The winding number of a positively oriented circle Γ with its center at a and its radius as r is

$$w(\Gamma, z) = \begin{cases} 1 & \text{if } |z - a| < r; \\ 0 & \text{if } |z - a| > r. \end{cases} \quad (5.3)$$

Proof. Parametrize Γ as $\gamma(t) = a + re^{it}$ with $t \in [0, 2\pi]$. By (5.1), we calculate $w(\Gamma, a) = 1$ and deduce the first clause from Lemma 5.4, which also implies the second clause. $\square$

Exercise 5.7. Denote $\mathbb{S}^1 = \{e^{it} : t \in [0, 2\pi]\}$ and consider a closed curve Γ as the image of $\mathbb{S}^1$ under the function

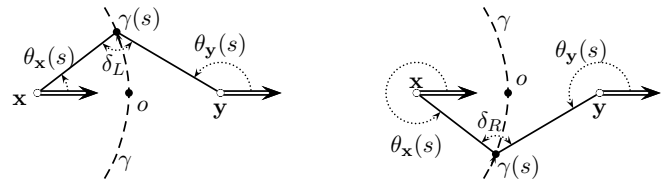
$$\gamma(t) = z + r(t)e^{i\theta(t)}, \quad (5.4)$$

where θ is the angle of $\gamma(t)$ in the polar coordinate system whose pole is at z . Show that $w(\Gamma, z) = \frac{1}{2\pi}(\theta(2\pi) - \theta(0))$. According to this result, what is the meaning of the winding number $w(\Gamma, z)$?

Definition 5.8 (Alexander [1928]). Let γ be a closed curve and $p(t)$ a particle moving continuously in the plane. Suppose p crosses γ at time t_c . A point $\gamma(s_c) = p(t_c)$ is called a *positive crossing* if, in a sufficiently small open neighborhood of s_c , γ is oriented from the left of the directed pathline of p to its right, and a *negative crossing* otherwise.

Lemma 5.9. Let Γ be a closed curve parametrized and oriented by $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$. Then a positive/negative crossing of a moving particle p increases/decreases $w(\Gamma, p(t))$ by 1, respectively.

Proof. We only prove the case of a negative crossing since the other case is symmetric.



Referring to the above figure, let $o = p(t_0) \in \gamma$ be the crossing point and denote $\mathbf{x} = p(t_0 - \tau)$, $\mathbf{y} = p(t_0 + \tau)$. As $\tau \rightarrow 0$, the direction of the ray $\mathbf{x} \rightarrow \mathbf{y}$ converges to that of $\mathbf{u}(o, t_0)$, which splits the plane into two halfplanes: $\theta_y(s) = \theta_x(s) + \delta_L$ in the left one and $\theta_y(s) = \theta_x(s) - \delta_R$ in the right one. Choose the basepoint of γ as $o = \gamma(0)$, $\gamma(2\pi) = o$, then $\lim_{s \rightarrow 0} \delta_L = \lim_{s \rightarrow 2\pi} \delta_R = \pi$ for a single traversal of γ . Finally, Exercise 5.7 yields

$$\begin{aligned} w_{\gamma}(\mathbf{y}) - w_{\gamma}(\mathbf{x}) &= \frac{1}{2\pi}(\theta_y(2\pi) - \theta_x(2\pi) + \theta_x(0) - \theta_y(0)) \\ &= \frac{1}{2\pi}(-\pi - \pi) = -1. \end{aligned} \quad \square$$

Lemma 5.10. If a particle crosses an oriented closed curve a finite number of times as it moves continuously from $\mathbf{x}$ to infinity, the winding number $w_{\gamma}(\mathbf{x})$ is equal to the number of negative crossings minus the number of positive crossings.

Exercise 5.11. Prove Lemma 5.10 by induction.

Theorem 5.12. Suppose the cycle decomposition of an oriented closed curve Γ as in Lemma 3.70 consists of a finite number of Jordan curves. Then for each $\mathbf{x} \notin \Gamma$, its winding number $w(\Gamma, \mathbf{x})$ is the number of counterclockwise Jordan curves containing $\mathbf{x}$ minus that of clockwise Jordan curves containing $\mathbf{x}$.

Exercise 5.13. Prove Theorem 5.12.

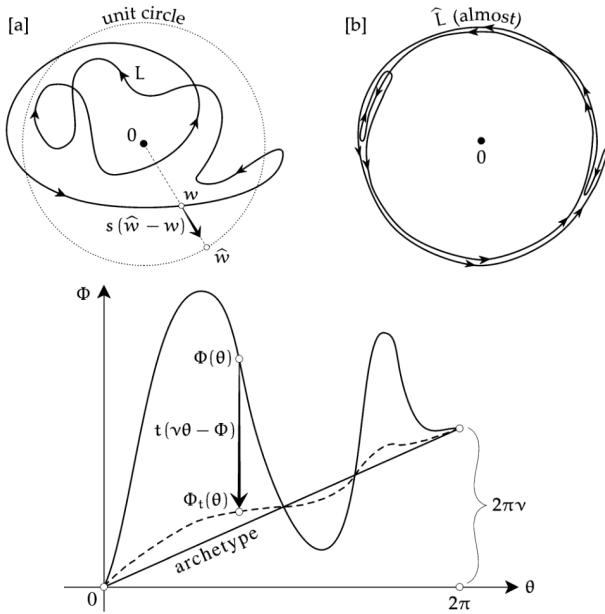
5.2 The Hopf degree theorem

Lemma 5.14. Any closed curve Γ that satisfies $0 \notin \Gamma$ and $w(\Gamma, 0) = \nu \neq 0$ can be continuously deformed into the archetypal loop $\hat{\gamma}_\nu = e^{i\nu\varphi}$ with $\varphi \in [0, 2\pi]$, and vice versa.

Proof. Parametrize Γ as $\gamma(\theta) = r(\theta)e^{i\Phi(\theta)}$ with $\theta \in [0, 2\pi]$ and define a normalized version of Γ as $\tilde{\gamma}(\theta) = e^{i\Phi(\theta)}$. Construct a free homotopy

$$H(s, t) := \begin{cases} (1 - 2t)\gamma(2\pi s) + 2t\tilde{\gamma}(2\pi s) & \text{if } t \in [0, \frac{1}{2}]; \\ \exp\{i\Phi_t(\theta)\} & \text{if } t \in [\frac{1}{2}, 1], \end{cases}$$

where $\Phi_t(\theta) := (2 - 2t)\Phi(2\pi s) + (2t - 1)2s\nu\pi$. Then Γ is continuously deformed to $\tilde{\gamma}([0, 2\pi])$ and then to $\hat{\gamma}_\nu([0, 2\pi])$, as shown below.



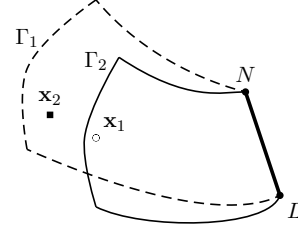
The free homotopy from the archetypal loop to Γ can be constructed similarly. $\square$

Exercise 5.15. For the proof of Lemma 5.14, give another free homotopy in which the modulus and the argument changes simultaneously from 0 to 1.

Theorem 5.16 (Hopf [1935]). For any given point $\mathbf{x} \in \mathbb{C}$, two closed curves Γ_1 and Γ_2 are freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}\}$ if and only if $w(\Gamma_1, \mathbf{x}) = w(\Gamma_2, \mathbf{x})$.

Proof. The sufficiency follows directly from Lemma 5.14 while the necessity from Lemma 5.4 and the free homotopy being a continuous function. $\square$

Example 5.17. Two loops Γ_1 (dashed lines) and Γ_2 (thin solid lines) share as their common arc $\overline{LN}$ (the thick line segment). In $\mathbb{C}$, they are of the same homotopy class $H(s, t) = t\Gamma_1(s) + (1 - t)\Gamma_2(s)$ with the fixed basepoint as $L = \Gamma_1(0) = \Gamma_1(1) = \Gamma_2(0) = \Gamma_2(1)$; they are also freely homotopic in $\mathbb{C}$.



Furthermore, they are freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}_1\}$ because of the Hopf theorem 5.16 and $w(\Gamma_1, \mathbf{x}_1) = w(\Gamma_2, \mathbf{x}_1)$. However, they are not freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}_2\}$ because $w(\Gamma_1, \mathbf{x}_2) \neq w(\Gamma_2, \mathbf{x}_2)$. In other words, there is no way Γ_1 can be continuously deformed to Γ_2 in $\mathbb{C} \setminus \{\mathbf{x}_2\}$, or, any continuous deformation of Γ_1 to Γ_2 in $\mathbb{C}$ must cross $\mathbf{x}_2$.

5.3 The argument principle

Notation 7. $\text{BCJ}(\Gamma)$ and $\text{UCJ}(\Gamma)$ denote the bounded and unbounded complements of a Jordan curve Γ , respectively.

Corollary 5.18. A positively oriented Jordan curve Γ has

$$w(\Gamma, p) = \begin{cases} 1 & \text{if } p \in \text{BCJ}(\Gamma); \\ 0 & \text{if } p \in \text{UCJ}(\Gamma). \end{cases} \quad (5.5)$$

Proof. This follows from Theorem 5.16 and Lemma 5.6. $\square$

Corollary 5.19. A negatively oriented Jordan curve Γ has

$$w(\Gamma, p) = \begin{cases} -1 & \text{if } p \in \text{BCJ}(\Gamma); \\ 0 & \text{if } p \in \text{UCJ}(\Gamma). \end{cases} \quad (5.6)$$

Proof. The winding number in Definition 5.2 depends on the orientation of Γ indicated by the parametrization γ . For the reciprocal curve Γ^- of Γ , we have $w(\Gamma^-, p) = -w(\Gamma, p)$. The rest follows from Corollary 5.18. $\square$

5.3.1 For polynomials

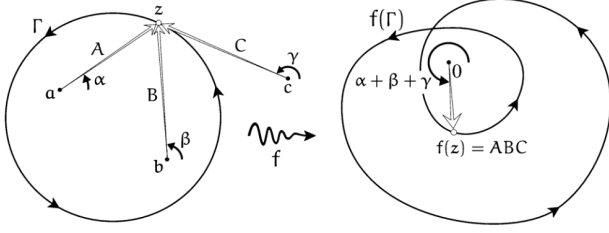
Lemma 5.20. Suppose a polynomial f has no roots on a positively oriented Jordan curve Γ . Then $w(f(\Gamma), 0)$ equals the number of roots of f in $\text{BCJ}(\Gamma)$, counted with their algebraic multiplicity.

Proof. Write $f(z) = \prod_{j=1}^{n_m} (z - z_j)^{m_j} \prod_{j=1}^{n_k} (z - y_j)^{k_j}$ where m_j and k_j are respectively the algebraic multiplicities of the roots z_j and y_j in $\text{BCJ}(\Gamma)$ and $\text{UCJ}(\Gamma)$. Definition 5.2 yields

$$\begin{aligned} w(f(\Gamma), 0) &= \frac{1}{2\pi i} \int_{f(\Gamma)} \frac{dz}{z} = \frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z) dz}{f(z)} \\ &= \frac{1}{2\pi i} \int_{\Gamma} \left(\sum_{j=1}^{n_m} \frac{m_j}{z - z_j} + \sum_{j=1}^{n_k} \frac{k_j}{z - y_j} \right) dz \\ &= \sum_{j=1}^{n_m} m_j w(\Gamma, z_j) + \sum_{j=1}^{n_k} k_j w(\Gamma, y_j) \\ &= \sum_{j=1}^{n_m} m_j, \end{aligned}$$

where the last step follows from Corollary 5.18. $\square$

Example 5.21. For $f(z) = (z - a)(z - b)(z - c)$, we have $w(f(\Gamma), 0) = 2$, as shown below.



Suppose instead we had $f(z) = (z-a)^2(z-b)(z-c)$. Then this multiplicity would be taken care of by $f'(z)$ so that $w(f(\Gamma), 0) = 3$.

Exercise 5.22 (Alternative proof of the fundamental theorem of algebra). Prove Theorem 3.49 via the argument principle in Lemma 5.20.

5.3.2 For analytic functions

Lemma 5.23. Let $f, g : \Omega \rightarrow \mathbb{C}$ be analytic functions where $\Omega \subset \mathbb{C}$ is a simply connected domain. Suppose $\Gamma \subset \Omega$ is an oriented closed curve such that $f(z)g(z) \neq 0$ for all $z \in \Gamma$. Then we have

$$w((fg)(\Gamma), 0) = w(f(\Gamma), 0) + w(g(\Gamma), 0). \quad (5.7)$$

Exercise 5.24. Prove Lemma 5.23.

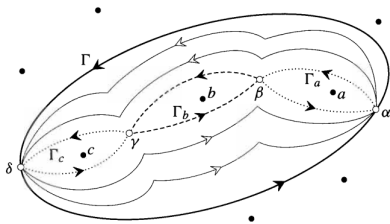
Theorem 5.25. Let Ω be a simply connected domain that contains a closed curve $\Gamma : [0, 1] \rightarrow \Omega$. For an analytic function $f : \Omega \rightarrow \mathbb{C}$ and a point $p \notin f(\Gamma)$, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} w(\Gamma, z_j) m_j, \quad (5.8)$$

where m_j is the algebraic multiplicity of the j th root z_j of the equation $f(x) = p$. In the particular case of Γ being a positively oriented Jordan curve, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p) \cap \text{BCJ}(\Gamma)} m_j. \quad (5.9)$$

Proof. Consider the case of Γ being a positively oriented Jordan curve. By the Hopf degree theorem 5.16 and the Seifert decomposition in Lemma 3.70, we can deform and decompose, without changing any $w(\Gamma, z_j)$, the Jordan curve Γ into a finite number of Jordan curves $\{\Gamma_j\}$ such that each Γ_j encircles at most one z_j and each z_j with $w(\Gamma, z_j) \neq 0$ is encircled by some Γ_j .



Then Definition 5.2 yields

$$(*) : \quad \begin{cases} w(\Gamma_j, z_j) &= w(\Gamma, z_j), \\ w(f(\Gamma), p) &= \sum_j w(f(\Gamma_j), p). \end{cases}$$

Since f is analytic on Ω , Lemma 4.20 dictates that, at a neighborhood of z_j ,

$$f(z) - p = K(z)g(z) \quad (5.10)$$

where $K(z) = \frac{f^{(m_j)}(z_j)}{m_j!} + \frac{f^{(m_j+1)}(z_j)}{(m_j+1)!}(z - z_j) + \dots$, $f^{(m_j)}(z_j) \neq 0$, and $g(z) = (z - z_j)^{m_j}$. It follows that

$$\begin{aligned} w(f(\Gamma_j), p) &= w(f(\Gamma_j) - p, 0) = w(K(\Gamma_j)g(\Gamma_j), 0) \\ &= w(K(\Gamma_j), 0) + w(g(\Gamma_j), 0) = w(g(\Gamma_j), 0) \\ &= m_j w(\Gamma_j, z_j) = m_j w(\Gamma, z_j) = m_j, \end{aligned}$$

where the first equality follows from Definition 5.2, the second from (5.10), the third from Lemma 5.23, the fourth from $f^{(m_j)}(z_j) \neq 0 \Rightarrow K(z_j) \neq 0$ and the Hopf theorem 5.16, the fifth from $g(z) = (z - z_j)^{m_j}$ and Lemma 5.20, the sixth from the first clause in (*), and the last from Corollary 5.18. Then the second clause in (*) implies (5.9).

Finally, (5.8) follows from (5.9) and Theorem 5.12. $\square$

Example 5.26. At a critical point z_c of an analytic function f , we have, from (5.10), $f(z) - f(z_c) = K(z)(z - z_c)^{m_c}$ where $m_c > 1$ and $K(z_c) \neq 0$. Hence f maps a positively oriented Jordan curve Γ around z_c to a closed curve $f(\Gamma)$ that winds m_c times around $f(z_c)$.

Lemma 5.27. For an analytic function $f : \Omega \rightarrow \mathbb{C}$ with $\Omega \subset \mathbb{C}$ open, the algebraic multiplicity of f at $a \in \Omega$ is 1 if and only if a is not a critical point of f .

Proof. Suppose $\det f'(a) \neq 0$. Then by the inverse function theorem f is a local homeomorphism: it maps an infinitesimal circle C_a centered at a to another infinitesimal circle centered at $p = f(a)$. Furthermore, Lemma 2.50 dictates that the two circles have the same orientation. Hence (5.8) yields $m_j(a) = 1$.

Suppose $\det f'(a) = 0$. Then f can be expressed in the form of (5.10), where, by Lemma 5.20, the algebraic multiplicity m_j of f at a does not equal 1. $\square$

Definition 5.28. The *signum function* or the *sign function* is the function $\text{sgn} : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$\text{sgn}(x) = \begin{cases} +1 & \text{if } x > 0; \\ 0 & \text{if } x = 0; \\ -1 & \text{if } x < 0. \end{cases} \quad (5.11)$$

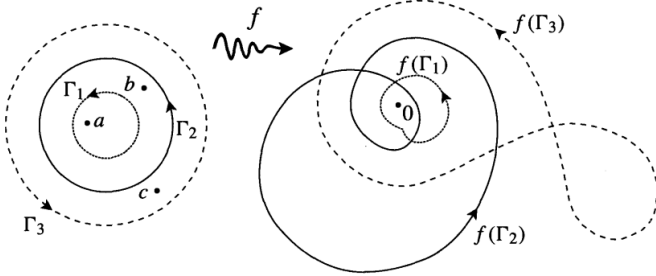
Corollary 5.29. For a positively oriented Jordan curve Γ , an analytic map $f : \text{BCJ}(\Gamma) \rightarrow \mathbb{C}$, and a regular value p of f satisfying $p \notin f(\Gamma)$, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} \text{sgn} \det f'(z_j). \quad (5.12)$$

Proof. For each preimage z_j of a regular value p of f , Lemma 2.50 states that $\det f'(z_j) > 0$. The rest follows from (5.9) and Lemma 5.27. $\square$

5.3.3 For meromorphic functions

Example 5.30. For constants $a, b, c \in \mathbb{C}$, consider the rational function $f(z) = AB\frac{1}{C}$ where $A = z - a$, $B = z - b$, and $C = z - c$. The plot below shows an expanding circle $\Gamma(t)$ at three stages of its growth Γ_1 , Γ_2 , and Γ_3 such that $\text{BCJ}(\Gamma_1) \subset \text{BCJ}(\Gamma_2) \subset \text{BCJ}(\Gamma_3)$.

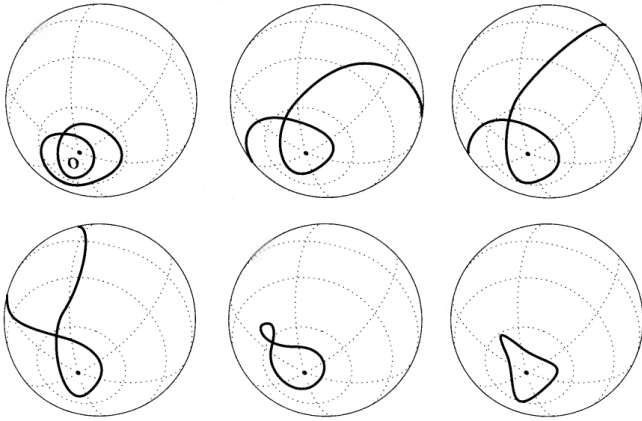


It can be verified that $w(f(\Gamma_1), 0) = 1$, $w(f(\Gamma_2), 0) = 2$, and $w(f(\Gamma_3), 0) = 1$, which invites two questions:

- (I) As $\Gamma(t)$ goes from Γ_2 to Γ_3 , why does $w(f(\Gamma(t)), 0)$ decrease from 2 to 1?
- (II) Exactly *how* does this decrease happen?

The answer to Question (I) is simple. As z traverses Γ_3 , the winding number of $f(z)$ is the sum of the number of revolutions executed separately by A , B , and $\frac{1}{C}$, cf. Lemma 5.23. As C rotates counterclockwise, $\frac{1}{C}$ rotates in the opposite direction, which decreases $w(f(\Gamma), 0)$.

Question (II) is best answered by illustrating the process on the Riemann sphere as below.



As shown in the top-middle, top-right, and bottom-left subplots, the expansion of $\Gamma(t)$ through c corresponds to the crossing of $f(\Gamma)$ of the north pole on the Riemann sphere, which leads to the unwinding of $f(\Gamma)$. We emphasize that this argument requires the homeomorphism of the extended complex plane $\overline{\mathbb{C}}$ and the Riemann sphere S^2 , cf. the stereographic projection in Definition 4.211.

Theorem 5.31. Suppose a positively oriented Jordan curve Γ in a simply connected domain $\mathcal{D}$ go through neither the zeros $a_1, a_2, \dots, a_n$ nor the poles $b_1, b_2, \dots, b_m$ of a meromorphic function $f : \mathcal{D} \rightarrow \overline{\mathbb{C}}$. Then we have

$$w(f(\Gamma), 0) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f'(z)}{f(z)} dz = N(f; \Gamma) - P(f; \Gamma), \quad (5.13)$$

where $N(f; \Gamma)$ and $P(f; \Gamma)$ are respectively the total number of zeros and poles (counting multiplicity) of f in $\text{BCJ}(\Gamma)$,

$$N(f; \Gamma) = \sum_{j=1}^n \text{ord}(f; a_j), \quad (5.14a)$$

$$P(f; \Gamma) = - \sum_{j=1}^m \text{ord}(f; b_j). \quad (5.14b)$$

Proof. The first equality follows from Definition 5.2 and the substitution rule for integrals. The second equality follows from Theorem 4.132 and Lemma 4.137 with $g \equiv 1$. $\square$

Example 5.32. For $f(z) = (z-1)(z-2)^2(z-6)$, the Cauchy integral formula gives

$$\frac{1}{2\pi i} \oint_{|z|=3} \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi i} \oint_{|z|=3} \left(\frac{1}{z-1} + \frac{2}{z-2} + \frac{1}{z-6} \right) dz = 3.$$

Alternatively, $f(z)$ has three zeros in $|z| = 3$ and Theorem 5.31 yields the result immediately. Yet another way to derive the result is to argue that $f(\{z : |z| = 3\})$ winds around the origin three times and then use the first equality in (5.13).

Corollary 5.33. Suppose a closed curve α in a simply connected domain $\mathcal{D}$ go through neither the zeros $a_1, a_2, \dots, a_n$ nor the poles $b_1, b_2, \dots, b_m$ of a meromorphic function $f : \mathcal{D} \rightarrow \overline{\mathbb{C}}$. Then we have

$$\begin{aligned} \frac{1}{2\pi i} \oint_{\alpha} \frac{f'(z)}{f(z)} dz &= \sum_{j=1}^n \text{ord}(f; a_j) w(\alpha, a_j) \\ &\quad + \sum_{j=1}^m \text{ord}(f; b_j) w(\alpha, b_j). \end{aligned} \quad (5.15)$$

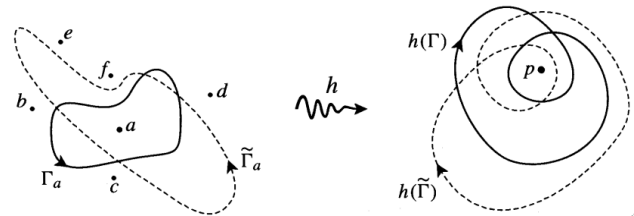
Proof. This follows directly from Theorem 5.31 and the Seifert decomposition (Lemma 3.70). $\square$

5.3.4 For continuous functions

Definition 5.34. The *topological multiplicity* or the *local degree* or the *index* of a continuous map $h : \Omega \rightarrow \mathbb{C}$ at a point $a \in \Omega$ is

$$\nu_h(a) := w(h(\Gamma_a), p), \quad (5.16)$$

where Γ_a is a positively oriented Jordan curve Γ_a such that $a \in \text{BCJ}(\Gamma_a) \subseteq \Omega$ and $\text{BCJ}(\Gamma_a)$ contains no other preimages of $p = h(a)$.



Example 5.35. The topological multiplicity $\nu_h(a)$ of the continuous function $h(z) = \bar{z}$ is -1 for any $a \in \mathbb{C}$.

Exercise 5.36. Is it appropriate to illustrate Example 5.26 by the plot below Definition 5.34? If yes, explain why; otherwise state how to modify the plot to illustrate Example 5.26.

Definition 5.37. The *topological degree* of a continuous map $h : \text{BCJ}(\Gamma) \rightarrow \mathbb{C}$ at $p \in \mathbb{C} \setminus h(\Gamma)$ is given by

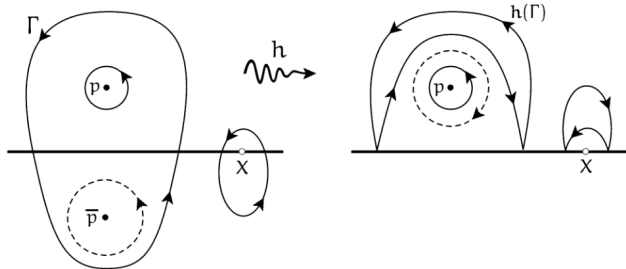
$$\deg(h, \text{BCJ}(\Gamma), p) := w(h(\Gamma), p). \quad (5.17)$$

Theorem 5.38. For a continuous map $h : \overline{\text{BCJ}(\Gamma)} \rightarrow \mathbb{C}$ and a point $p \in \mathbb{C} \setminus h(\Gamma)$, the topological degree of h at p equals the number of preimages of p in $\text{BCJ}(\Gamma)$, counted with their topological multiplicities,

$$\deg(h, \text{BCJ}(\Gamma), p) = \sum_{a \in h^{-1}(p)} \nu_h(a). \quad (5.18)$$

Proof. This follows from Definitions 5.34 and 5.37 and the first paragraph in the proof of Theorem 5.25. $\square$

Example 5.39. Depending on the choice of Γ , the topological degree of the continuous function $h(x + iy) = x + i|y|$ can be 0 or 1; see the plots below.



In particular, both the topological multiplicity $\nu_h(X)$ and the topological degree $\deg(h, \text{BCJ}(\Gamma), X)$ are always zero, regardless of Γ ; this curious fact can be understood by considering X as a coalescence of p and $\bar{p}$ at which their topological multiplicities cancel.

Exercise 5.40. For the continuous map $h(z) = |z|^2 - i\bar{z}$:

- Find the roots of h .
- For each root a of h , choose a positively oriented Jordan curve Γ_a and compute the topological multiplicity $\nu_h(a)$.
- Let $\Gamma : [0, 1] \rightarrow \mathbb{C}$, $\Gamma(t) = 2e^{2\pi it}$. Find the image curve $h(\Gamma)$ and compute the topological degree of h at 0.
- Use (b,c) to verify (5.18). Gain a better understanding of this equality by observing that $h(z) = \bar{z}(z - i)$, and then mimicking the analysis of Example 5.21.

5.4 Some consequences

5.4.1 The Darboux–Picard theorem

Theorem 5.41 (Darboux–Picard). Suppose $f : \Omega \rightarrow \mathbb{C}$ is an analytic function where Ω is a domain that contains a Jordan curve $\Gamma : [0, 1] \rightarrow \Omega$. If the restriction $f|_{\Gamma}$ is injective, then so is $f|_{\text{BCJ}(\Gamma)}$.

Proof. Since Γ is Jordan and $f|_{\Gamma}$ is injective and continuous, $f(\Gamma)$ is also a Jordan curve. For any $p \in \text{BCJ}(f(\Gamma))$, (5.8) gives $w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} w(\Gamma, z_j)m_j$. Suppose Γ is positively oriented. Then Lemma 5.4 and Lemma 5.6 imply $w(\Gamma, z_j) = 0$ for any $z_j \in \text{UCJ}(\Gamma)$ and $w(\Gamma, z_j) = 1$ for any $z_j \in \text{BCJ}(\Gamma)$. Consequently, $w(f(\Gamma), p)$ must be positive and $f^{-1}(p) \cap \text{BCJ}(\Gamma)$ must contain exactly one point; otherwise it would contradict (5.8). $\square$

Exercise 5.42. Show that Γ and $f(\Gamma)$ in Theorem 5.41 have the same orientation.

Corollary 5.43. Suppose an analytic function $f : \Omega \rightarrow \mathbb{C}$ satisfies the conditions in Theorem 5.41. Then the analytic function $\tilde{f} := f|_{\text{BCJ}(\Gamma)}$ is a globally conformal map whose inverse φ satisfies $\varphi' = \frac{1}{\tilde{f}'}$.

Proof. This follows from Theorem 5.41, Definition 2.38, and Theorem 2.74. $\square$

5.4.2 The Rouché theorem

Theorem 5.44 (Rouché). Let Γ be a positively oriented Jordan curve and $\mathcal{D} \subseteq \mathbb{C}$ an open set that contains $\text{BCJ}(\Gamma)$. If two analytic functions $f, g : \mathcal{D} \rightarrow \mathbb{C}$ satisfy

$$\forall z \in \Gamma, |f(z)| > |g(z)|, \quad (5.19)$$

then f and $f + g$ have the same number of zeros in $\text{BCJ}(\Gamma)$, i.e., $N(f; \Gamma) = N(f + g; \Gamma)$, cf. (5.14a).

Proof. For $t \in [0, 1]$, define a homotopy

$$F(z, t) = f(z) + tg(z)$$

so that $F(z, 0) = f(z)$ and $F(z, 1) = f(z) + g(z)$. Denote by $n(t) := N(F(z, t); \Gamma)$ the number of zeros of $F(z, t)$ in $\text{BCJ}(\Gamma)$. By (5.19), both f and $f + g$ are nonzero on Γ . Then the argument principle (Theorem 5.31) yields

$$n(t) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f'(z) + tg'(z)}{f(z) + tg(z)} dz.$$

$n(t)$ is continuous with respect to t , but on the other hand $n(t)$ is an integer. Hence $n(t)$ must be a constant; otherwise it would contradict the intermediate value theorem B.74. $\square$

Corollary 5.45. A polynomial $p(z) = \sum_{j=0}^n a_j z^j$ with $a_n \neq 0$ has t zeros inside the unit circle if

$$|a_t| > \sum_{j \neq t; j=0}^n |a_j|. \quad (5.20)$$

Proof. Let $f(z) = a_t z^t$ and $g(z) = p(z) - f(z)$. Then (5.20) implies $|f(z)| > |g(z)|$ for all z on the unit circle. Hence Rouché's theorem 5.44 implies that $p(z)$ and $f(z)$ both have t zeros inside the unit circle. $\square$

Example 5.46. Sometimes it is extremely easy to determine the number of roots of a polynomial equation inside the unit circle, as exemplified below.

equation	# of roots inside $ z = 1$
$z^7 - 6z^5 - 2z + 1 = 0$	5
$z^8 + 5z^4 - 2z^2 + 1 = 0$	4
$z^4 - 8z + 1 = 0$	1
$z^4 + 6z - 10 = 0$	0

Exercise 5.47. Show that all roots of the equation

$$p(z) = z^7 - z^3 + 12 = 0$$

are inside the annulus $1 < |z| < 2$.

Exercise 5.48. Show that the equation

$$p(z) = z^4 - 5z + 1 = 0$$

has one root in $|z| < \frac{1}{4}$ and three roots inside the annulus $\frac{3}{2} < |z| < \frac{9}{5}$.

5.4.3 The Hurwitz theorem

Theorem 5.49 (Hurwitz 1889). For a domain $\mathcal{D}$, suppose $(f_n : \mathcal{D} \rightarrow \mathbb{C})_{n \in \mathbb{N}}$ is a sequence of analytic functions that converge locally uniformly to a nonzero analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$. Then for any compact subset $K \subset \mathcal{D}$ that satisfies $f(z) \neq 0$ for any $z \in \partial K$, there exists $N \in \mathbb{N}$ such that, for all $n > N$, the functions f_n and f have the same number of zeros (counted with multiplicity) inside K .

Proof. Since any compact subset $K \subset \mathbb{C}$ must be the closure of the union of a finite number of disks with holes, cf. Definition 3.71, we can assume WLOG that K is a single disk without holes. The compactness of ∂K and the condition of f not vanishing on ∂K imply that

$$\exists \delta > 0 \text{ s.t. } \forall z \in \partial K, |f| \geq \delta.$$

The uniform convergence of $(f_n)_{n \in \mathbb{N}}$ to f over K gives

$$\exists N \in \mathbb{N} \text{ s.t. } \forall n > N, \forall z \in \partial K, |f_n(z) - f(z)| < \delta$$

and the proof is completed by Rouché's theorem 5.44. $\square$

Corollary 5.50. For a domain $\mathcal{D}$, suppose $(f_n : \mathcal{D} \rightarrow \mathbb{C})_{n \in \mathbb{N}}$ is a sequence of injective analytic functions that converge locally uniformly to an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$. Then f is either constant or injective.

Proof. Suppose f is neither constant nor injective. Then there exist distinct $x, y \in \mathcal{D}$ such that $f(x) = f(y) = f_0$. Define a new function $g = f - f_0$ and a new sequence (g_n) by $g_n(z) = f_n(z) - f_n(x)$. Since f_n converges uniformly to f on any compact subset K of $\mathcal{D}$, g_n converges uniformly to g on K . Choose K so that $y \in K$ but $x \notin K$.

Since each f_n is injective, $g_n(z) \neq 0$ for any $z \in \mathcal{D} \setminus \{x\}$. In other words, g_n has no zeros in K for any $n \in \mathbb{N}$. However, $g(z) = f(z) - f_0$, Theorem 4.27, and the assumption of f being nonconstant dictate that f has an isolated zero at $y \in K$, contradicting the Hurwitz theorem 5.49. $\square$

5.4.4 Root finding for an analytic function

Lemma 5.51. Let Ω be a simply connected domain that contains a positively oriented Jordan curve Γ . For analytic functions $f, h : \Omega \rightarrow \mathbb{C}$ and a point $p \notin f(\Gamma)$, we have

$$\frac{1}{2\pi i} \int_{\Gamma} h(z) \frac{f'(z)}{f(z) - p} dz = \sum_{z_j \in f^{-1}(p) \cap \text{BCJ}(\Gamma)} h(z_j) m_j, \quad (5.21)$$

where m_j is the multiplicity of the root z_j of $f(x) = p$.

Proof. The proof is similar to that of Theorem 5.25. The only difference is that, instead of deriving $w(f(\Gamma_j), p) = m_j$, we compute

$$\begin{aligned} \int_{\Gamma_j} h(z) \frac{f'(z)}{f(z) - p} dz &= \int_{\Gamma_j} h(z) \left(\frac{K'}{K} + \frac{g'}{g} \right) dz \\ &= \int_{\Gamma_j} h(z) \frac{g'}{g} dz = \int_{\Gamma_j} h(z) \frac{m_j}{z - z_j} dz = 2\pi i h(z_j) m_j, \end{aligned}$$

where the second step follows from $K(z_j) \neq 0$ and the Cauchy integral theorem 3.59, and the last from Cauchy integral formula 3.75. The rest of the proof follows from $\int_{\Gamma} h(z) \frac{f'(z)}{f(z) - p} dz = \sum_j \int_{\Gamma_j} h(z) \frac{f'(z)}{f(z) - p} dz$. $\square$

Corollary 5.52. Let Ω be a domain, $f : \Omega \rightarrow \mathbb{C}$ an analytic function, and Γ a positively oriented Jordan curve such that $\text{BCJ}(\Gamma) \subseteq \Omega$. If each of the n roots of $f(z) - p$ in $\text{BCJ}(\Gamma)$ is simple, then we have

$$\forall k \in 1, 2, \dots, n, \quad \sum_{j=1}^n z_j^k = \frac{1}{2\pi i} \int_{\Gamma} \frac{\zeta^k f'(\zeta)}{f(\zeta) - p} d\zeta. \quad (5.22)$$

Proof. Set $h(\zeta) = \zeta^k$ in Lemma 5.51 and we have (5.22). $\square$

5.5 Topological degrees in $\mathbb{R}^n$

5.5.1 Uniqueness of the degree

Theorem 5.53. There is at most one function $\deg : M \rightarrow \mathbb{Z}$,

$$M := \left\{ (f, \Omega, y) : \begin{cases} \Omega \subset \mathbb{R}^n \text{ open and bounded;} \\ f : \overline{\Omega} \rightarrow \mathbb{R}^n \text{ continuous;} \\ y \in \mathbb{R}^n \setminus f(\partial\Omega) \end{cases} \right\}, \quad (5.23)$$

that satisfies normalization, additivity, and homotopy, i.e.,

(TPD-1) $\deg(I, \Omega, y) = 1$ for all $y \in \Omega$ and the identity I ;

(TPD-2) $\deg(f, \Omega, y) = \deg(f, \Omega_1, y) + \deg(f, \Omega_2, y)$ whenever Ω_1 and Ω_2 are disjoint open subsets of Ω such that $y \notin f(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$;

(TPD-3) $\deg(h(t, \cdot), \Omega, y(t))$ is independent of $t \in [0, 1]$ if $y : [0, 1] \rightarrow \mathbb{R}^n$ is continuous, $y(t) \notin h(t, \partial\Omega)$ for all $t \in [0, 1]$, and $h : [0, 1] \times \overline{\Omega} \rightarrow \mathbb{R}^n$ is continuous.

Proof. See [Deimling, 2010, §1]. $\square$

5.5.2 Construction of the degree

Notation 8. $\mathcal{C}^k(\Omega)$ denotes the set of $f : \Omega \rightarrow \mathbb{R}^n$ that are k -times continuously differentiable in Ω and

$$\bar{\mathcal{C}}^k(\Omega) := \mathcal{C}^k(\Omega) \cap C(\overline{\Omega}). \quad (5.24)$$

Definition 5.54. For $(f, \Omega, y) \in M$ with $f \in \mathcal{C}^1(\Omega)$ and $y \notin S_f$, the *degree function* is defined as

$$\deg(f, \Omega, y) := \sum_{x \in f^{-1}(y)} \text{sgn } J_f(x). \quad (5.25)$$

In particular, $\deg(f, \Omega, y) = 0$ if $f^{-1}(y) = \emptyset$.

Definition 5.55. For $(f, \Omega, y) \in M$ with $f \in \mathcal{C}^2(\Omega)$, the *degree function* is defined as $\deg(f, \Omega, y) := \deg(f, \Omega, y^1)$ where y^1 is any regular value of f such that $|y^1 - y| < \rho(y, f(\partial\Omega))$, $\rho(y, S)$ is the distance from y to the set S , and $\deg(f, \Omega, y^1)$ is given by Definition 5.54.

Definition 5.56. For $(f, \Omega, y) \in M$, the *degree function* is defined as $\deg(f, \Omega, y) := \deg(g, \Omega, y)$ where $g \in \bar{\mathcal{C}}^2(\Omega)$ is any map such that $\|g - f\|_{\infty} < \rho(y, f(\partial\Omega))$ and $\deg(g, \Omega, y)$ is given by Definition 5.55.

5.5.3 Further properties

Theorem 5.57. The degree function $d : M \rightarrow \mathbb{Z}$ given by Definition 5.56 has the following properties

- (TPD-1) $\deg(I, \Omega, y) = 1$ for all $y \in \Omega$ and the identity I ;
- (TPD-2) $\deg(f, \Omega, y) = \deg(f, \Omega_1, y) + \deg(f, \Omega_2, y)$ whenever Ω_1 and Ω_2 are disjoint open subsets of Ω such that $y \notin f(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$;
- (TPD-3) $\deg(h(t, \cdot), \Omega, y(t))$ is independent of $t \in [0, 1]$ if $y : [0, 1] \rightarrow \mathbb{R}^n$ is continuous, $y(t) \notin h(t, \partial\Omega)$ for all $t \in [0, 1]$, and $h : [0, 1] \times \overline{\Omega} \rightarrow \mathbb{R}^n$ is continuous.
- (TPD-4) $\deg(f, \Omega, y) \neq 0$ implies $f^{-1}(y) \neq \emptyset$.
- (TPD-5) $\deg(f, \Omega, \cdot)$ and $\deg(\cdot, \Omega, y)$ are constant on $\mathcal{U}_r(y) \subset \mathbb{R}^n$ and $\{g \in \mathcal{C}(\overline{\Omega}) : \|g - f\|_\infty < r\}$, respectively, where $r := \rho(y, f(\partial\Omega))$. Furthermore, $\deg(f, \Omega, \cdot)$ is a constant on each connected component of $\mathbb{R}^n \setminus f(\partial\Omega)$.
- (TPD-6) $\deg(f, \Omega, y) = \deg(g, \Omega, y)$ whenever $f|_{\partial\Omega} = g|_{\partial\Omega}$.
- (TPD-7) $\deg(f, \Omega, y) = \deg(f, \Omega_1, y)$ for every open subset Ω_1 of Ω satisfying $y \notin f(\overline{\Omega} \setminus \Omega_1)$.

5.5.4 The product formula

Theorem 5.58. Suppose $\Omega \subset \mathbb{R}^n$ is open and bounded, $f \in \mathcal{C}(\Omega)$, $g \in \mathcal{C}(\mathbb{R}^n)$, $y \notin (g \circ f)(\partial\Omega)$, and K_i are the bounded connected components of $\mathbb{R}^n \setminus f(\partial\Omega)$. Then

$$\deg(g \circ f, \Omega, y) = \sum_i \deg(f, \Omega, K_i) \deg(g, K_i, y), \quad (5.26)$$

where $\deg(f, \Omega, K_i) = \deg(f, \Omega, y_i)$ for any $y_i \in K_i$ and the summation has a finite number of nonzero terms.

Definition 5.59. For $f \in \mathcal{C}(\overline{\mathcal{U}_r(x_0)})$ and $y = f(x_0)$ with $y \neq f(x)$ in $\overline{\mathcal{U}_r(x_0)} \setminus \{x_0\}$, the degree $\deg(f, \mathcal{U}_q(x_0), y)$ for $q \in (0, r)$ is called the *index of f at x_0* and denoted by $j(f, x_0, y)$.

Corollary 5.60. For $f \in \mathcal{C}(\overline{\Omega})$, $g \in \mathcal{C}(\mathbb{R}^n)$, $y \notin (g \circ f)(\partial\Omega)$, and $g^{-1}(y) = \{z_1, \dots, z_p\}$, we have

$$\deg(g \circ f, \Omega, y) = \sum_{j=1}^p \deg(f, \Omega, z_j) j(g, z_j, y). \quad (5.27)$$

In particular, $\deg(g \circ f, \Omega, y) = \sum_{j=1}^p \deg(f, \Omega, z_j)$ if each point in $g^{-1}(y)$ is a regular value of f .

Corollary 5.61. For an analytic function $f : \overline{\mathcal{U}_r(z_0)} \rightarrow \mathbb{C}$ that satisfies $f(z_0) = 0$ and $f(z) \neq 0$ for any $z \in \mathcal{U}_r(z_0) \setminus \{z_0\}$, its index equals its algebraic multiplicity m_0 .

Proof. Construct a function $g(z) = z_0 + qz$ with $q > 0$ sufficiently small. Then we have

$$\begin{aligned} j(f, z_0, 0) &= \deg(f, \mathcal{U}_q(z_0), 0) = \deg(f \circ g, \mathcal{U}_1(0), 0) \\ &= w(f(g(\partial\mathcal{U}_1(0))), 0) = w(f(\partial\mathcal{U}_q(0))), 0 \\ &= m_0, \end{aligned}$$

where the first step follows from Definition 5.59, the second from Theorem 5.58, the third from Definition 5.56, and the last from Theorem 5.25. $\square$

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